

Large Language Model-Guided Discovery of Weight-Five Bivariate Bicycle Codes

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Building on our earlier program-evolution workflow guided by large language models (LLMs), we study weight-five bivariate bicycle (BB) and perturbed bivariate bicycle (PBB) codes. The resulting catalogue contains 1,142 distinct code proposals, including 1,081 nonbaseline proposals attributable to LLM-generated programs. Across the catalogue, we certify connected Calderbank–Shor–Steane (CSS) realizations $[[96, 4, 10]]$, $[[140, 6, 10]]$, and $[[180, 4, 14]]$. A post-search comparison certifies seven imported Lin–Pryadko archive constructions. For leading parameter triples also represented in that archive, we provide exact distance evidence, explicit bivariate presentations, and verified component reductions. A basis-independent connectivity analysis identifies 409 of the 1,142 catalogue entries as disconnected and shows that 73.1% of the classes with exact distance certificates contain repeated connected components. Algebraic analysis organizes the connected CSS classes into order-3, order-7, and order-15 cyclotomic-kernel strata. The strongest exact connected PBB parameter point is $[[216, 4, 10]]$, attained by two distinct component classes. Among the 936 distinct CSS proposals from the LLM-guided campaign with a recorded positive distance, 816 (87.18%) are certified at $d \geq 5$. For comparison, three random-search controls each sample 6,444 CSS code proposals uniformly without replacement, using the same per-lattice and encoded-dimension sample counts as the LLM-guided campaign. In these controls, 4,672–4,785 proposals (72.50–74.26%) meet the same criterion. The LLM-guided campaign has the higher certified yield, while the random controls cover more connected classes. Together, these results extend LLM-guided discovery to a more constrained code family and provide a reproducible structural and exact-distance account of its strongest candidates.

I. INTRODUCTION

Bivariate bicycle (BB) codes are a prominent family of Calderbank–Shor–Steane (CSS) quantum low-density parity-check (qLDPC) codes. The most extensively studied BB codes use two trinomials, yielding weight-six stabilizer generators [1]. Our previous work introduced and benchmarked a program-evolution workflow guided by large language models (LLMs) for BB and perturbed bivariate bicycle (PBB) codes [2], including comparisons with uniform random sampling and genetic algorithms, and applied it to weight-six and higher-weight CSS and PBB codes. Here we extend that workflow to weight-five codes, a more constrained polynomial family, and combine candidate discovery with systematic distance certification, structural reduction, and algebraic classification. “Weight five” denotes the Pauli weight of each stabilizer generator. For our regular binomial–trinomial $(2 + 3)$ CSS constructions, the data-qubit degree also equals five; circuit connectivity and gauge-check weight remain distinct quantities. We apply a stochastic LLM-guided program-evolution search to bivariate codes with a $2 + 3$ support split and block lengths $72 \leq n \leq 432$. The CSS campaigns optimize kd^2/n across complementary profiles of small and large periodic lattices, whereas the PBB campaigns prioritize non-CSS candidates with $d \geq 5$. The targeted search provides a finite-length view of these families. The six CSS distance-evaluation lattices contain 13,656,867 normalized binomial–trinomial pairs. The logged positive-distance output contains 936 distinct CSS proposals, of which 912 remain after the $d \leq 2$ filter. A three-replicate stratified-uniform control complements the general uniform-random and genetic-

algorithm comparisons in the earlier work by probing transfer within the weight-five family. A post-search comparison establishes that our $[[96, 4, 10]]$ realization is equivalent to a Lin–Pryadko archive construction over the cyclic group C_{48} of order 48; several additional results match archive parameters without an equivalence claim. For these overlaps, we provide exact certificates, explicit bivariate realizations, and verified parent–component relations. The resulting catalogue links the discovered proposals to distance evidence, connected components, equivalence classes, algebraic structure, and finite-length comparison frontiers.

II. RELATED WORK AND COMPARISON SCOPE

Recent artificial-intelligence-assisted quantum error correction (QEC) code discovery spans reinforcement-learning searches for low-weight and decoder-tailored BB codes [3, 4], Bayesian optimization of BB codes [5], and LLM- or agent-guided exploration of lifted-product and broader balanced-product constructions and code–circuit–decoder co-design [6–8]. The present work belongs to this emerging methodological line while focusing on a comparatively less explored structural regime: weight-five BB and related qLDPC constructions.

In this regime, general weight-reduction methods can transform any qLDPC code into one with stabilizer weight at most five, but need not preserve the bicycle algebraic form [9]. Because the finite-length overhead depends on the construction, we include a transformed code as a weight-five comparator only when its checks have

been constructed explicitly and its parameters certified. Voss *et al.* reported the trivariate bicycle code [30, 4, 5] and the additional weight-five examples [72, 4, 8] and [96, 4, 8] [10]. We include them in the bivariate cohort because the third shift in their encoding satisfies $z = xy$, reducing each presentation to an ordinary BB code—or, equivalently, an abelian two-block group-algebra (2BGA) code. Khesin and Lu tabulated the weight-five mirror code [36, 4, 6], which is locally Clifford equivalent to a CSS code [11]. Through the mirror–2BGA correspondence, we likewise include its bivariate $2 + 3$ representation. The [30, 4, 5] construction tabulated by Khesin and Lu is equivalent to the Voss code and is counted only once.

The generalized-bicycle and two-block group-algebra literature provides a broader family-level comparison. Panteleev and Kalachev introduced generalized-bicycle (GB) codes over cyclic groups and demonstrated their finite-length performance [12]. Lin and Pryadko exhaustively enumerated connected 2BGA presentations of stabilizer weight at most eight, considering codes over abelian ambient groups through block length $n = 100$ and codes over nonabelian ambient groups through $n = 200$ [13]. In their archive tuple (G, k, d) , G is the ambient group, k the encoded dimension, and d the distance; the associated 2BGA code has block length $n = 2|G|$. Their accompanying machine-readable archive [14] retains the first construction encountered for each such tuple. It is more detailed than the parameter tables in the article and includes the weight-five constructions used in our comparison below. They estimated distances with the probabilistic `DistRandCSS` routine from `QDistRnd` [15], using 10^5 trials. We treat these archived values as randomized estimates and distinguish them from exact distances in the comparison.

The archive serves as a post hoc comparison set and was not used during the campaign. Its exhaustive abelian range ends at block length $n = 100$, below the six searched abelian lattices, whose parent block lengths satisfy $n \geq 120$. Moreover, retaining one construction per (G, k, d) tuple supports parameter-level comparison but not same-lattice recall or code equivalence. Sections IV B and IV D report the equivalence and distance evidence used for individual constructions.

The literature at adjacent check weights is much broader. At weight six, it includes the BB codes of Bravyi *et al.* [1], the exact mirror code [60, 4, 10] [11], the coprime constructions of Wang and Mueller [16], and the generalized-toric enumeration of Liang *et al.* [17]. Qian and Li report coset-orbit balanced-product codes with distances certified by mixed-integer linear programming (MILP), including [336, 12, 20] at overall weight six and [224, 22, 16], [288, 24, 18], and [378, 32, 19] at overall weight eight [8]. From their published construction supports, we assign these four to the same maximum-stabilizer-weight classes used here; their overall-weight convention additionally bounds the total number of checks incident on each qubit. Other weight-eight

comparators include the covering-graph construction [144, 14, 14] [18] and the catalogue from our previous search [2].

For a broader benchmark, we use the August 28, 2026 repository snapshot (accessed August 29, 2026) of the Unitary Foundation QEC Challenge [19], a public leaderboard of verifier-checked submissions. We use its submitted family labels as provenance metadata rather than equivalence classifications. We assign each entry a check weight equal to the maximum nonzero row weight of its submitted H_X and H_Z matrices. The comparison set comprises all 97 entries labelled BB in that revision and the five weight-five entries outside that family. We analyze their evidence and connectivity in Sec. IV D.

Weight-five abelian 2BGA codes admit a three-dimensional local representation; Ref. [20] derives the conditional finite-size upper bound $d = O(n^{2/3})$. Accordingly, the conventional metric kd^2/n summarizes finite-length performance. For the literature comparison, we construct separate exact Pareto frontiers by check weight for two pairs of objectives: rate and distance, both maximized; and block length and kd^2/n , with n minimized and kd^2/n maximized. The plotted frontiers use exact distances, with nonexact evidence retained in the accompanying dataset.

III. SEARCH SPACE AND METHODS

A. Code families and search objective

For a periodic lattice with side lengths (ℓ, m) , let

$$R_{\ell,m} = \mathbb{F}_2[x, y]/(x^\ell - 1, y^m - 1). \quad (1)$$

This is the binary group algebra of the lattice translation group $C_\ell \times C_m$, where C_r denotes the cyclic group of order r . We represent each polynomial by its $\ell m \times \ell m$ binary circulant matrix. A CSS BB code has check matrices

$$H_X = \begin{pmatrix} A & B \end{pmatrix}, \quad H_Z = \begin{pmatrix} B^\top & A^\top \end{pmatrix}, \quad (2)$$

with block length $n = 2\ell m$. Commutativity of $R_{\ell,m}$ gives $H_X H_Z^\top = AB + BA = 0$. We impose

$$\{|\text{supp}(A)|, |\text{supp}(B)|\} = \{2, 3\}. \quad (3)$$

The $2 + 3$ and $3 + 2$ orientations differ only by exchanging the two qubit sectors. We exclude the $1 + 4$ split because a monomial block is a unit in $R_{\ell,m}$, making both CSS check matrices full row rank; the resulting BB code therefore encodes no logical qubits.

The non-CSS PBB ansatz from Ref. [2] is

$$H = \left(\begin{array}{cc|cc} A & B & C & D \\ 0 & 0 & B^\top & A^\top \end{array} \right), \quad (4)$$

where the vertical bar separates the symplectic X and Z coordinates. We require

$$\begin{aligned} \text{supp}(C) &\subseteq \text{supp}(A), & \text{supp}(D) &\subseteq \text{supp}(B), \\ (C, D) &\neq (0, 0). \end{aligned} \quad (5)$$

Symplectic commutativity further requires $AC^\top + BD^\top$ to be symmetric over $\mathbb{F}_2$. Support containment ensures that the mixed rows have Pauli weight $|\text{supp}(A) \cup \text{supp}(C)| + |\text{supp}(B) \cup \text{supp}(D)| = 5$; each generator in the second block row also has weight five.

For the CSS family in Eq. (2), every data qubit also has Tanner degree $|\text{supp}(A)| + |\text{supp}(B)| = 5$. Stabilizer weight and qubit degree happen to coincide for these regular constructions; they need not coincide for the literature comparators or for general weight-reduction procedures.

We compare codes using the finite-length figure of merit (FOM)

$$\text{FOM} = \frac{kd^2}{n}, \quad (6)$$

following our earlier work [2]. Equation (6) serves as a finite-length comparison metric without assuming two-dimensional locality. Its normalization echoes the two-dimensional local-code tradeoff $kd^2 = O(n)$ [21]. For an idealized baseline of k independent rotated surface-code patches, each using d^2 data qubits, Eq. (6) is exactly the ratio of surface-code data qubits to BB-code data qubits at the same nominal (k, d) . This data-qubit ratio isolates one finite-length resource measure; circuit and hardware costs are evaluated separately where corresponding data are available.

B. Search protocol

Four 300-iteration OpenEvolve campaigns [22] evolved Python functions that generate polynomial tuples across several lattices. The CSS fitness function favored high-FOM weight-five 2 + 3 BB candidates, and the PBB fitness function prioritized non-CSS candidates with $d \geq 5$. Each family used complementary small- and large-lattice profiles and a staged protocol that reserved distance calculations for promising candidates while sampling low-, medium-, and high- k bands. An ensemble of Claude Opus 5 and GPT-5.6 Sol generated program variants, with model labels retained as provenance. The small profile used stage-1 lattices (6, 6), (6, 9), (9, 8) and stage-2 lattices (6, 10), (9, 9), (10, 9); the large profile used (12, 9), (12, 12), (15, 12) and (16, 12), (15, 14), (18, 12), respectively. CSS distance evaluation used the three stage-2 lattices of each profile, six in total. For candidates that passed Stage 1, PBB distance evaluation covered all six lattices of the corresponding profile.

Generated tuples were normalized, deduplicated within each batch, filtered by exact $\mathbb{F}_2$ rank and the family constraints, and stratified into low-, medium-, and high- k bands before distance evaluation. We call a positive-distance record an *observation* and a distinct normalized (ℓ, m, A, B, C, D) tuple a *proposal*. Repeated observations can therefore map to the same proposal. Proposals with a proved value or reported endpoint $d \leq 2$

are excluded from the final catalogue. The Supplemental Material gives the per-campaign accounting.

C. Distance certification and catalogue reduction

Initial CSS ranking used belief propagation with ordered-statistics decoding (BP-OSD), followed for the strongest candidates by combination-sweep OSD and an exact-search attempt. PBB candidates first underwent exhaustive low-weight search and then size-dependent symplectic MILP and BP-OSD. These stages rank candidates and produce distance estimates or solver bounds; exact-distance status is assigned using the criteria below.

We call d exact only when exhaustive enumeration or an analytic argument provides a lower bound matching a logical witness, or when all required MILP problems terminate with proven optima. Otherwise, an explicit logical witness or a nonoptimal MILP incumbent supplies an upper bound $d \leq d_{\text{ub}}$; a decoder scalar whose operator was not retained is reported only as an estimated upper endpoint. The MILP uses the Landahl-Anderson-Rice binary formulation [23], implemented with `scipy.optimize.milp` [24] and HiGHS, with one problem for each of the $2k$ logical objectives. Every reported witness is checked for weight, zero stabilizer syndrome, and nontrivial logical action. Whenever weight-four exclusion and a weight-five witness jointly establish $d = 5$, we reconstruct explicit weight-five logical operators in both sectors.

Fixed follow-up rules selected retained CSS proposals with estimated $\text{FOM} \geq 5$, all retained PBB-small proposals with $d = 6$, and PBB-large proposals evaluated by both MILP and BP-OSD. The high- k CSS tail received componentwise analysis. These rules concentrate certification on the strongest decoder-ranked candidates.

To identify repeated presentations, we map every retained proposal to a colored Tanner graph and canonicalize it with BLISS [25]. CSS graph colors distinguish qubits, X checks, and Z checks. For PBB codes, separate vertices represent the X and Z supports of each stabilizer row and a tying edge preserves their pairing. Equal canonical graph signatures define a presentation class under qubit and stored-generator permutations, within which compatible distance evidence is transferred.

For each CSS presentation class containing at least one proposal without a certified direct lower bound, we exhaustively searched both logical sectors for nontrivial operators of weight at most four. We enumerated all supports of at most two qubits and combined pairs whose syndromes cancel, rejecting combinations that are stabilizers. To identify disconnected components independently of the chosen generators, we row-reduced the binary symplectic check matrix and grouped qubits that co-occur in its unique reduced row echelon form. This gives the finest decomposition of the stabilizer group into disjoint qubit sets. We then canonicalized one connected component of each parent and used these compo-

nent classes, rather than replicated parents, in the Pareto comparison.

D. Stratified-uniform transfer control

Our preceding paper benchmarks the LLM-guided workflow against uniform random sampling and genetic algorithms in Sec. VI F and Supplemental Appendix B of Ref. [2]. Here we ask the narrower question of how the workflow transfers to weight-five CSS codes. On each of the six CSS distance-evaluation lattices, three control replicates sample normalized pairs $A = \{1, a\}$ and $B = \{1, b, c\}$ uniformly without replacement. Sampling is stratified by lattice over the $k = 4$ and $k \geq 6$ bands and continues until each stratum matches the corresponding count of logged positive-distance campaign observations. The control therefore inherits the campaign’s lattice–band allocation and tests selection within strata, not allocation among them. Each selected pair follows the same BP–OSD cascade, followed by an exhaustive search for nontrivial logical operators of weight at most four. Exact evidence is transferred only between proposals in the same certified presentation class.

The primary endpoints are the certified- $d \geq 5$ fraction, connected certified-class coverage, and highest-FOM campaign-certified class sampled. The Supplemental Material summarizes campaign and solver budgets. Full prompts, configurations, canonicalization details, and validation artifacts are available in the code repository [26].

IV. RESULTS

A. Catalogue certification and structure

The four campaigns produced 13,923 positive-distance observations representing 1,297 distinct proposals. After excluding 155 proposals with an exact value or reported endpoint $d < 2$, the retained catalogue contains 12,036 observations across 1,142 proposals: 912 CSS and 230 PBB. Of these, 1,081 are nonbaseline proposals attributable to LLM-generated programs. The remaining 61 comprise 35 seed/null-only proposals, six without a retained attribution event, and 20 independently supplied fixed baselines; the baselines are excluded from the 1,081 even when evaluated by model-generated programs. The Supplemental Material gives the per-campaign accounting.

Distance certification began with the targeted follow-up summarized in Table I, which established exact distances for 106 of 175 retained targets. Its 50 exact CSS results, together with six retained high- k component certificates, yielded exact distances for 56 CSS proposals before the catalogue-wide low-weight search.

For the 96 prespecified CSS follow-up targets, including two later excluded at $d = 2$, the ratio of the initial

TABLE I. Certification outcomes for the targeted follow-up after excluding proposals with $d \leq 2$. “Nonexact” combines a certified lower bound with either a rigorous upper bound or a decoder estimate.

Campaign	Targets	Exact	Nonexact
CSS-small	16	12	4
CSS-large	78	38	40
PBB-small	30	30	0
PBB-large	51	26	25
Total	175	106	69

estimated endpoint to the final reported endpoint had medians 1.80 in the small profile and 2.08 in the large profile, with maximum 14.5. Class-level evidence also materially tightens individual presentations: five presentations of one exact $\llbracket 432, 4, 10 \rrbracket$ class comprised one exact result at 10 and four decoder estimates at 10, 20, 22, and 23; the exact result applies to all five.

The catalogue-wide low-weight search examined 492 CSS classes containing proposals without direct lower-bound evidence. It found two $d = 2$ proposals, which were excluded from the catalogue; among retained entries it resolved 62 classes and 92 proposals at $d = 3$ or 4, and certified $d \geq 5$ in 428 classes and 764 proposals. Of the latter, 45 proposals also have explicit weight-five logical operators in both sectors and are therefore exact. Thus every retained CSS entry has a certified positive lower bound.

The 1,142 retained proposals form 630 presentation classes: 535 CSS and 95 PBB. Before the final presentation-class merge, 398 proposals have exact row-level status. Merging class evidence adds 38, giving 436 proposals in 227 exact classes; the remaining 403 classes retain certified lower bounds paired with either rigorous upper bounds or decoder estimates. No class contains conflicting distances or incompatible bounds.

The basis-independent connectivity partition agrees with the stored-generator Tanner partition for all 1,142 proposals. It identifies 307 of 912 CSS proposals (33.7%) and 102 of 230 PBB proposals (44.3%) as disconnected, for 409 of 1,142 overall (35.8%). At the class level, the corresponding counts are 206 of 535 CSS (38.5%) and 62 of 95 PBB (65.3%), or 268 of 630 overall (42.5%). Among those exact classes, 166 are disconnected (73.1%): 108 of 146 CSS and 58 of 81 PBB. For each disconnected proposal, the components have equal size, their dimensions sum to the parent k , and their colored Tanner graphs are pairwise isomorphic under BLISS canonical labelling. Accordingly, $m \times \llbracket n_0, k_0, d \rrbracket$ reports the decomposition inferred from the parent construction, not an independently generated smaller lattice presentation. Explicit connected realizations for three CSS and four PBB component classes are given in Secs. IV B and IV E. Pareto selection uses one representative of each component class, so replication cannot create a frontier point.

Componentwise analysis of seven high- k CSS parents finds that each is a direct sum of pairwise-isomorphic components. One parent, $18 \times \llbracket 24, 4, 2 \rrbracket$, is excluded by the $d \leq 2$ rule; the other six are retained exact catalogue entries tabulated in the Supplemental Material. Full component certificates are available in the repository [26]. In particular, all four retained CSS proposals with $k \geq 40$ are disconnected direct sums with exact distances $d \leq 5$. More generally, 408 of the 409 disconnected proposals have component dimension $k_0 \in \{2, 4, 6, 8\}$; the sole exception is $\llbracket 420, 20, 5 \rrbracket = 2 \times \llbracket 210, 10, 5 \rrbracket$. No retained connected CSS proposal has $k \geq 20$: the largest connected dimension is 16, attained by the exact $\llbracket 384, 16, 4 \rrbracket$ code, while the largest connected PBB dimension is four. Thus most of the catalogue's large parent dimensions arise through replication.

Repeating the canonical labelling after replacing each parent by one connected component reduces the 630 parent classes to 608 component classes: 528 CSS and 80 PBB, with 206 exact and 402 nonexact classes. This component quotient is the catalogue used for comparisons between connected constructions. The shortest components with certified exact distances are a PBB $\llbracket 12, 2, 3 \rrbracket$ code and a CSS $\llbracket 24, 4, 3 \rrbracket$ code, both with FOM 1.50.

The following subsections examine individual constructions. Table II summarizes the provenance of the CSS, PBB, and archive results discussed below.

B. CSS codes and leading candidates

Three exact disconnected parent codes have components that admit explicit shorter BB presentations. The $\llbracket 384, 16, 10 \rrbracket$ parent decomposes into four isomorphic components; the connected BB presentation on $(\ell, m) = (16, 3)$,

$$A = 1 + y + x^2y^2, \quad B = 1 + x^3$$

is BLISS-isomorphic to one component and therefore defines an exact $\llbracket 96, 4, 10 \rrbracket$ code. Likewise, the three components of $\llbracket 420, 18, 10 \rrbracket$ admit the connected BB presentation on $(\ell, m) = (5, 14)$,

$$A = 1 + y^{13} + x^4y^{11}, \quad B = 1 + xy^7,$$

giving $\llbracket 140, 6, 10 \rrbracket$. Their FOMs are 4.17 and 4.29. In particular, under the element ordering of the GAP (Groups, Algorithms, Programming) computer-algebra system used by the Lin-Pryadko archive, the archived $\llbracket 96, 4, 10 \rrbracket$ construction over C_{48} uses the group-algebra generator pair $(1 + z^3, 1 + z^{16} + z^2)$. This is the pair in Table IV, up to exchanging the two polynomial blocks. The exact component certificate obtained here also certifies that archived construction and identifies it with the explicit connected BB presentation on $(\ell, m) = (16, 3)$ given above. This adds an explicit bivariate realization and parent-component relation for that archive construction. Relative to the Voss $\llbracket 96, 4, 8 \rrbracket$ benchmark, the $\llbracket 96, 4, 10 \rrbracket$ parameter point has larger distance at the same (n, k) .

The exact $\llbracket 180, 8, 9 \rrbracket = 2 \times \llbracket 90, 4, 9 \rrbracket$ parent similarly gives the connected BB presentation on $(\ell, m) = (5, 9)$,

$$A = 1 + x^2y^8 + x^3y^7, \quad B = 1 + x^2y^6.$$

Its component FOM is 3.60. We retain this explicit construction and compare it with the compiled benchmarks below.

Among the connected CSS codes in the present catalogue, $\llbracket 180, 4, 14 \rrbracket$ has the largest exact FOM, 4.356; $\llbracket 180, 4, 13 \rrbracket$ provides another exact result, with FOM 3.76. Among connected CSS classes with nonexact distance, the largest supported FOM ceiling is $\llbracket 432, 4, 5 \leq d \leq 23 \rrbracket$, at 4.90. Two non-isomorphic singleton classes tie next at $\llbracket 420, 4, 5 \leq d \leq 22 \rrbracket$, with ceiling 4.61. The exhaustive low-weight search certifies the lower endpoints. All three presentations were included among the original 96 CSS targets. The two length-420 upper endpoints are nonoptimal MILP incumbents; an independently verified logical operator tightens the length-432 upper bound from the MILP incumbent 24 to 23. These interval classes are distinct from the exact $\llbracket 432, 4, 10 \rrbracket$ and $\llbracket 420, 4, 5 \rrbracket$ presentation classes with the same respective (n, k) values. Table III gives their generators alongside selected exact constructions. The supplement contains the compact catalogue, and the full machine-readable catalogue is available in the repository. At $n = 120$, the connected $\llbracket 120, 4, 10 \rrbracket$ and disconnected $\llbracket 120, 16, 5 \rrbracket = 4 \times \llbracket 30, 4, 5 \rrbracket$ codes both have FOM 3.33; the comparison uses the connected code.

The codes on the large-profile CSS lattices have block lengths beyond the exhaustive abelian regime of Ref. [13]. The exactly certified connected codes in this profile have parameters $\llbracket 384, 16, 4 \rrbracket$, $\llbracket 384, 4, 4 \rrbracket$, $\llbracket 420, 10, 5 \rrbracket$, $\llbracket 420, 6, 5 \rrbracket$, $\llbracket 420, 4, 5 \rrbracket$, $\llbracket 420, 6, 3 \rrbracket$, $\llbracket 432, 4, 10 \rrbracket$, and $\llbracket 432, 4, 4 \rrbracket$. The best FOM among them is 0.93 for $\llbracket 432, 4, 10 \rrbracket$.

C. Algebraic family classification

The exact connected CSS constructions classified below share a common quotient-character structure. Writing the lattice translation group additively, let $G = \mathbb{Z}_\ell \times \mathbb{Z}_m$ and let $\phi : G \rightarrow \mathbb{Z}_q$ be a surjective homomorphism. For a primitive q th root of unity ω in an extension field of $\mathbb{F}_2$, evaluate a monomial by

$$x^i y^j \mapsto \omega^{\phi(i,j)}. \quad (7)$$

If the two terms of the binomial have the same image and the three terms of the trinomial sum to zero, both BB polynomials vanish on this character. Its Frobenius orbit is then a common-kernel sector. A multiplicity-one order-three orbit has degree two and contributes four logical dimensions; a multiplicity-one order-seven orbit has degree three and contributes six. In a non-semisimple group algebra, a repeated common factor contributes according to its multiplicity, so character vanishing alone

TABLE II. Provenance of selected exact results. “Parameter match” records a shared (n, k, d) ; equivalence is listed where an explicit certificate is available.

Parameters	Role in this study	Prior-art relationship
$\llbracket 96, 4, 10 \rrbracket$	Connected realization of a generated CSS parent	Equivalent to archive construction over C_{48}
$\llbracket 140, 6, 10 \rrbracket$	Connected realization of a generated CSS parent	Archive parameter match
$\llbracket 180, 4, 14 \rrbracket$	Generated directly as a connected CSS presentation	Archive parameter match
$\llbracket 132, 4, 12 \rrbracket$	Post-search archive construction	Distance certified here
$\llbracket 216, 4, 10 \rrbracket$	Two connected PBB component classes	—

TABLE III. Selected connected weight-five CSS BB presentations. Exact distances are certified; interval rows combine a certified lower bound with a supported upper endpoint, so the corresponding displayed FOMs are upper bounds. M denotes exact MILP certification over all logical sectors of the displayed presentation; M+class transfers such a certificate from an isomorphic presentation in the same presentation class; M+component denotes a connected BB realization shown by BLISS canonicalization to be isomorphic to a component of an MILP-certified parent. B denotes a decoder estimate whose operator was not retained, I a nonoptimal MILP incumbent, W an independently reconstructed and verified logical operator, and L4 exhaustive exclusion through weight four. Each row specifies a presentation through its lattice and generators; matching (n, k) values do not imply equivalence.

Parameters	(ℓ, m)	$A(x, y)$	$B(x, y)$	FOM	Evidence
$\llbracket 90, 4, 9 \rrbracket$	(5, 9)	$1 + x^2y^8 + x^3y^7$	$1 + x^2y^6$	3.60	M+component
$\llbracket 96, 4, 10 \rrbracket$	(16, 3)	$1 + y + x^2y^2$	$1 + x^3$	4.17	M+component
$\llbracket 120, 4, 10 \rrbracket$	(6, 10)	$1 + xy + x^5y^5$	$1 + y$	3.33	M
$\llbracket 140, 6, 10 \rrbracket$	(5, 14)	$1 + y^{13} + x^4y^{11}$	$1 + xy^7$	4.29	M+component
$\llbracket 180, 4, 13 \rrbracket$	(10, 9)	$1 + x^2y^3$	$1 + x^5y + x^7y^8$	3.76	M
$\llbracket 180, 4, 14 \rrbracket$	(10, 9)	$1 + x^2y^2 + x^5y$	$1 + x^3y^3$	4.356	M
$\llbracket 432, 4, 10 \rrbracket$	(18, 12)	$1 + x^4y^3$	$1 + xy^{11} + x^2y^{10}$	0.93	M+class
$\llbracket 420, 4, 5 \leq d \leq 22 \rrbracket$	(15, 14)	$1 + x^3y$	$1 + x^{10}y^6 + x^{11}y^2$	≤ 4.61	L4+I+B
$\llbracket 420, 4, 5 \leq d \leq 22 \rrbracket$	(15, 14)	$1 + x^4y^3 + x^8y^{11}$	$1 + x^3y^3$	≤ 4.61	L4+I+B
$\llbracket 432, 4, 5 \leq d \leq 23 \rrbracket$	(18, 12)	$1 + x^4y^3$	$1 + x^5y^{11} + x^{10}y$	≤ 4.90	L4+I+W+B

does not determine k . For every selected construction below, direct rank calculations—and exact gcds for the constructions over cyclic ambient groups—verify completeness and multiplicity one.

This criterion is constructive. Translate one trinomial exponent to the identity. For $q = 3$, its other two exponents must map to residues 1 and 2, while the binomial exponent difference lies in $\ker \phi$. Thus a code in the order-three stratum is generated by choosing a quotient map and placing the three trinomial terms in the three distinct cosets of its kernel. For $q = 7$, choose ω with $1 + \omega + \omega^3 = 0$; the unordered residue pairs are (1, 3), (2, 6), and (4, 5), with unit multiples of ϕ covering the conjugate primitive factor. The same kernel condition applies to the binomial. These rules generate the order-three and order-seven strata before any distance calculation; direct rank calculation still determines multiplicity and the resulting k .

No analogous order-five stratum can exist. Because 2 has multiplicative order four modulo five, a primitive fifth root ω has minimal polynomial $\Phi_5(z) = 1 + z + z^2 + z^3 + z^4$ over $\mathbb{F}_2$. Under a surjective $\phi : G \rightarrow \mathbb{Z}_5$, the trinomial evaluates to $t(\omega)$, where $t(z) = z^{r_1} + z^{r_2} + z^{r_3}$ is reduced in $\mathbb{F}_2[z]/(z^5 - 1)$ and $r_i \in \mathbb{Z}_5$ are the images of its three exponents. If two residues coincide, those terms cancel and t is a single monomial; otherwise t has exactly three terms. In either case t is nonzero of degree at most four, so $t(\omega) = 0$ would force $t = \Phi_5$, which has five terms.

Hence the trinomial of a weight-five $2 + 3$ code never vanishes on an order-five character, for any lattice. The same argument excludes every prime $q \geq 5$ for which 2 is a primitive root modulo q ($q = 5, 11, 13, \dots$), since Φ_q is then irreducible with q terms. The argument does not apply to the orders that occur: Φ_3 is itself a trinomial, and Φ_7 and Φ_{15} each factor over $\mathbb{F}_2$ into two trinomials.

There is also a simple restriction specific to the $2 + 3$ support split. For an abelian BB code, $k/2$ is the dimension of the common kernel of the two regular-representation blocks. If that kernel were one-dimensional, its translation action would be trivial because $\text{GL}(1, \mathbb{F}_2) = \{1\}$, so it would be spanned by the all-ones vector. The even-weight binomial annihilates that vector, whereas the odd-weight trinomial does not. Hence a nontrivial code in this family cannot have $k = 2$. Independently, $k = 2 \dim(\ker A \cap \ker B)$ is even and therefore excludes $k = 3$. Its smallest possible nonzero dimension is consequently $k = 4$, explaining why the campaign’s $k \in \{2, 3\}$ sampling quota was empty.

When $\gcd(\ell, m) = 1$, the ambient group is cyclic. A change of generator based on the Chinese remainder theorem (CRT) identifies the code with a generalized-bicycle code over $\mathbb{F}_2[z]/(z^L - 1)$, where $L = \ell m$. If its two polynomials are $a(z)$ and $b(z)$, the GB/copprime-BB dimension formula [12, 16] is

$$k = 2 \deg h, \quad h = \gcd(a, b, z^L - 1). \quad (8)$$

Table IV identifies five exact constructions over cyclic ambient groups as members of the known cyclic GB/coprime-BB subclass. The two constructions over noncyclic ambient groups share the same order-three quotient mechanism, connecting all seven through a common design rule.

For connected 2 + 3 presentations over cyclic ambient groups, h is necessarily squarefree. Translating each polynomial by a monomial changes neither h nor connectivity, so write the binomial as $1 + z^e$ and the trinomial as $1 + z^f + z^g$ with $0 < f < g < L$. Suppose an irreducible p satisfies $p^2 \mid h$. Since $h = \gcd(z^L - 1, 1 + z^e, 1 + z^f + z^g)$, p^2 divides each of the three polynomials individually, hence so does p ; a repeated irreducible factor of a polynomial also divides its formal derivative, so p divides the formal derivatives of $z^L - 1$, $1 + z^e$, and $1 + z^f + z^g$, and $p \neq z$ because $z \nmid z^L - 1$. The first two conditions force L and e to be even. If exactly one of f and g were odd, p would divide a monomial; if both were odd, p would divide $z^{f-1}(1 + z^{g-f})$, hence $1 + z^{g-f}$, and therefore $1 + z^f(1 + z^{g-f}) \equiv 1 \pmod{p}$. Both cases are impossible, so e, f, g , and L are all even. Both translated supports then lie in the index-two subgroup $2\mathbb{Z}_L$, each check acts on a single coset of $2\mathbb{Z}_L$ in both blocks, and the Tanner graph splits into two parity components, contradicting connectivity.

The chosen lattice profile strongly influences which low-order quotient characters can occur; the observed pattern is not a consequence of weight five alone. All six CSS distance-evaluation translation groups, of orders 60, 81, 90, 192, 210, and 216, admit an order-three quotient, whereas only the order-210 group admits an order-seven quotient. Orders 60, 90, and 210 admit an order-15 quotient; the observed order-15 classes occur only at orders 90 and 210.

For the support-subgroup comparisons below, let

$$k_S = \dim \ker A + \dim \ker B - \dim \ker(AB). \quad (9)$$

The substantive relation is

$$k_S \leq \dim(\ker A \cap \ker B) = k/2, \quad (10)$$

with equality if and only if $\ker(AB) = \ker A + \ker B$. Equality holds in the semisimple case but can fail when $|G|$ is even.

The two selected exact classes presented over noncyclic ambient groups exhibit a second structural pattern. After translating one term of each support to the identity, the supports of both polynomials generate cyclic subgroups even though the ambient group is noncyclic. For $[[120, 4, 10]]$, the two directions may be chosen as $u = (1, 1)$ of order 30 and $v = (0, 1)$ of order 10 in $C_6 \times C_{10}$, with $u^6 = v^6$ and an intersection of order five. For $[[432, 4, 10]]$, $u = (4, 3)$ and $v = (1, 11)$ both have order 36 in $C_{18} \times C_{12}$, with $u^6 = v^6$ and an intersection of order six. For $[[120, 4, 10]]$, the common subgroup $N = \langle u \rangle \cap \langle v \rangle$ satisfies $N \cong C_5$; for $[[432, 4, 10]]$, it satisfies $N \cong C_6$. For $[[120, 4, 10]]$, both cyclic support-group extensions over N

split, and the quasi-abelian lifted-product reduction of Lin and Pryadko [13] identifies it with a hypergraph-product code over the binary group algebra $\mathbb{F}_2[C_5]$. For $[[432, 4, 10]]$, the corresponding C_{36} extensions over C_6 do not split. In the support-subgroup rank decomposition, its k_S lower bound vanishes even though the common-kernel dimension is two, so Eq. (10) is strict. The normal cyclic overlap alone therefore does not justify assigning this code the same lifted-product reduction.

We refer to this broader pattern as the noncyclic two-direction cyclic-support stratum. It contains 28 of the 114 noncyclic connected CSS presentation classes; the other 86 do not have cyclic support subgroups in any retained presentation. The split or nonsplit character further divides this stratum: 18 of the 28 classes are split and 10 are nonsplit, with the same classification across all retained presentations. Eight of the 10 nonsplit classes, including $[[432, 4, 10]]$, form a BB stratum characterized by nonsplit cyclic overlap and a strict bound: $k = 4$ and $k_S = 0 < k/2$. The remaining two saturate Eq. (10), so nonsplitting alone does not imply strictness. The $[[120, 4, 10]]$ and $[[432, 4, 10]]$ codes are the two high-distance exact members highlighted here. Seven of the 28 classes are exact (five at $d = 4$ and these two at $d = 10$), compared with nine of the other 86 noncyclic classes (two at $d = 3$, four at $d = 4$, and three at $d = 5$). Certified lower bounds of at least five occur in 23 of 28 and 80 of 86 classes, respectively. Because certification targeted selected candidates and nonexact distances are censored, we interpret these counts as structural coverage statistics for the stratum rather than as performance estimates.

Low-order quotient characters appear throughout the sampled lattices, not only among the leading codes. Enumerating every surjective homomorphism to $\mathbb{Z}_3, \mathbb{Z}_5, \mathbb{Z}_7$, and $\mathbb{Z}_{15}$ yields an order-three certificate in 298 of the 329 connected CSS presentation classes and an order-seven certificate in 21; two classes have both. As the argument above requires, no class has an order-five certificate. Thus 317 of 329 classes lie in those two low-order strata. The remaining 12 are precisely classes with cyclic presentations and one of the two primitive order-15 quartic factors. Hence all 329 connected CSS classes fall into order-three, order-seven, or order-15 cyclotomic-kernel strata. No order-15 class appears among the selected high-distance exact constructions. Of the 215 connected classes with a coprime lattice presentation, the exact cyclic gcd, including multiplicity, is $z^2 + z + 1$ for 182 classes, one of the two primitive order-seven cubics for 19, one of the two primitive order-15 quartics for 12, and the product of the order-three factor with an order-seven cubic for two. The remaining 114 classes are presented over noncyclic ambient groups, and all have an order-three quotient certificate. Every computed cyclic gcd is squarefree, as the connectivity argument above requires; because the gcds were computed with multiplicity, this agreement is an independent check rather than an assumption of Eq. (8). The calculation can be reproduced using code in the qcode-discovery repository [26].

TABLE IV. Algebraic classification of the exact connected CSS constructions. The cyclic forms use a CRT generator z ; C_r denotes the cyclic group of order r . The ordering of a and b matches the corresponding presentation in Table III, which need not be the representative selected for the supplemental class summary. For noncyclic ambient groups, the last column gives an explicit quotient character.

Parameters	Ambient group	Cyclic form $(a(z), b(z))$	Complete common-kernel sector
$[[90, 4, 9]]$	C_{45}	$(1 + z^{17} + z^{43}, 1 + z^{42})$	$h = z^2 + z + 1$ (order 3)
$[[96, 4, 10]]$	C_{48}	$(1 + z^2 + z^{16}, 1 + z^3)$	$h = z^2 + z + 1$ (order 3)
$[[120, 4, 10]]$	$C_6 \times C_{10}$	—	$\phi(i, j) = i \pmod{3}$
$[[140, 6, 10]]$	C_{70}	$(1 + z^{39} + z^{55}, 1 + z^{21})$	$h = z^3 + z^2 + 1$ (order 7)
$[[180, 4, 13]]$	C_{90}	$(1 + z^{12}, 1 + z^{17} + z^{55})$	$h = z^2 + z + 1$ (order 3)
$[[180, 4, 14]]$	C_{90}	$(1 + z^2 + z^{55}, 1 + z^3)$	$h = z^2 + z + 1$ (order 3)
$[[432, 4, 10]]$	$C_{18} \times C_{12}$	—	$\phi(i, j) = j \pmod{3}$

The same low-order quotient structure appears in earlier weight-five constructions. Substituting $z = xy$ in the three Voss constructions gives ordinary bivariate BB codes, each with an order-three quotient character; the CSS presentation of the $[[36, 4, 6]]$ Khesin–Lu mirror code has the same property. We therefore use “low-order quotient character” and “cyclotomic kernel” as a structural taxonomy within the known abelian 2BGA/BB family. The order-seven $[[140, 6, 10]]$ code realizes the factor-first coprime-BB approach at weight five. The leading interval classes—one $[[432, 4, 5 \leq d \leq 23]]$ class and two non-isomorphic $[[420, 4, 5 \leq d \leq 22]]$ classes—are further members of the order-three stratum; exact resolution would enable a performance comparison.

D. Pareto comparison across check weights

Figure 1 compares exact per-weight Pareto frontiers in the compiled dataset: this study’s weight-five catalogue, our previous weight-six and weight-eight results, published benchmarks, and the Challenge comparison set defined above. Each included construction is assigned to a cohort by its maximum stabilizer weight, producing disjoint fixed-weight cohorts rather than cumulative “weight at most w ” envelopes. Their ordering at short block lengths is interpreted in that fixed-weight sense. Only exact distances enter the figure. For this study, the figure uses one connected representative per component class rather than unreduced parent presentations. For earlier-campaign codes with polynomial supports, a translation-subgroup test computes the index in $\mathbb{Z}_\ell \times \mathbb{Z}_m$ of the subgroup generated by differences within $\text{supp}(A) \cup \text{supp}(C)$ and within $\text{supp}(B) \cup \text{supp}(D)$, with $C = D = 0$ for CSS codes; this index gives the number of translation components of the stored presentation. Every Challenge entry is rebuilt from its submitted H_X, H_Z matrices and reduced using the same invariant row-space decomposition as the present catalogue. Certified intervals, decoder estimates, witness-only ceilings, and disconnected parents remain available in the accompanying comparison dataset.

Challenge entries without exact certificates are treated only as witness-based upper bounds. When different sources report identical parameters, the comparison plots that parameter point once using the strongest available

evidence; this aggregation does not assert code equivalence. The component reduction changes several entries used to construct the frontiers: a prior weight-six $[[144, 16, 8]]$ parent reduces to $2 \times [[72, 8, 8]]$, a prior weight-eight $[[144, 44, 4]]$ parent to $2 \times [[72, 22, 4]]$, and a prior weight-eight PBB $[[180, 12, 8]]$ parent to $3 \times [[60, 4, 8]]$. The corresponding components, rather than their replicated parents, enter the frontiers. The Challenge analysis similarly reduces $[[96, 12, 4]]$ and $[[192, 24, 4]]$ to six and twelve copies, respectively, of the same connected $[[16, 2, 4]]$ component. Table V lists every exact Pareto point in each panel so that the figure need not be densely labelled.

Exhaustive enumeration independently verifies the reported distances of the Voss $[[30, 4, 5]]$ code and the Khesin–Lu $[[36, 4, 6]]$ mirror/2BGA code; their FOMs are 3.33 and 4.00, respectively [10, 11]. Voss also reports $[[72, 4, 8]]$ and $[[96, 4, 8]]$, with FOMs 3.56 and 2.67; these remain literature-reported values. The connected constructions from this study extend the range of distances with exact certificates in this comparison. This is a check-weight-constrained comparison. For unrestricted context, CodeTables gives the exact optimum $[[12, 2, 4]]$ and known constructions $[[30, 4, 8]]$, $[[36, 4, 9]]$, $[[48, 4, 11]]$, and $[[60, 4, 14]]$ [27]. These provide unrestricted reference points for the distance gap associated jointly with qLDPC sparsity, the bicycle algebraic form, and weight five at matched (n, k) . In the post-search comparison, MILP over all logical sectors certified five Lin–Pryadko archive constructions over cyclic ambient groups, with parameters $[[48, 4, 7]]$, $[[60, 4, 8]]$, $[[66, 4, 8]]$, $[[78, 4, 9]]$, and $[[84, 4, 9]]$. The exact component certificate and explicit CRT equivalence certify the archived $[[96, 4, 10]]$ construction over C_{48} , while MILP certifies the archived $[[132, 4, 12]]$ construction over a nonabelian ambient group. These results preserve the original attribution while allowing the constructions to enter the exact-distance comparison. The resulting exact length–FOM envelope reaches 4.27 at $[[60, 4, 8]]$ and 4.364 at $[[132, 4, 12]]$. The latter narrowly exceeds our $[[180, 4, 14]]$ code’s FOM of 4.356. Thus, $[[132, 4, 12]]$ has the highest FOM on the exact length–FOM envelope, while $[[140, 6, 10]]$ and $[[180, 4, 14]]$ add points to the exact rate–distance frontier. The Voss $[[72, 4, 8]]$ value remains in the comparison dataset as a literature-reported estimate; the exact-only frontier uses certified points. At the re-

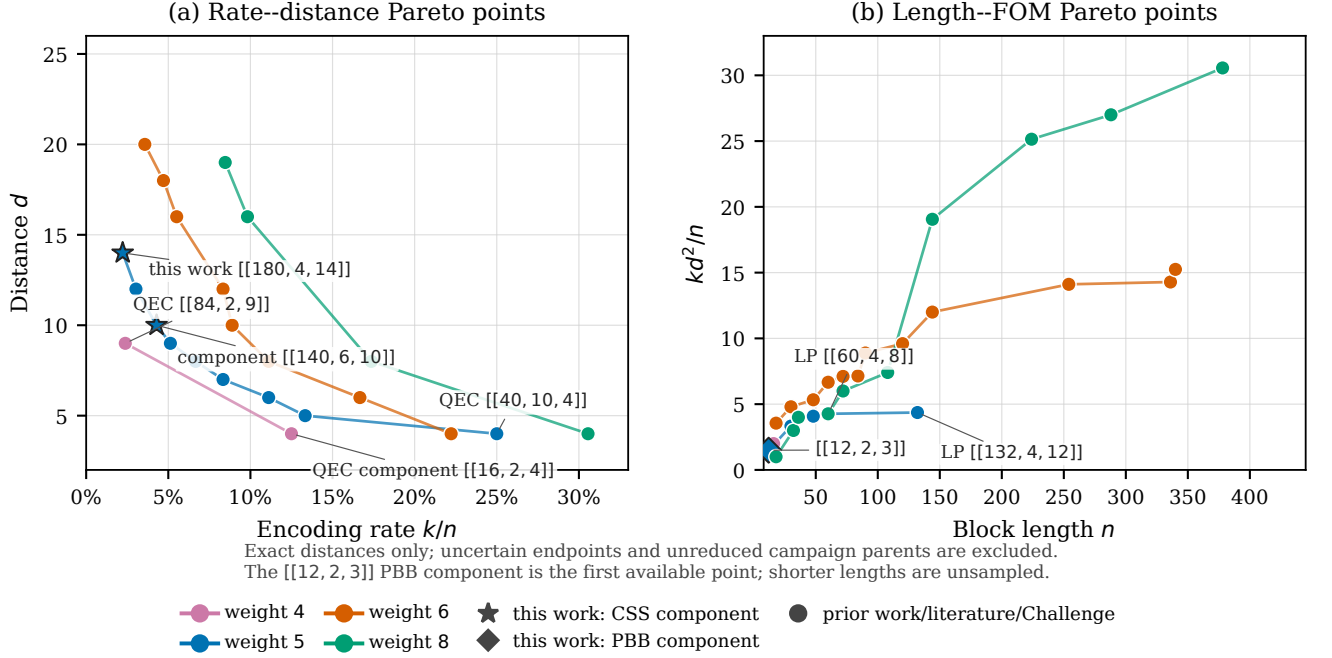


FIG. 1. Exact-distance Pareto frontiers within the compiled dataset, grouped by maximum stabilizer weight after excluding $d \leq 2$. Panel (a) shows the nondominated rate–distance frontier; panel (b) shows the idealized data-qubit ratio kd^2/n relative to k independent rotated surface-code patches as block length increases. Stars and diamonds denote component-class points originating from CSS and PBB codes in this study, respectively; circles denote prior constructions and Challenge entries; color denotes maximum stabilizer weight. The cohorts are noncumulative, and the compiled dataset is not an exhaustive census of the corresponding design spaces. The $[[12, 2, 3]]$ label marks the first available point in the compiled weight-five series; shorter lengths are currently unsampled. Lines are guides to the eye. Every current or prior campaign entry with stored polynomial supports is reduced to one translation-connected component before frontier selection, and all Challenge entries receive the basis-independent connectivity test.

TABLE V. Exact-distance points defining the per-weight frontiers in the compiled dataset shown in Fig. 1, listed in plot order. Source superscripts are O (this work, connected CSS catalogue proposal), O_c (this work, CSS component-class point, with a separately supplied explicit polynomial realization when available), O_p (this work, PBB component-class point), Q (connected QEC Challenge entry), Q_c (QEC Challenge component), V (Voss *et al.* [10]), K (Khesin–Lu [11]), B (Bravyi *et al.* [1]), LP (Lin–Pryadko archive construction, with distance independently certified here), L (Liang *et al.* [17]), P (our previous work [2]), QL (Qian–Li [8]), and S (Symons *et al.* [18]).

Weight	Panel	Pareto points, in plot order
4	(a)	$[[16, 2, 4]]^{Q_c}, [[84, 2, 9]]^Q$
4	(b)	$[[16, 2, 4]]^{Q_c}$
5	(a)	$[[180, 4, 14]]^O, [[132, 4, 12]]^{LP}, [[140, 6, 10]]^{O_c}, [[78, 4, 9]]^{LP}, [[60, 4, 8]]^{LP}, [[48, 4, 7]]^{LP}, [[36, 4, 6]]^K, [[30, 4, 5]]^V, [[40, 10, 4]]^Q$
5	(b)	$[[12, 2, 3]]^{O_p}, [[30, 4, 5]]^V, [[36, 4, 6]]^K, [[48, 4, 7]]^{LP}, [[60, 4, 8]]^{LP}, [[132, 4, 12]]^{LP}$
6	(a)	$[[336, 12, 20]]^{Q_L}, [[340, 16, 18]]^L, [[254, 14, 16]]^L, [[144, 12, 12]]^B, [[90, 8, 10]]^B, [[72, 8, 8]]^P, [[72, 12, 6]]^B, [[18, 4, 4]]^Q$
6	(b)	$[[18, 4, 4]]^Q, [[30, 4, 6]]^Q, [[48, 4, 8]]^Q, [[60, 4, 10]]^K, [[72, 8, 8]]^P, [[84, 6, 10]]^Q, [[90, 8, 10]]^B, [[120, 8, 12]]^L, [[144, 12, 12]]^B, [[254, 14, 16]]^L, [[336, 12, 20]]^{Q_L}, [[340, 16, 18]]^L$
8	(a)	$[[378, 32, 19]]^{Q_L}, [[224, 22, 16]]^{Q_L}, [[288, 50, 8]]^P, [[72, 22, 4]]^P$
8	(b)	$[[18, 2, 3]]^P, [[32, 6, 4]]^P, [[36, 4, 6]]^P, [[60, 4, 8]]^P, [[72, 12, 6]]^P, [[108, 8, 10]]^P, [[144, 14, 14]]^S, [[224, 22, 16]]^{Q_L}, [[288, 24, 18]]^{Q_L}, [[378, 32, 19]]^{Q_L}$

ported value, the Voss point is dominated by the exact Lin–Pryadko $[[60, 4, 8]]$ archive point, while the exact Lin–Pryadko $[[78, 4, 9]]$ archive point dominates our derived $[[90, 4, 9]]$ component. Both dominated points remain available in the comparison data.

The archive also contains constructions over non-abelian ambient groups with the same parameter

triples as our $[[120, 4, 10]]$, $[[140, 6, 10]]$, $[[180, 4, 13]]$, and $[[180, 4, 14]]$ codes. We establish explicit code equivalence for $[[96, 4, 10]]$ above; for these four constructions, the archive relationship is currently a parameter match. Our $[[432, 4, 10]]$ construction lies beyond the archive’s enumeration over nonabelian ambient groups through block length $n = 200$.

The archive also reports constructions over nonabelian ambient groups with parameters $[[168, 4, 14]]$ (FOM 4.67), $[[180, 4, 15]]$ (5.00), and $[[192, 4, 16]]$ (5.33). Their archived distances were obtained using a randomized routine. Our all-logical MILP attempts found matching weight-14, weight-15, and weight-16 witnesses, respectively, and proved optimality for 1 of 8, 0 of 8, and 1 of 8 logical objectives within the per-logical time limit. Every MILP run found an incumbent at the reported weight, establishing the three values as upper bounds in the comparison dataset. If all three reported distances were eventually certified, $[[168, 4, 14]]$ would dominate the campaign's $[[180, 4, 14]]$ point, while $[[140, 6, 10]]$ would remain on the rate-distance frontier.

Analysis of their support subgroups further separates these three presentations. The archived presentation with $(n, k) = (180, 4)$ and upper bound $d \leq 15$ has a central overlap $N \cong C_{15}$ with split support-group extensions and admits the quasi-abelian reduction to a hypergraph-product code over $\mathbb{F}_2[C_{15}]$; here $k_S = k/2 = 2$. For the archived presentations with $(n, k) = (168, 4)$ and $(192, 4)$ and upper bounds 14 and 16, respectively, the central overlaps are isomorphic to C_{14} and C_8 , respectively; one support-group extension is nonsplit in each case, and direct rank calculations give the strict inequality $k_S = 0 < k/2 = 2$. This presentation-level classification is structural; distance certification and equivalence to an abelian BB code are separate questions.

The exact $[[132, 4, 12]]$ construction has ambient group $C_{11} \times S_3$, where $S_3 = \langle r, s \mid r^3 = s^2 = 1, srs = r^{-1} \rangle$ is the symmetric group on three letters. Its binomial and trinomial supports generate cyclic groups C_{22} and C_{33} with common central overlap C_{11} ; modulo that overlap, the corresponding group-algebra elements are $1 + s$ and $1 + r + r^2$ in $\mathbb{F}_2[S_3]$. It is therefore a quasi-abelian lifted-product code over $\mathbb{F}_2[C_{11}]$, but no abelian 2BGA presentation has been established. Under the idealized comparison of Eq. (6), the ratios of data qubits in k independent rotated surface-code patches to those in the $[[96, 4, 10]]$, $[[140, 6, 10]]$, and $[[180, 4, 14]]$ codes are 4.17, 4.29, and 4.356, respectively. These ratios compare data-qubit counts at the same nominal (k, d) ; a total-hardware comparison additionally requires circuit-level resources.

Higher-weight checks can yield substantially larger exact FOMs. Exact weight-six reference points include the mirror code $[[60, 4, 10]]$ with FOM 6.67 [11], $[[288, 12, 18]]$ with FOM 13.5 [1], $[[336, 12, 20]]$ with FOM 14.29 [8], and $[[340, 16, 18]]$ with FOM 15.25 [17]. Within the compiled dataset, the Qian-Li $[[336, 12, 20]]$ point replaces $[[294, 10, 20]]$ on the exact weight-six rate-distance frontier and joins the length-FOM frontier. The weight-six $[[360, 12, \leq 24]]$ benchmark is currently known only through the ceiling FOM ≤ 19.20 [1]. At weight eight, the MILP-certified $[[144, 14, 14]]$ has exact FOM 19.06 [18]. Within the compiled dataset, Qian and Li extend the exact length-FOM frontier through $[[224, 22, 16]]$, $[[288, 24, 18]]$, and $[[378, 32, 19]]$, with FOMs 25.14, 27.00, and 30.56, respectively [8]; the first and

third also enter the rate-distance frontier. These four Qian-Li points enter at their published parameters and exact MILP distances; their component structure was not independently evaluated here. Exact codes from our previous work include $[[144, 12, 12]]$ with FOM 12.00 and $[[288, 50, 8]]$ with FOM 11.11 [2]. Repeated parameter triples in different maximum-weight cohorts correspond to distinct presentations; parameter equality here does not imply equivalence. This applies to $[[36, 4, 6]]$ at weights five and eight, $[[60, 4, 8]]$ at weights five and eight, $[[72, 12, 6]]$ at weights six and eight, and $[[144, 12, 12]]$ at weights six and eight. Two additional short weight-eight Pareto points result from the same component reduction: the prior parents $[[144, 16, 3]]$ and $[[288, 54, 4]]$ reduce to $8 \times [[18, 2, 3]]$ and $9 \times [[32, 6, 4]]$, respectively. Matched block lengths make the tradeoff associated with lower check weight especially clear. Liang *et al.* exactly certify weight-six $[[96, 4, 12]]$ and $[[140, 6, 14]]$ codes, compared with our weight-five $[[96, 4, 10]]$ and $[[140, 6, 10]]$. At $n = 180$, their exact $[[180, 8, 16]]$ improves both k and d over our $[[180, 4, 14]]$. These length-matched comparisons clarify the stabilizer-weight tradeoff. As a scheduling proxy, the combined CSS incidence graph of every $2 + 3$ construction here is bipartite and 5-regular, so König's line-coloring theorem fixes the graph-theoretic minimum at five conflict-free check-data interaction layers. The analogous minimum is six for regular $3 + 3$ weight-six BB codes. These minima are not fault-tolerant circuit depths; for example, Ref. [1] gives a seven-round extraction schedule for weight-six BB codes. Ancilla preparation and measurement, hook-error ordering, routing, and hardware constraints can add further layers. For the family-relevant scale suggested by the conditional $d = O(n^{2/3})$ bound, the ratios $d/n^{2/3}$ for $[[96, 4, 10]]$, $[[140, 6, 10]]$, and $[[180, 4, 14]]$ are 0.477, 0.371, and 0.439, respectively. The corresponding ratios for the exact weight-five comparators $[[30, 4, 5]]$, $[[60, 4, 8]]$, and $[[132, 4, 12]]$ are 0.518, 0.522, and 0.463. These are finite-size descriptors and do not establish an asymptotic trend.

Of the 97 Challenge entries labelled bivariate bicycle, 55 have exact certificates and 42 have witness-only upper bounds. Under our row-weight definition, five have weight four, 76 weight six, 10 weight eight, one weight 10, and five weight greater than 25; none has weight five. The weight-four entries are the exact $[[84, 2, 9]]$, $[[96, 12, 4]]$, and $[[192, 24, 4]]$ codes and the witness-only $[[176, 22, \leq 4]]$ and $[[192, 6, \leq 8]]$ entries. The comparison set also includes five non-BB weight-five entries: exact $[[40, 10, 4]]$, $[[40, 6, 4]]$, and $[[45, 9, 3]]$ codes, and witness-only $[[18, 4, \leq 3]]$ and $[[676, 71, \leq 4]]$ entries.

The basis-independent connectivity test covers all 102 entries in the Challenge comparison set: 81 are connected, 12 decompose into multiple pairwise-isomorphic positive- k components, and nine decompose into one positive- k component together with $k = 0$ stabilizer-state factors. Submitted logical witnesses were checked to lie in the retained component before distance evidence was transferred. The Challenge entries supply

the short-block points on the exact weight-six length-FOM curve: $\llbracket 18, 4, 4 \rrbracket$ (3.56), $\llbracket 30, 4, 6 \rrbracket$ (4.80), $\llbracket 48, 4, 8 \rrbracket$ (5.33), and $\llbracket 84, 6, 10 \rrbracket$ (7.14); all four are connected. A weight-eight $\llbracket 154, 8, \leq 17 \rrbracket$ submission has verified logical witnesses, setting the certified ceiling $\text{FOM} \leq 15.01$; the exact $\llbracket 144, 14, 14 \rrbracket$ code remains the preceding point on the length-FOM frontier. At weight four, $\llbracket 84, 2, 9 \rrbracket$ is connected, while the exact $\llbracket 96, 12, 4 \rrbracket$ and $\llbracket 192, 24, 4 \rrbracket$ parents reduce to $\llbracket 16, 2, 4 \rrbracket$ as described above; the component has FOM 2.00; the two remaining weight-four entries retain only witness-based ceilings. Among the five non-BB weight-five entries, the exact $\llbracket 40, 10, 4 \rrbracket$ code lies on the rate-distance frontier; the other two exact entries and two witness-only entries do not change either exact Pareto frontier. Within the exact subset of the compiled dataset, our constructions extend both the weight-five distance envelope and its rate-distance frontier. The higher-weight cohorts retain the larger FOM envelope, defining a stabilizer-weight tradeoff whose practical value depends on circuit and hardware costs.

E. PBB codes

The PBB search tested the support-containment ansatz in Eq. (4), with fitness downweighting estimated distances below five. Exhaustive search certified every small-profile proposal. Excluding the 106 with $d = 2$, the retained 151 comprise 31 with $d = 3$, 90 with $d = 4$, and 30 with $d = 6$. Thus, 30 retained proposals meet the $d \geq 5$ target, while 121 have $d = 3$ or 4. These counts are specific to the ansatz and search objective. The largest small-profile FOM is 1.50; one retained proposal attaining it has parameters $\llbracket 72, 12, 3 \rrbracket$. Among the small-profile proposals meeting the $d \geq 5$ target, the $\llbracket 144, 4, 6 \rrbracket$ parameter point has the highest FOM, 1.00, and is attained by five connected presentation classes. Across both profiles, reducing certified parents to connected components gives a largest exact nonbaseline PBB FOM of 1.85, attained by two distinct connected component classes with parameters $\llbracket 216, 4, 10 \rrbracket$.

After excluding 25 exact $d = 2$ proposals, the large profile contains 79 retained proposals with direct evidence for 54 exact distances and 25 certified intervals. Combining evidence within presentation classes adds five exact distances across three classes, for 59 exact proposals and 20 proposals in 14 interval classes. Exhaustive search gives every interval the lower endpoint $d \geq 5$; their upper endpoints range from seven to twelve across $n = 288$ (2), $n = 360$ (10), and $n = 432$ (8). Among the 25 initial interval targets, 112 of 208 logical MILP instances exhausted their per-instance time limits. At the retained-proposal level, the largest exact distance is 11, attained by a $\llbracket 360, 4, 11 \rrbracket$ proposal, while the largest exact FOM is 1.85, attained by two $\llbracket 432, 8, 10 \rrbracket$ proposals. All three are disconnected: the former decomposes into two copies of a $\llbracket 180, 2, 11 \rrbracket$ component, and each latter proposal decomposes into two copies of its respective

$\llbracket 216, 4, 10 \rrbracket$ component class.

We also give connected PBB presentations of these components. The $\llbracket 180, 2, 11 \rrbracket$ presentation matches a component of $\llbracket 360, 4, 11 \rrbracket$, while two $\llbracket 216, 4, 10 \rrbracket$ presentations with distinct colored-Tanner canonical labels match components of the two $\llbracket 432, 8, 10 \rrbracket$ parent presentation classes. A connected $\llbracket 144, 4, 8 \rrbracket$ presentation similarly matches a component of a certified $\llbracket 432, 12, 8 \rrbracket = 3 \times \llbracket 144, 4, 8 \rrbracket$ parent. Each distance transfer is supported by an exact parent MILP certificate, a decomposition into pairwise-isomorphic components, and matching canonical graph signatures for the explicit connected presentation and a parent component. Thus the largest connected exact PBB FOM with $d \geq 5$ is 1.85, attained by the two distinct $\llbracket 216, 4, 10 \rrbracket$ component classes, and the largest connected exact distance is 11, attained by the displayed $\llbracket 180, 2, 11 \rrbracket$ presentation. Among retained large-profile PBB proposals that are themselves connected under the row-space analysis, the largest exact distance is 10, attained at both $n = 360$ and $n = 420$, and the largest exact FOM is 1.11, attained by $\llbracket 360, 4, 10 \rrbracket$. The connected exact catalogue proposals extend through $n = 420$; the exact $n = 432$ proposals reduce to smaller connected components. The proposals with parameters $\llbracket 216, 2, 8 \rrbracket$, $\llbracket 360, 4, 10 \rrbracket$, and $\llbracket 420, 4, 10 \rrbracket$ are members of a fixed baseline family supplied independently of model output.

For reference, one of the two disconnected parent proposals with parameters $\llbracket 432, 8, 10 \rrbracket$ is defined on $(\ell, m) = (18, 12)$ by

$$\begin{aligned} A &= 1 + x^2 y^2, & B &= 1 + xy^6 + x^2, \\ C &= 1 + x^2 y^2, & D &= 1 + x^2. \end{aligned} \quad (11)$$

All four explicit PBB component presentations fail the catalogue's tested local-Clifford-to-CSS screens; classification under their complete local-Clifford orbits remains open. The shorter $\llbracket 12, 2, 3 \rrbracket$ PBB point is represented by four distinct component classes. For these small codes, an exhaustive search over all 6^{12} local Pauli-axis permutations proves that none is local-Clifford-equivalent to CSS. This point begins the compiled weight-five length-FOM curve.

Table VI summarizes the selected PBB presentations and their evidence.

The PBB presentations selected in Table VI share a compact algebraic form, which we call the order-three geometric PBB subfamily. Up to exchanging the two sectors, write

$$\begin{aligned} A &= 1 + v, & B &= 1 + u + u^2, \\ C &\in \{0, A\}, & D &\in \{u, 1 + u^2\}. \end{aligned} \quad (12)$$

Here u and v are lattice monomials. Both $AC^\top$ and $BD^\top$ are inversion-symmetric, so the PBB commutativity condition holds identically. Four parents in this ansatz yield the explicit connected presentations above: the two distinct $2 \times \llbracket 216, 4, 10 \rrbracket$ parents, the $3 \times \llbracket 144, 4, 8 \rrbracket$ parent, and the $2 \times \llbracket 180, 2, 11 \rrbracket$ parent. The same form also contains all four derived presentations. For selected (A, B)

TABLE VI. Selected weight-five PBB presentations. Exact rows are certified. Interval rows combine exclusion through weight four with a supported upper endpoint, so their FOMs are upper bounds. Evidence labels are Exh.+M (exhaustive search corroborated by all-logical MILP), M (all-logical MILP), M+component (a connected presentation matched by colored-BLISS to a component of a homogeneous M-certified parent), and L4+I+B (certified $d \geq 5$ plus both a nonoptimal MILP incumbent and a decoder estimate). Baseline rows identify fixed-family controls supplied by the campaign evaluators independently of model output. Each row specifies a particular polynomial presentation, and repeated parameter triples need not denote the same class. In particular, the two $\llbracket 216, 4, 10 \rrbracket$ rows belong to distinct component classes, and the $\llbracket 144, 4, 6 \rrbracket$ component class underlying the displayed $\llbracket 288, 8, 6 \rrbracket$ parent is distinct from that of the displayed connected $\llbracket 144, 4, 6 \rrbracket$ row. “conn.” denotes connected.

Parameters	(ℓ, m)	(A, B)	(C, D)	FOM	Structure	Evidence
$\llbracket 144, 4, 6 \rrbracket$	(9, 8)	$(1 + x^6y, 1 + x^2y^4 + x^4)$	$(0, x^2y^4)$	1.00	conn.	Exh.+M
$\llbracket 144, 4, 8 \rrbracket$	(18, 4)	$(1 + x^3y, 1 + xy^2 + x^2)$	$(1 + x^3y, xy^2)$	1.78	conn.	M+component
$\llbracket 180, 2, 11 \rrbracket$	(15, 6)	$(1 + x^2y^2, 1 + x^2y^3 + x^4)$	$(1 + x^2y^2, 1 + x^4)$	1.34	conn.	M+component
$\llbracket 216, 2, 8 \rrbracket$	(12, 9)	$(1 + x + x^2, 1 + y)$	$(x, 1 + y)$	0.59	conn.	M; baseline
$\llbracket 216, 4, 10 \rrbracket$	(18, 6)	$(1 + x^2y, 1 + xy^3 + x^2)$	$(1 + x^2y, 1 + x^2)$	1.85	conn.	M+component
$\llbracket 216, 4, 10 \rrbracket$	(18, 6)	$(1 + x^4y, 1 + xy^3 + x^2)$	$(0, 1 + x^2)$	1.85	conn.	M+component
$\llbracket 288, 8, 6 \rrbracket$	(12, 12)	$(1 + x^4y^2, 1 + xy^6 + x^2)$	$(1 + x^4y^2, 1 + x^2)$	1.00	$2 \times \llbracket 144, 4, 6 \rrbracket$	M
$\llbracket 288, 4, 5 \leq d \leq 8 \rrbracket$	(12, 12)	$(1 + x^3y, 1 + xy^6 + x^2)$	$(1 + x^3y, 1 + x^2)$	≤ 0.89	conn.	L4+I+B
$\llbracket 360, 4, 11 \rrbracket$	(15, 12)	$(1 + x^2y^4, 1 + x^2y^6 + x^4)$	$(1 + x^2y^4, 1 + x^4)$	1.34	$2 \times \llbracket 180, 2, 11 \rrbracket$	M
$\llbracket 360, 4, 10 \rrbracket$	(15, 12)	$(1 + x + x^2, 1 + y)$	$(x, 1 + y)$	1.11	conn.	M; baseline
$\llbracket 360, 4, 5 \leq d \leq 12 \rrbracket$	(15, 12)	$(1 + x^4y, 1 + xy^6 + x^2)$	$(0, 1 + x^2)$	≤ 1.60	conn.	L4+I+B
$\llbracket 420, 4, 10 \rrbracket$	(15, 14)	$(1 + x + x^2, 1 + y)$	$(x, 1 + y)$	0.95	conn.	M; baseline
$\llbracket 432, 12, 8 \rrbracket$	(18, 12)	$(1 + x^3y^3, 1 + xy^6 + x^2)$	$(1 + x^3y^3, xy^6)$	1.78	$3 \times \llbracket 144, 4, 8 \rrbracket$	M
$\llbracket 432, 8, 10 \rrbracket$	(18, 12)	$(1 + x^2y^2, 1 + xy^6 + x^2)$	$(1 + x^2y^2, 1 + x^2)$	1.85	$2 \times \llbracket 216, 4, 10 \rrbracket$	M
$\llbracket 432, 8, 10 \rrbracket$	(18, 12)	$(1 + x^4y^2, 1 + xy^6 + x^2)$	$(0, 1 + x^2)$	1.85	$2 \times \llbracket 216, 4, 10 \rrbracket$	M
$\llbracket 432, 4, 5 \leq d \leq 12 \rrbracket$	(18, 12)	$(1 + x^2y^5, 1 + xy^6 + x^2)$	$(0, 1 + x^2)$	≤ 1.33	conn.	L4+I+B

pairs, the search evaluated all 31 nonempty support-contained (C, D) choices; over the broader set, it sampled only sparse perturbations. Equation (12) characterizes the retained subfamily; exhaustive enumeration of all support-contained choices is a separate problem. Each of the 33 connected catalogue PBB presentation classes has an underlying (A, B) pair with an order-three quotient-character certificate. Thus, all connected PBB classes sampled by the search share this order-three backbone. This common structure motivates an order-three geometric PBB subfamily within the support-containment ansatz; broader sampling can test its reach across non-CSS weight-five codes.

These four explicit PBB components connect this pattern to the CSS classification above. In each case, the supports of A and B lie along two cyclic directions that jointly generate a noncyclic ambient group, while C and D are supported on those same directions. For $\llbracket 144, 4, 8 \rrbracket$, the two cyclic direction subgroups have orders 12 and 18 and intersect in a subgroup isomorphic to C_3 . The corresponding orders are 15 and 30, with intersection isomorphic to C_5 , for $\llbracket 180, 2, 11 \rrbracket$; for both $\llbracket 216, 4, 10 \rrbracket$ presentations, the orders are 18 and 18 and the intersection is isomorphic to C_3 . The CSS BB codes defined by the corresponding (A, B) pairs therefore belong to the same cyclic-support, cyclic-overlap stratum, and the PBB terms are support-contained symplectic perturbations of those codes. A lifted-product reduction would additionally require the relevant overlap extensions to split, which is not implied by these support data alone.

The fixed baseline family is obtained from the sector-swapped form of Eq. (12) by setting $u = x$ and $v = y$:

$$(A, B, C, D) = (1 + x + x^2, 1 + y, x, 1 + y). \quad (13)$$

For $3 \mid \ell$ and $m \geq 2$, the dimension of this family is exactly

$$k = 2 \gcd(m, 2). \quad (14)$$

To see this, put $N = \ell m$ and $h = 1 + y + \dots + y^{m-1}$. The mixed top-row map has rank N because $C = x$ is a unit, whereas the CSS-type bottom-row map has rank $N - 2$ because $\text{Ann}(A^*) \cap \text{Ann}(B^*)$ has dimension two, where Ann denotes the annihilator in $R_{\ell, m}$. The two row spaces intersect in dimension two when m is even and trivially when m is odd. Indeed, the image of multiplication by $1 + y$ is the kernel of evaluation at $y = 1$; because $h(1) = m \bmod 2$, it contains h precisely when m is even. Subtracting the stabilizer rank from $2N$ gives Eq. (14).

There is also a general analytic distance upper bound. With

$$Q = (1 + x) \sum_{j=0}^{\ell/3-1} x^{3j},$$

the operators

$$L_x = (0, Q \mid 0, Q), \quad L_y = (0, 0 \mid 0, h)$$

are nontrivial logicals of Pauli weights $2\ell/3$ and m , respectively. Here $AQ = 0$ and $Bh = 0$ establish commutation. Nontriviality of L_x follows because a stabilizer representation would require a coefficient p satisfying $pB = Q$, which is impossible after evaluating at $y = 1$. For L_y , a representation would require a coefficient q satisfying $qB = 0$; because $D = B$, this also gives $qD = 0$, leaving $rA^* = h$ for another coefficient r . Coefficientwise in y , this would make the zero divisor A^*

invertible. Consequently,

$$d \leq \min\left(m, \frac{2\ell}{3}\right). \quad (15)$$

Every campaign lattice in this sequence has $m \geq 2\ell/3$. The exact baseline-family presentations on these lattices saturate the $2\ell/3$ branch: they include $[[72, 4, 4]]$, $[[120, 4, 4]]$, $[[144, 4, 6]]$, $[[216, 2, 8]]$, $[[288, 4, 8]]$, $[[360, 4, 10]]$, and $[[420, 4, 10]]$, together with the odd- m members $[[108, 2, 4]]$ and $[[162, 2, 6]]$. The baseline $[[144, 4, 6]]$ presentation is distinct from the same-parameter connected row in Table VI. For the interval-certified baseline presentation on $(\ell, m) = (18, 12)$, which has parameters $[[432, 4, 5 \leq d \leq 12]]$, Eq. (15) supplies the upper endpoint. This presentation is likewise distinct from the same-parameter row in Table VI. Equality with the $2\ell/3$ branch is not universal: the unsearched lattice $(\ell, m) = (18, 2)$ has $n = 72$, within the stated block-length range, and gives the exact $[[72, 4, 2]]$ code that saturates the m branch. This sequence is a fixed baseline supplied independently of model output. A released bit-level verifier checks the stabilizer ranks, both logical constructions, and the counterexamples [26].

At $n = 288$ and $n = 432$, the interval ceilings 0.89 and 1.33 lie below the largest exact FOMs at those same lengths, 1.00 and 1.85; the $n = 360$ intervals are the sole remaining possibilities for a same-length improvement. Across all lengths, every interval ceiling is at most 1.85, so the explicit $[[216, 4, 10]]$ realizations define the connected PBB FOM envelope under the current evidence.

Across the tested support-containment ansatz and parameter range, the certified CSS frontier remains above every PBB candidate passing the implemented non-CSS screens. Broader non-CSS weight-five families provide a natural next test. The two weight-five mirror-code examples tabulated in Ref. [11] are equivalent to CSS codes under the authors’ Hadamard test; the non-CSS mirror-code constructions emphasized in that work have weight at least six.

F. Weight-five transfer control

Table VII compares 6,444 logged positive-distance campaign observations with three control replicates of 6,444 completed selected evaluations.

At the distinct-proposal level, the campaign output has a 12.9–14.7 percentage-point higher certified- $d \geq 5$ fraction. The without-replacement control replicates contain 6,444 distinct proposals versus 936 in the campaign output and therefore cover more connected certified classes; this absolute coverage difference partly reflects the suppression of repeats.

The control included low-weight certification through weight four but no separate high-distance exact follow-up. The last row therefore records sampling of a presentation in a campaign-certified class, not the control’s maximum. Of 385 pooled classes meeting the finalist

rule (estimated FOM ≥ 5 and certified $d \geq 5$), 359 remain without exact evidence; every replicate samples the campaign-certified $[[180, 4, 14]]$ class. Each replicate evaluated the ranks of 94,775–96,487 sampled pairs to fill its quotas. Because each stratum stops when its quota is filled, 6,444 divided by this count is not a full-space sampling fraction. Per-lattice and per-replicate summaries, all sampled draws and distance records, the low-weight audits, and reproduction and validation scripts are available in the code repository [26].

V. LIMITATIONS

The 630 presentation classes quotient symmetries of the stored generator graphs, not arbitrary stabilizer-basis changes or general local-Clifford operations, and are therefore an upper bound on code-equivalence classes. The PBB label “non-CSS” likewise refers to the tested transformation set, except for the four $[[12, 2, 3]]$ component classes whose complete local-Clifford orbits were checked. The algebraic taxonomy in Sec. IV C also classifies the displayed group-algebra presentations rather than all equivalent realizations.

The stratified-uniform experiment fixes the campaign’s allocation by lattice and dimension band and tests selection within those strata, not allocation among them. Its campaign denominator contains logged positive-distance observations, whereas the control denominator contains all completed selected evaluations. Because failed and zero-distance calls are unavailable at the required granularity, the reported campaign fraction can exceed its attempted-call yield; the experiment therefore does not compare attempted-call yield or total compute. Campaign observations can also repeat proposals, whereas the control samples are drawn without replacement, and the control did not receive separate high-distance exact certification. The broader uniform-random and genetic-algorithm benchmarks remain those of our preceding work [2].

The cyclic-kernel formula in Eq. (8) and the subgroup test make algebraic enumeration and symmetry reduction of the $2 + 3$ sector over cyclic ambient groups practical. A complete census also requires an exact distance or verified upper-bound witness for every surviving orbit; this remains open over C_{45} , C_{48} , C_{70} , C_{90} , and C_{210} .

The release supports reproducing the analysis from the retained catalogue onward through solver inputs, reconstructed logical witnesses, campaign observations, and all uniform-control draws. Repository documentation records operational provenance and the limits of reconstructing model-generation history.

The present analysis covers algebraic code parameters, stabilizer connectivity, and an ideal interaction-layer schedule. Full syndrome-extraction circuits, circuit-level noise, decoding thresholds, and logical-error rates form the next layer of physical evaluation.

The connectivity analysis is complete for the re-

TABLE VII. Weight-five transfer control. Control entries are ranges over three stratified-uniform replicates.

Endpoint	LLM-guided output	Stratified-uniform control
Records/evaluations	6,444 logged positive-distance observations	6,444 completed selected evaluations
Distinct proposals	936	6,444
Distinct certified- $d \geq 5$ proposals	816/936 (87.18%)	4,672–4,785/6,444 (72.50–74.26%)
Connected certified- $d \geq 5$ classes	311	566–572
Highest-FOM campaign-certified class sampled	$\llbracket 180, 4, 14 \rrbracket$; 4.356	$\llbracket 180, 4, 14 \rrbracket$; 4.356

tained weight-five catalogue and every entry in the Challenge comparison set. Earlier campaign entries used in Fig. 1 were reduced with the corresponding translation-subgroup test wherever their polynomial supports were available; a full row-space analysis of those earlier campaigns is outside the present scope.

VI. DATA AND CODE AVAILABILITY

The Supplemental Material gives a compact catalogue containing every proposal in an exact class and one representative of each remaining nonexact class. The complete machine-readable catalogue and reproduction code are available in the qcode-discovery repository [26]. The repository also contains campaign configurations, archived QEC Challenge comparison data, certification inputs and witnesses, equivalence and connectivity analyses, interaction schedules, uniform-control draws, and validation scripts. A manifest records configurations, versions, hashes, run metadata, and validation history.

VII. CONCLUSION

Building on our earlier LLM-guided program-evolution workflow, we discover weight-five BB and PBB codes and subject the resulting catalogue to exact distance certification and structural reduction. The leading connected CSS realizations are $\llbracket 96, 4, 10 \rrbracket$, $\llbracket 140, 6, 10 \rrbracket$, and $\llbracket 180, 4, 14 \rrbracket$, with FOMs 4.17, 4.29, and 4.356; the latter two lie on the exact rate–distance frontier compiled here. At parameter triples shared with the Lin–Pryadko archive, we contribute exact certificates, explicit bivariate presentations, and verified parent–component relations. The comparison also certifies seven archive constructions, including the $\llbracket 132, 4, 12 \rrbracket$ construction over the nonabelian group $C_{11} \times S_3$, whose FOM of 4.364 is the highest compiled exact weight-five value.

The three-replicate weight-five transfer control reveals complementary search behavior. Of 936 distinct CSS proposals recorded with positive distance, 816 (87.18%) are certified at $d \geq 5$, compared with 4,672–4,785 of 6,444 proposals per stratified-uniform replicate

(72.50–74.26%). Without-replacement sampling covers more connected classes, and every replicate samples the campaign-certified $\llbracket 180, 4, 14 \rrbracket$ class. This weight-five experiment complements the broader uniform-random and genetic-algorithm benchmarks in our earlier work.

The structural analysis identifies 409 of 1,142 retained proposals as disconnected and reduces 630 presentation classes to 608 component classes. A parallel analysis of the 102-entry Unitary Foundation QEC Challenge comparison set identifies $\llbracket 16, 2, 4 \rrbracket$ as the connected component underlying two weight-four frontier parents. The algebraic analysis explains the absence of $k = 2$ in the sampled CSS family, organizes the connected classes into common-kernel sectors associated with quotient characters of orders 3, 7, and 15, and yields a 28-class taxonomy of two-direction presentations within the sampled lattice orders.

For PBB codes, the strongest connected exact parameter point is $\llbracket 216, 4, 10 \rrbracket$, attained by two distinct component classes, and a complete group-level calculation proves that none of the four $\llbracket 12, 2, 3 \rrbracket$ component classes is local-Clifford-equivalent to CSS. Across the investigated parameter range and current exact evidence, the CSS discoveries define the stronger finite-length envelope. The component results motivate testing connectivity during generation and optimizing a canonical connected component directly in future campaigns. Together, the exact certificates, structural reductions, and algebraic classification establish a reproducible foundation for exhaustive enumeration over cyclic ambient groups, broader equivalence testing, and circuit-level evaluation.

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Supplemental Material for “Large Language Model-Guided Discovery of Weight-Five Bivariate Bicycle Codes”

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Throughout, BB denotes bivariate bicycle, PBB denotes perturbed bivariate bicycle, and CSS denotes Calderbank–Shor–Steane. We use BP–OSD for belief propagation with ordered-statistics decoding, MILP for mixed-integer linear programming, and FOM for the figure of merit kd^2/n .

CAMPAIGN AND CERTIFICATION SETTINGS

Campaign	Iterations	Population	Islands	Workers	Eval. timeout (s)
CSS-small	300	500	5	6	1,200
CSS-large	300	750	6	24	900
PBB-small	300	300	4	6	1,200
PBB-large	300	450	5	16	1,800

All campaigns used random seed 42, temperature 1.0, and model-selection probabilities 0.49 for Claude Opus 5 and 0.51 for GPT-5.6 Sol. CSS ranking used 1,000 BP–OSD trials, three 500-trial refinements, 200 combination-sweep ordered-statistics decoding (OSD-CS) trials, and a 300-s exact-search attempt; post-search MILP allowed 1,800 s per logical objective and 14,400 s per code. PBB ranking used low-weight enumeration, adaptive MILP budgets of 15/90, 30/180, or 60/360 s per logical objective/per code, and 1,000+500 BP–OSD trials. Initial post-search all-logical PBB MILP passes allowed 180 s per logical objective for the small profile and 1,800 s for the large profile; unresolved objectives were rerun with limits up to 3,600 and 14,400 s, respectively. Because the profiles used different time and parallelism budgets, their output counts are descriptive rather than equal-budget yields. Further configuration details are available in the code repository [1].

I. WEIGHT-FIVE CATALOGUE AND SUPPORTING AUDITS

A. Catalogue construction

The source logs contain 1,297 proposals (13,923 observations). After duplicate observations are merged, the catalogue retains 1,142 proposals (12,036 observations), omitting exact $d \leq 2$ rows and rows with a reported endpoint $d_{\text{rep}} \leq 2$. Canonical labelling groups the retained normalized tuples into 630 presentation classes, defined here as stored-generator colored-Tanner isomorphism classes: 535 CSS and 95 PBB. This classification captures sector exchanges, monomial shifts, and lattice automorphisms that preserve the colored generator graph. It does not capture equivalences requiring stabilizer-basis changes or local Clifford operations, so 630 remains an upper bound on the number of inequivalent codes. ID prefixes CS, CL, PS, and PL denote CSS-small, CSS-large, PBB-small, and PBB-large; the C or P label after the slash identifies the presentation class. The four campaign tables print all 436 proposals in exact classes and one representative of each nonexact class (839 rows total); the complete 1,142-row catalogue and supporting evidence are available through the project repository [1].

B. Distance evidence and class merging

Duplicate observations are merged by requiring exact values to agree, maximizing certified lower bounds, and retaining the smallest reported endpoint together with whether that endpoint is rigorous or decoder-only. For one representative of each of 492 presentation classes containing all 858 proposals formerly lacking direct lower-bound evidence, we exhaustively searched both CSS logical sectors for nontrivial operators of weight at most four. The search enumerates supports of at most two qubits and combines pairs whose syndromes cancel, rejecting combinations that are stabilizers. The audit resolved 64 classes (94 proposals) at low weight: two $d = 2$ proposals were omitted, and 62 classes (92 retained proposals) have $d = 3$ or 4. For the other 428 classes (764 proposals), it certified $d \geq 5$. Every retained row has either an exact distance or a certified positive lower bound paired with a reported endpoint. Before the final class merge, 398 proposals have exact row-level status and 744 have unresolved row-level status. Of the latter, 64 have rigorous upper bounds and 680 retain decoder estimates. 30 proposals with exact row-level status inherit an exact $d \leq 4$ result from an audited class representative. A separate set of 15 proposals with exact row-level status combines a transferred $d \geq 5$ result with its own weight-five witness. These transferred-evidence subsets measure a different property from the 45 proposals whose decoder estimates of 5 were matched by reconstructed and independently verified weight-five witnesses.

Merging distance evidence within each presentation class gives 227 exact classes and 403 classes with a certified lower bound and reported endpoint: 53 rigorous intervals and 350 lower-bound-plus-estimate records. The exact classes cover 436 proposals: 398 already had exact row-level status before this merge and 38 acquire it by class isomorphism. Class merging tightens without establishing an exact value for another 99 rows. The campaign tables separate each proposal’s own evidence from the class effect. Where a class effect is shown, the displayed d and FOM use the class-level evidence, whereas Evid. lists only evidence attached to that proposal. The figure of merit is $\text{FOM} = kd^2/n$; values are rounded to two decimals unless three are needed to distinguish a close comparison.

C. Connectivity and component classes

The basis-independent connectivity test finds that 409 of 1,142 retained proposals are disconnected, including 235 of the 398 proposals with exact row-level status. By family, 307 of the 912 CSS proposals and 102 of the 230 PBB proposals are disconnected. The component partition agrees with the generator-graph partition for every retained row. At the class level, 268 of 630 classes are disconnected, including 166 of 227 exact classes.

The high- k component audit resolves six retained direct sums: $10 \times \llbracket 42, 6, 3 \rrbracket$, $16 \times \llbracket 24, 4, 3 \rrbracket$, $12 \times \llbracket 36, 4, 3 \rrbracket$, $10 \times \llbracket 42, 4, 5 \rrbracket$, $8 \times \llbracket 48, 4, 3 \rrbracket$, and $8 \times \llbracket 48, 4, 6 \rrbracket$. The seventh, $18 \times \llbracket 24, 4, 2 \rrbracket$, is excluded from the retained catalogue because $d = 2$.

Selecting one connected component from each parent presentation and repeating the canonical labelling reduces 630 parent classes to 608 component classes: 528 CSS and 80 PBB, of which 206 are exact and 402 are nonexact. Because this comparison uses presentation-graph isomorphism rather than complete stabilizer-code equivalence, 608 remains an upper bound on the number of inequivalent component codes.

D. Local-Clifford screen

A separate complete local-Clifford-to-CSS audit covers all four exact PBB component classes with parameters $\llbracket 12, 2, 3 \rrbracket$. An exhaustive branch-and-bound search over the 6^{12} assignments of Pauli-axis permutations to the 12 qubits shows that none is locally Clifford equivalent to CSS. The machine-readable certificates are available through the project repository [1].

E. Table conventions

Evidence codes are Exh. (exact exhaustive search), H_c (exact component enumeration plus isomorphism and dimension additivity), M (MILP optimality for every logical sector), L_w (exhaustive exclusion of logical operators through weight w), I (MILP incumbent), W (independently reconstructed and verified logical operator), and B (BP-OSD/decoder estimate; operator not retained). For a nonexact row, $[L, U]$ denotes a certified lower bound and a rigorous upper bound, whereas $L; \hat{U}$ pairs the certified lower bound with a decoder estimate; the FOM column uses the same notation. Own evidence labels these cases Interval and Estimate, respectively. Class effect identifies an exact value, a tighter endpoint, or a rigorous endpoint supplied by an isomorphic presentation. Obs. is the number of raw-log observations, not an equivalence-class multiplicity. Decomp. gives the connectivity decomposition computed from the displayed polynomials: **conn.** for one component, or $r \times \llbracket n_0, k_0 \rrbracket$ for r pairwise-isomorphic disjoint components with parameters $\llbracket n_0, k_0 \rrbracket$. Each component has the same true distance as the direct sum, so the row’s exact distance or its certified lower bound and reported endpoint carry over unchanged.

F. Generator attribution and artifacts

Attr. uses G for GPT-5.6 Sol, O for Claude Opus 5, N for the seed/null source, F for the fixed PBB baseline evaluated in every run, and – when no attribution record exists. These labels record associated generator programs, not independent discovery or the distance-certification method. At the proposal level, the retained CSS association counts are G=739 and O=313, with 181 in both sets, 35 seed/null-only rows, and six rows without an event. For PBB-small, the attribution counts are 90 G, 51 G/O, 2 G/N, and 8 G/O/N; for PBB-large, they are 8 G, 56 G/O, 11 G/O/N, and 4 O. F marks 20 retained fixed-baseline proposals and is orthogonal to the G/O/N partition: it can be appended to any model-association label and is not included in those partition counts. Twenty-five PBB-large bursts remain ambiguous at the individual-program level, although their sets of associated models are fixed.

For the 51 retained PBB-large targets, the completed audit certifies 26 exact distances and leaves 25 ranges with certified lower bounds and supported upper endpoints. Further details, the complete machine-readable catalogue, and supporting evidence are available through the project repository [1].

TABLE S1: Catalogue accounting after filtering. “Logged obs.” counts all observations with a positive reported distance; “Kept obs.” restricts this count to retained proposals. “All” and “Kept” give proposal counts before and after filtering, and “Classes” gives the number of retained presentation classes. Exact and unresolved counts are proposal-level statuses before the final class merge and can include low-weight evidence transferred from a verified representative. Unresolved rows pair a certified lower bound with either a rigorous upper bound or a decoder estimate.

Campaign	Logged obs.	Kept obs.	All	Kept	Classes	Removed	Exact	Unresolved
CSS-small	5,610	5,269	382	365	164	17	99	266
CSS-large	834	824	554	547	371	7	94	453
PBB-small	4,890	3,625	257	151	39	106	151	0
PBB-large	2,589	2,318	104	79	56	25	54	25
Total	13,923	12,036	1,297	1,142	630	155	398	744

TABLE S2: Explicit connected BB presentations of components of certified disconnected parents. Canonical labelling verifies that the corresponding presentation graphs are isomorphic, transferring the component’s certified distance to each presentation.

Code	(ℓ, m)	A	B	FOM	Certified parent
[[90, 4, 9]]	(5, 9)	$1 + x^2y^8 + x^3y^7$	$1 + x^2y^6$	3.60	[[180, 8, 9]]
[[96, 4, 10]]	(16, 3)	$1 + y + x^2y^2$	$1 + x^3$	4.17	[[384, 16, 10]]
[[140, 6, 10]]	(5, 14)	$1 + y^{13} + x^4y^{11}$	$1 + xy^7$	4.29	[[420, 18, 10]]

TABLE S3: Explicit connected PBB presentations of components of certified disconnected parents. Canonical labelling verifies that the corresponding presentation graphs are isomorphic, transferring the component’s certified distance to each presentation.

Code	(ℓ, m)	(A, B)	(C, D)	FOM	Certified parent
[[144, 4, 8]]	(18, 4)	$(1 + x^3y, 1 + xy^2 + x^2)$	$(1 + x^3y, xy^2)$	1.78	[[432, 12, 8]]
[[180, 2, 11]]	(15, 6)	$(1 + x^2y^2, 1 + x^2y^3 + x^4)$	$(1 + x^2y^2, 1 + x^4)$	1.34	[[360, 4, 11]]
[[216, 4, 10]]	(18, 6)	$(1 + x^2y, 1 + xy^3 + x^2)$	$(1 + x^2y, 1 + x^2)$	1.85	[[432, 8, 10]]
[[216, 4, 10]]	(18, 6)	$(1 + x^4y, 1 + xy^3 + x^2)$	$(0, 1 + x^2)$	1.85	[[432, 8, 10]]

TABLE S4: Certified parameter summary after combining evidence within presentation classes. For each exact parameter triple, Props. and Cls. respectively count retained proposals and presentation classes; Conn./disc. splits the classes into connected and disconnected. The displayed distance and FOM reflect class-level evidence. Rep. is chosen deterministically, preferring an exact proposal-level status and then a connected presentation. The representative columns give its lattice, generators (with $-$ in (C, D) for CSS), and component decomposition. The selected representative need not supply the decisive certificate. An asterisk marks a row containing both connected and disconnected classes; in such a row Decomp. still describes only the representative. The seven explicit connected-component presentations in Tables S2 and S3 are excluded from these counts.

Fam. Code	Rep.	(ℓ, m)	(A, B)	(C, D)	Decomp.	FOM Props.	Cls.	Conn./disc.
CSS [120, 4, 4]	CS-d74f23a3 (6, 10)	$(1 + y^3, 1 + x + x^2)$	—	conn.	0.53	1	1	1/0
CSS [120, 4, 10]	CS-07d0fcb6 (6, 10)	$(1 + xy + x^5y^5, 1 + y)$	—	conn.	3.33	6	1	1/0
CSS [120, 8, 3]	CS-183fa7f4 (6, 10)	$(1 + xy^2 + x^2y^4, 1 + x^3y^6)$	—	$2 \times [60, 4]$	0.60	2	1	0/1
CSS [120, 8, 4]	CS-ab5fa0e9 (6, 10)	$(1 + x^2y^2 + x^4y^4, 1 + x^3y^8)$	—	$2 \times [60, 4]$	1.07	5	3	0/3
CSS [120, 8, 5]	CS-999229aa (6, 10)	$(1 + y^2, 1 + x + x^5y^3)$	—	conn.*	1.67	17	9	3/6
CSS [120, 16, 3]	CS-c600e4d1 (6, 10)	$(1 + x^2y^2 + x^4y^6, 1 + y^2)$	—	$4 \times [30, 4]$	1.20	1	1	0/1
CSS [120, 16, 4]	CS-9dfccec1 (6, 10)	$(1 + y^4, 1 + x^2 + x^4y^4)$	—	$4 \times [30, 4]$	2.13	2	1	0/1
CSS [120, 16, 5]	CS-435681c1 (6, 10)	$(1 + x^2y^2 + x^4, 1 + y^6)$	—	$4 \times [30, 4]$	3.33	1	1	0/1
CSS [162, 4, 3]	CS-b50a397b (9, 9)	$(1 + y^3, 1 + y^4 + x^4y^7)$	—	conn.	0.22	9	2	2/0
CSS [162, 12, 3]	CS-a8c58a3b (9, 9)	$(1 + y^3, 1 + x^4y^4 + x^8y^5)$	—	$3 \times [54, 4]$	0.67	10	4	0/4
CSS [162, 12, 4]	CS-df3ce049 (9, 9)	$(1 + x^2, 1 + y^6 + x^2y^3)$	—	$3 \times [54, 4]$	1.19	2	1	0/1
CSS [162, 12, 5]	CS-7701034d (9, 9)	$(1 + xy^6 + x^8y^3, 1 + x)$	—	$3 \times [54, 4]$	1.85	3	1	0/1
CSS [162, 12, 6]	CS-5aa2af5d (9, 9)	$(1 + y^7 + x^3y^4, 1 + x^3y^5)$	—	$3 \times [54, 4]$	2.67	21	3	0/3
CSS [180, 4, 3]	CS-d501d19e (10, 9)	$(1 + y^2 + x^3y^4, 1 + y^6)$	—	conn.	0.20	6	5	5/0
CSS [180, 4, 5]	CS-712a2f67 (10, 9)	$(1 + xy + x^2y^2, 1 + x^6)$	—	conn.	0.56	11	8	8/0
CSS [180, 4, 13]	CS-6a4be7ad (10, 9)	$(1 + x^2y^3, 1 + x^5y + x^7y^8)$	—	conn.	3.76	1	1	1/0
CSS [180, 4, 14]	CS-1530f6fa (10, 9)	$(1 + x^2y^2 + x^5y, 1 + x^2y^3)$	—	conn.	4.356	1	1	1/0
CSS [180, 8, 3]	CS-7223b27c (10, 9)	$(1 + y^3, 1 + x^6y^2 + x^9y^5)$	—	conn.*	0.40	3	3	1/2
CSS [180, 8, 4]	CS-54e115ab (10, 9)	$(1 + x^6y + x^8y^5, 1 + x^2y^3)$	—	$2 \times [90, 4]$	0.71	1	1	0/1
CSS [180, 8, 5]	CS-770cc2a8 (10, 9)	$(1 + y^2 + x^8y, 1 + x^6)$	—	$2 \times [90, 4]$	1.11	10	7	0/7
CSS [180, 8, 9]	CS-d473da9b (10, 9)	$(1 + x^4y^8 + x^6y^7, 1 + x^4y^6)$	—	$2 \times [90, 4]$	3.60	1	1	0/1
CSS [180, 12, 4]	CS-359b0899 (10, 9)	$(1 + y^3 + x^5y^6, 1 + x^2)$	—	$3 \times [60, 4]$	1.07	2	2	0/2
CSS [180, 12, 5]	CS-6d6a31f9 (10, 9)	$(1 + x^6, 1 + xy^3 + x^5y^6)$	—	$3 \times [60, 4]$	1.67	2	2	0/2
CSS [180, 16, 3]	CS-4496a7f0 (10, 9)	$(1 + y^3, 1 + x^2y^7 + x^4)$	—	$2 \times [90, 8]$	0.80	2	1	0/1
CSS [180, 20, 5]	CS-31aff634 (10, 9)	$(1 + x^5y^2 + x^5y^4, 1 + x^5y^3)$	—	$5 \times [36, 4]$	2.78	1	1	0/1
CSS [180, 24, 3]	CS-fb880f70 (10, 9)	$(1 + x^2, 1 + x^2y^6 + x^6y^3)$	—	$6 \times [30, 4]$	1.20	2	1	0/1
CSS [180, 24, 5]	CS-6a1b1dfe (10, 9)	$(1 + x^2y^3 + x^2y^6, 1 + x^6)$	—	$6 \times [30, 4]$	3.33	1	1	0/1

TABLE S4 (continued)

Fam. Code	Rep.	(ℓ, m)	(A, B)	(C, D)	Decomp.	FOM Props.	Cls.	Conn./disc.	
CSS [384, 4, 4]	CL-6e950816	(16, 12)	$(1 + y^3, 1 + x^3y^{10} + x^{13}y^5)$	–	conn.	0.17	11	6	6/0
CSS [384, 8, 4]	CL-bea18d22	(16, 12)	$(1 + x, 1 + y^2 + y^4)$	–	$2 \times [192, 4]$	0.33	5	3	0/3
CSS [384, 8, 10]	CL-cbea6bcb	(16, 12)	$(1 + xy^3, 1 + x^4y^{10} + x^{11}y^5)$	–	$2 \times [192, 4]$	2.08	3	1	0/1
CSS [384, 16, 3]	CL-64686929	(16, 12)	$(1 + x^5y^4 + x^{11}y^8, 1 + x)$	–	$4 \times [96, 4]$	0.38	1	1	0/1
CSS [384, 16, 4]	CL-f1c90545	(16, 12)	$(1 + x^4, 1 + x^2y + x^{15}y^5)$	–	conn.*	0.67	9	8	1/7
CSS [384, 16, 6]	CL-52dcc64d	(16, 12)	$(1 + x^2y^9, 1 + x^2y^{11} + x^{14}y)$	–	$4 \times [96, 4]$	1.50	3	2	0/2
CSS [384, 16, 7]	CL-9fc416af	(16, 12)	$(1 + x, 1 + y^4 + x^3y^8)$	–	$4 \times [96, 4]$	2.04	2	2	0/2
CSS [384, 16, 8]	CL-49103759	(16, 12)	$(1 + x, 1 + x^3y^4 + x^7y^8)$	–	$4 \times [96, 4]$	2.67	6	4	0/4
CSS [384, 16, 10]	CL-885dfa58	(16, 12)	$(1 + y^4 + x^2y^8, 1 + x^3y^6)$	–	$4 \times [96, 4]$	4.17	1	1	0/1
CSS [384, 32, 3]	CL-d9f65a7e	(16, 12)	$(1 + x^4y^2 + x^{10}y^7, 1 + x^2y^3)$	–	$8 \times [48, 4]$	0.75	1	1	0/1
CSS [384, 32, 4]	CL-a94de289	(16, 12)	$(1 + x^2y^5 + x^4y^{10}, 1 + x^4y^6)$	–	$8 \times [48, 4]$	1.33	2	2	0/2
CSS [384, 32, 5]	CL-83dd7b5d	(16, 12)	$(1 + x^2y^6, 1 + x^2y^2 + x^4y^4)$	–	$8 \times [48, 4]$	2.08	1	1	0/1
CSS [384, 32, 6]	CL-a25b2baf	(16, 12)	$(1 + x^4y^2 + x^4y^{10}, 1 + x^2y^9)$	–	$8 \times [48, 4]$	3.00	1	1	0/1
CSS [384, 64, 3]	CL-929c2a70	(16, 12)	$(1 + x^4y^4 + x^{12}y^8, 1 + x^4)$	–	$16 \times [24, 4]$	1.50	1	1	0/1
CSS [420, 4, 5]	CL-89dd6c68	(15, 14)	$(1 + x^2y^{10} + x^{10}y^5, 1 + x^6)$	–	conn.	0.24	3	3	3/0
CSS [420, 6, 3]	CL-01ad3134	(15, 14)	$(1 + x^2y^2 + x^3y^3, 1 + x^5)$	–	conn.	0.13	1	1	1/0
CSS [420, 6, 5]	CL-c065aeaf	(15, 14)	$(1 + xy^7, 1 + x^9y^9 + x^{13}y^3)$	–	conn.	0.36	1	1	1/0
CSS [420, 8, 4]	CL-ae04e166	(15, 14)	$(1 + xy^2 + x^2y^4, 1 + x^6y^4)$	–	$2 \times [210, 4]$	0.30	1	1	0/1
CSS [420, 8, 5]	CL-37987300	(15, 14)	$(1 + xy^6 + x^{11}y^2, 1 + x^3y^6)$	–	$2 \times [210, 4]$	0.48	2	1	0/1
CSS [420, 10, 5]	CL-94086b72	(15, 14)	$(1 + x^2y + x^{10}y^5, 1 + x^6)$	–	conn.	0.60	1	1	1/0
CSS [420, 12, 3]	CL-7a5b9b2b	(15, 14)	$(1 + x^5, 1 + xy^4 + x^8y^{12})$	–	$2 \times [210, 6]$	0.26	1	1	0/1
CSS [420, 12, 9]	CL-84c084df	(15, 14)	$(1 + x, 1 + x^2y^2 + x^8y^6)$	–	$2 \times [210, 6]$	2.31	1	1	0/1
CSS [420, 12, 10]	CL-97118c06	(15, 14)	$(1 + x^4, 1 + y^4 + x^9y^6)$	–	$2 \times [210, 6]$	2.86	1	1	0/1
CSS [420, 16, 7]	CL-e6216eab	(15, 14)	$(1 + y^2, 1 + x^3 + x^4y^4)$	–	$2 \times [210, 8]$	1.87	3	3	0/3
CSS [420, 18, 9]	CL-0286b883	(15, 14)	$(1 + x^3y^7, 1 + x^6y^5 + x^9y)$	–	$3 \times [140, 6]$	3.47	1	1	0/1
CSS [420, 18, 10]	CL-f8ef7ae2	(15, 14)	$(1 + y^{13} + x^{12}y^{11}, 1 + x^3y^7)$	–	$3 \times [140, 6]$	4.29	1	1	0/1
CSS [420, 20, 5]	CL-253c2c53	(15, 14)	$(1 + x^6, 1 + x^7y^2 + x^{14}y^6)$	–	$2 \times [210, 10]$	1.19	1	1	0/1
CSS [420, 20, 6]	CL-e2ab51c4	(15, 14)	$(1 + y^2, 1 + x^5y^3 + x^{10}y^{13})$	–	$5 \times [84, 4]$	1.71	2	2	0/2
CSS [420, 20, 7]	CL-0eaa9430	(15, 14)	$(1 + y^4, 1 + x^5y^{10} + x^{10}y^5)$	–	$5 \times [84, 4]$	2.33	1	1	0/1
CSS [420, 20, 8]	CL-1830493f	(15, 14)	$(1 + x^5y^2 + x^{10}y^6, 1 + y^3)$	–	$5 \times [84, 4]$	3.05	1	1	0/1
CSS [420, 30, 3]	CL-abc3d2fe	(15, 14)	$(1 + x^5, 1 + y^2 + x^5y^3)$	–	$5 \times [84, 6]$	0.64	1	1	0/1
CSS [420, 40, 5]	CL-437b6e5f	(15, 14)	$(1 + x^5y^4 + x^{10}y^6, 1 + y^6)$	–	$10 \times [42, 4]$	2.38	1	1	0/1
CSS [420, 60, 3]	CL-4b943aa2	(15, 14)	$(1 + x^5y^{12} + x^{10}y^8, 1 + x^{10})$	–	$10 \times [42, 6]$	1.29	1	1	0/1
CSS [432, 4, 4]	CL-2de98d42	(18, 12)	$(1 + x^4y^3, 1 + x^3y^4 + x^6y^8)$	–	conn.	0.15	1	1	1/0
CSS [432, 4, 10]	CL-afd67a34	(18, 12)	$(1 + x^{14}y^5, 1 + xy^9 + x^{17}y^3)$	–	conn.	0.93	5	1	1/0
CSS [432, 8, 4]	CL-7ae78520	(18, 12)	$(1 + x, 1 + y^2 + y^4)$	–	$2 \times [216, 4]$	0.30	1	1	0/1
CSS [432, 12, 4]	CL-db6df8c9	(18, 12)	$(1 + y^3, 1 + x + x^2)$	–	$3 \times [144, 4]$	0.44	1	1	0/1
CSS [432, 12, 10]	CL-f85559a3	(18, 12)	$(1 + x^3y^3, 1 + x^5y^7 + x^{16}y^{11})$	–	$3 \times [144, 4]$	2.78	1	1	0/1
CSS [432, 16, 3]	CL-aa2abeb9	(18, 12)	$(1 + x^{12}y^8, 1 + xy^4 + x^5)$	–	$4 \times [108, 4]$	0.33	1	1	0/1
CSS [432, 16, 4]	CL-8157fe5b	(18, 12)	$(1 + x^4, 1 + y^2 + y^4)$	–	$4 \times [108, 4]$	0.59	3	2	0/2
CSS [432, 16, 6]	CL-c9d42fcb	(18, 12)	$(1 + y^2, 1 + x^2 + x^4)$	–	$4 \times [108, 4]$	1.33	2	2	0/2
CSS [432, 16, 7]	CL-d72254a8	(18, 12)	$(1 + x^2 + x^3y^8, 1 + xy^4)$	–	$4 \times [108, 4]$	1.81	2	1	0/1
CSS [432, 16, 8]	CL-c5ca6b9a	(18, 12)	$(1 + xy^6, 1 + x^{11}y^{10} + x^{15}y^2)$	–	$4 \times [108, 4]$	2.37	2	2	0/2
CSS [432, 16, 9]	CL-8fe7fff8	(18, 12)	$(1 + x^5y^4, 1 + x^5 + x^{17}y^8)$	–	$4 \times [108, 4]$	3.00	1	1	0/1
CSS [432, 24, 3]	CL-fedc4d95	(18, 12)	$(1 + x^3y^{11} + x^{12}y, 1 + x^6)$	–	$3 \times [144, 8]$	0.50	1	1	0/1
CSS [432, 24, 4]	CL-2c508b86	(18, 12)	$(1 + y^3, 1 + x^2 + x^4)$	–	$6 \times [72, 4]$	0.89	3	3	0/3
CSS [432, 24, 5]	CL-7827219c	(18, 12)	$(1 + x^6y^6, 1 + xy + x^2y^2)$	–	$6 \times [72, 4]$	1.39	1	1	0/1
CSS [432, 24, 6]	CL-63ba351e	(18, 12)	$(1 + y + x^{12}y^{11}, 1 + x^6y^6)$	–	$6 \times [72, 4]$	2.00	1	1	0/1
CSS [432, 24, 7]	CL-f93732aa	(18, 12)	$(1 + x^3y^3, 1 + y^{10} + x^3y^5)$	–	$6 \times [72, 4]$	2.72	1	1	0/1
CSS [432, 32, 3]	CL-5e6f356a	(18, 12)	$(1 + y^4, 1 + x^2 + x^4)$	–	$8 \times [54, 4]$	0.67	1	1	0/1
CSS [432, 32, 6]	CL-21ad24a2	(18, 12)	$(1 + x^4 + x^{12}y^4, 1 + x^4y^4)$	–	$8 \times [54, 4]$	2.67	1	1	0/1
CSS [432, 48, 3]	CL-b9275593	(18, 12)	$(1 + y^4, 1 + x^6y^2 + x^{12}y^4)$	–	$12 \times [36, 4]$	1.00	1	1	0/1
PBB [72, 4, 3]	PS-46a007cb	(6, 6)	$(1 + x^2y^2, 1 + xy^3 + x^2)$	$(1 + x^2y^2, 1 + x^2)$	$2 \times [36, 2]$	0.50	7	3	0/3
PBB [72, 4, 4]	PS-bb41fc22	(6, 6)	$(1 + y, 1 + x + x^2)$	$(1 + y, x)$	conn.*	0.89	41	7	4/3
PBB [72, 12, 3]	PS-5feb6ebb	(6, 6)	$(1 + x^3y^3, 1 + xy^3 + x^2)$	$(0, 1 + x^2)$	$6 \times [12, 2]$	1.50	8	4	0/4
PBB [108, 2, 4]	PS-dcbd4997	(6, 9)	$(1 + y, 1 + x + x^2)$	$(1 + y, x)$	conn.	0.30	2	1	1/0
PBB [120, 4, 4]	PS-9828cd17	(6, 10)	$(1 + y, 1 + x + x^2)$	$(1 + y, x)$	conn.*	0.53	36	7	4/3
PBB [120, 20, 3]	PS-194f5d29	(6, 10)	$(1 + x^3y^5, 1 + xy^5 + x^2)$	$(0, 1 + x^2)$	$10 \times [12, 2]$	1.50	8	4	0/4
PBB [144, 4, 6]	PS-b9bd7fa4	(9, 8)	$(1 + y, 1 + x + x^2)$	$(1 + y, x)$	conn.	1.00	28	5	5/0
PBB [144, 8, 3]	PS-c2455d37	(9, 8)	$(1 + x^3y^4, 1 + x + x^5y^4)$	$(0, 1 + x)$	$4 \times [36, 2]$	0.50	8	3	0/3
PBB [144, 8, 4]	PS-85429f03	(9, 8)	$(1 + x^3y^2, 1 + x + x^5y^4)$	$(0, 1 + x)$	$2 \times [72, 4]$	0.89	11	4	0/4
PBB [162, 2, 6]	PS-aacbee20	(9, 9)	$(1 + y, 1 + x + x^2)$	$(1 + y, x)$	conn.	0.44	2	1	1/0
PBB [216, 2, 8]	PL-ba29a9df	(12, 9)	$(1 + y, 1 + x + x^2)$	$(1 + y, x)$	conn.	0.59	2	1	1/0
PBB [288, 4, 8]	PL-592409d9	(12, 12)	$(1 + x + x^2, 1 + y)$	$(x, 1 + y)$	conn.	0.89	6	3	3/0
PBB [288, 8, 4]	PL-605688ff	(12, 12)	$(1 + x^2y, 1 + x^2y^6 + x^4)$	$(0, 1 + x^4)$	$2 \times [144, 4]$	0.44	6	2	0/2
PBB [288, 8, 6]	PL-adccb3fc	(12, 12)	$(1 + x^4y^2, 1 + xy^6 + x^2)$	$(1 + x^4y^2, 1 + x^2)$	$2 \times [144, 4]$	1.00	1	1	0/1
PBB [288, 16, 4]	PL-ad1cc828	(12, 12)	$(1 + xy, 1 + x^2y^6 + x^4)$	$(0, 1 + x^4)$	$4 \times [72, 4]$	0.89	6	4	0/4
PBB [288, 24, 4]	PL-78c3e7bf	(12, 12)	$(1 + x^3y^6, 1 + x^2y^6 + x^4)$	$(0, 1 + x^4)$	$6 \times [48, 4]$	1.33	1	1	0/1
PBB [288, 48, 3]	PL-9b97a02b	(12, 12)	$(1 + x^6y^6, 1 + x^2y^6 + x^4)$	$(1 + x^6y^6, 1 + x^4)$	$24 \times [12, 2]$	1.50	1	1	0/1
PBB [360, 4, 5]	PL-6b21188d	(15, 12)	$(1 + x^2y^2, 1 + x^2y^6 + x^4)$	$(0, x^2y^6)$	$2 \times [180, 2]$	0.28	3	2	0/2
PBB [360, 4, 7]	PL-48f3faa4	(15, 12)	$(1 + x^3y^4, 1 + xy^6 + x^2)$	$(1 + x^3y^4, xy^6)$	$2 \times [180, 2]$	0.54	1	1	0/1
PBB [360, 4, 8]	PL-93708752	(15, 12)	$(1 + xy, 1 + x^2y^6 + x^4)$	$(1 + xy, 1 + x^4)$	conn.	0.71	4	2	2/0

TABLE S4 (continued)

Fam. Code	Rep.	(ℓ, m)	(A, B)	(C, D)	Decomp.	FOM Props.	Cls.	Conn./disc.	
PBB [360, 4, 10]	PL-4602a8fa	(15, 12)	$(1 + x + x^2, 1 + y)$	$(x, 1 + y)$	conn.	1.11	2	1	1/0
PBB [360, 4, 11]	PL-8ee754a5	(15, 12)	$(1 + x^2y^4, 1 + x^2y^6 + x^4)$	$(1 + x^2y^4, 1 + x^4)$	$2 \times [180, 2]$	1.34	1	1	0/1
PBB [360, 12, 4]	PL-93be42eb	(15, 12)	$(1 + x^3y^3, 1 + x^2y^6 + x^4)$	$(1 + x^3y^3, x^2y^6)$	$3 \times [120, 4]$	0.53	2	2	0/2
PBB [420, 4, 10]	PL-ce4ae900	(15, 14)	$(1 + y, 1 + x + x^2)$	$(1 + y, x)$	conn.	0.95	2	1	1/0
PBB [432, 8, 3]	PL-232490ff	(18, 12)	$(1 + x^6y^4, 1 + x^2y^6 + x^4)$	$(0, x^2y^6)$	$4 \times [108, 2]$	0.17	1	1	0/1
PBB [432, 8, 7]	PL-2b5f2b8d	(18, 12)	$(1 + x^2y^4, 1 + xy^6 + x^2)$	$(0, 1 + x^2)$	$4 \times [108, 2]$	0.91	4	3	0/3
PBB [432, 8, 8]	PL-cdaa4f44	(18, 12)	$(1 + xy, 1 + x^2y^6 + x^4)$	$(1 + xy, x^2y^6)$	$2 \times [216, 4]$	1.19	3	2	0/2
PBB [432, 8, 10]	PL-36c5ce3b	(18, 12)	$(1 + x^2y^2, 1 + xy^6 + x^2)$	$(1 + x^2y^2, 1 + x^2)$	$2 \times [216, 4]$	1.85	2	2	0/2
PBB [432, 12, 4]	PL-597f6b12	(18, 12)	$(1 + x^3y^3, 1 + x^3y^6 + x^6)$	$(0, x^3y^6)$	$3 \times [144, 4]$	0.44	2	2	0/2
PBB [432, 12, 8]	PL-aefa6773	(18, 12)	$(1 + x^3y^3, 1 + xy^6 + x^2)$	$(1 + x^3y^3, xy^6)$	$3 \times [144, 4]$	1.78	1	1	0/1
PBB [432, 24, 3]	PL-88c05b17	(18, 12)	$(1 + x^3y^6, 1 + xy^6 + x^2)$	$(1 + x^3y^6, 1 + x^2)$	$12 \times [36, 2]$	0.50	5	5	0/5
PBB [432, 24, 4]	PL-5f3d871a	(18, 12)	$(1 + x^3y^2, 1 + x^3y^6 + x^6)$	$(0, x^3y^6)$	$12 \times [36, 2]$	0.89	3	3	0/3

TABLE S5: CSS-small compact catalogue (224 displayed of 365 retained proposals; 164 presentation classes). The table includes every proposal in an exact class and one deterministic representative of every nonexact class. Own evidence and Evid. describe the proposal before class merging; Class effect reports when merging makes the displayed d and FOM exact, tighter, or rigorously supported. Column abbreviations, filtering, and evidence conventions are defined above.

ID/class	(ℓ, m)	$[n, k]$	d	Own evidence	Class effect	FOM	A	B	Evid.	Obs.	Attr.	Decomp.
CS-435681c1/C040	(6, 10)	[120, 16]	5	Exact	—	3.33	$1 + x^2y^2 + x^4$	$1 + y^6$	M	9	G	$4 \times [30, 4]$
CS-9dfccec1/C038	(6, 10)	[120, 16]	4	Exact	—	2.13	$1 + y^4$	$1 + x^2 + x^4y^4$	Exh.	1	G	$4 \times [30, 4]$
CS-8146fb41/C038	(6, 10)	[120, 16]	4	Exact	—	2.13	$1 + x^2y^4 + x^4$	$1 + y^6$	Exh.	1	G	$4 \times [30, 4]$
CS-c600e4d1/C039	(6, 10)	[120, 16]	3	Exact	—	1.20	$1 + x^2y^2 + x^4y^6$	$1 + y^2$	Exh.	121	G/O	$4 \times [30, 4]$
CS-999229aa/C022	(6, 10)	[120, 8]	5	Exact	—	1.67	$1 + y^2$	$1 + x + x^5y^3$	L4+W+B 2		G	conn.
CS-fc4dc8ca/C025	(6, 10)	[120, 8]	5	Exact	—	1.67	$1 + y^2$	$1 + xy^2 + x^2y^6$	L4+W+B 1		G	$2 \times [60, 4]$
CS-461d5f82/C031	(6, 10)	[120, 8]	5	Exact	—	1.67	$1 + y^4$	$1 + xy + x^5y^4$	L4+W+B 7		G	conn.
CS-83ce21aa/C031	(6, 10)	[120, 8]	5	Exact	—	1.67	$1 + y^8$	$1 + xy^4 + x^5y^7$	L4+W+B 1		O	conn.
CS-373a98a4/C029	(6, 10)	[120, 8]	5	Exact	—	1.67	$1 + y^8$	$1 + x^2y^8 + x^4y$	L4+W+B 2		O	$2 \times [60, 4]$
CS-654080d0/C030	(6, 10)	[120, 8]	5	Exact	—	1.67	$1 + xy^4 + x^2y^8$	$1 + x^3y^4$	L4+W+B 5		G/O	$2 \times [60, 4]$
CS-95b2af5c/C031	(6, 10)	[120, 8]	5	Exact	—	1.67	$1 + xy^4 + x^5y^3$	$1 + y^6$	L4+W+B 7		G	conn.
CS-ba902f48/C025	(6, 10)	[120, 8]	5	Exact	—	1.67	$1 + xy^6 + x^2y^8$	$1 + y^4$	L4+W+B 1		O	$2 \times [60, 4]$
CS-56984308/C026	(6, 10)	[120, 8]	5	Exact	—	1.67	$1 + xy^6 + x^5y^6$	$1 + y^6$	L4+W+B 5		G/O	$2 \times [60, 4]$
CS-81c206b7/C027	(6, 10)	[120, 8]	5	Exact	—	1.67	$1 + x^2y + x^4$	$1 + y^2$	L4+W+B 3		G/O	$2 \times [60, 4]$
CS-75b1d537/C025	(6, 10)	[120, 8]	5	Exact	—	1.67	$1 + x^2y^3 + x^4y^9$	$1 + y^8$	L4+W+B 1		G	$2 \times [60, 4]$
CS-f0ceea98/C029	(6, 10)	[120, 8]	5	Exact	—	1.67	$1 + x^2y^4 + x^4y^3$	$1 + y^4$	L4+W+B 13		G/O	$2 \times [60, 4]$
CS-2aa84247/C029	(6, 10)	[120, 8]	5	Exact	—	1.67	$1 + x^2y^4 + x^4y^3$	$1 + y^6$	L4+W+B 2		G	$2 \times [60, 4]$
CS-49a65af6/C036	(6, 10)	[120, 8]	5	Exact	—	1.67	$1 + x^3y^2$	$1 + x + x^2y^6$	L4+W+B 2		O/N	$2 \times [60, 4]$
CS-1363c9b9/C036	(6, 10)	[120, 8]	5	Exact	—	1.67	$1 + x^3y^4$	$1 + x^4y^2 + x^5y^2$	L4+W+B 1		G	$2 \times [60, 4]$
CS-48842317/C034	(6, 10)	[120, 8]	5	Exact	—	1.67	$1 + x^4y + x^5y$	$1 + y^2$	L4+W+B 1		G	conn.
CS-ff5ab961/C036	(6, 10)	[120, 8]	5	Exact	—	1.67	$1 + x^4y^4 + x^5y^5$	$1 + x^3y^3$	L4+W+B 2		G	$2 \times [60, 4]$
CS-ab5fa0e9/C028	(6, 10)	[120, 8]	4	Exact	—	1.07	$1 + x^2y^2 + x^4y^4$	$1 + x^3y^8$	Exh.	1	G	$2 \times [60, 4]$
CS-139f4bc6/C019	(6, 10)	[120, 8]	4	Exact	—	1.07	$1 + x^3y^6$	$1 + x^4 + x^5$	Exh.	162	G/O	$2 \times [60, 4]$
CS-35bf8221/C021	(6, 10)	[120, 8]	4	Exact	—	1.07	$1 + x^3y^7$	$1 + xy^3 + x^2$	Exh.	5	G	$2 \times [60, 4]$
CS-f0816ca6/C028	(6, 10)	[120, 8]	4	Exact	—	1.07	$1 + x^3y^8$	$1 + x^2y^8 + x^4y^2$	Exh.	5	G	$2 \times [60, 4]$
CS-3a584232/C021	(6, 10)	[120, 8]	4	Exact	—	1.07	$1 + x^4 + x^5y^6$	$1 + x^3y^4$	Exh.	1	G	$2 \times [60, 4]$
CS-183fa7f4/C035	(6, 10)	[120, 8]	3	Exact	—	0.60	$1 + xy^2 + x^2y^4$	$1 + x^3y^6$	Exh.	1	G	$2 \times [60, 4]$
CS-3aa743db/C035	(6, 10)	[120, 8]	3	Exact	—	0.60	$1 + x^2y^6 + x^4y^3$	$1 + y^9$	Exh.	1	G	$2 \times [60, 4]$
CS-b1622590/C023	(6, 10)	[120, 8]	5; $\widehat{7}$	Estimate	Tighter	$1.67; \widehat{3.27}$	$1 + y$	$1 + x^2y + x^4y^3$	L4+B	170	G/O	$2 \times [60, 4]$
CS-61381a4a/C032	(6, 10)	[120, 8]	5; $\widehat{8}$	Estimate	—	$1.67; \widehat{4.27}$	$1 + x^3y^2$	$1 + x^2y^8 + x^4$	L4+B	2	G	$2 \times [60, 4]$
CS-c6d700d2/C020	(6, 10)	[120, 8]	5; $\widehat{7}$	Estimate	Tighter	$1.67; \widehat{3.27}$	$1 + x^3y^3$	$1 + xy + x^2$	L4+B	4	G/O/N	$2 \times [60, 4]$
CS-4ae19507/C024	(6, 10)	[120, 8]	5; $\widehat{6}$	Estimate	Tighter	$1.67; \widehat{2.40}$	$1 + xy + x^5y^5$	$1 + x^3y$	L4+B	2	G/O	$2 \times [60, 4]$
CS-5f82ded4/C037	(6, 10)	[120, 8]	5; $\widehat{7}$	Estimate	—	$1.67; \widehat{3.27}$	$1 + xy^6 + x^2y^8$	$1 + x^3y^8$	L4+B	2	G	$2 \times [60, 4]$
CS-960aef72/C018	(6, 10)	[120, 8]	5; $\widehat{6}$	Estimate	—	$1.67; \widehat{2.40}$	$1 + y^7$	$1 + x^2y^2 + x^4y^6$	L4+B	2	G	$2 \times [60, 4]$
CS-f5976984/C033	(6, 10)	[120, 8]	5; $\widehat{6}$	Estimate	—	$1.67; \widehat{2.40}$	$1 + x^2y^2 + x^4y^2$	$1 + x^3y^9$	L4+B	1	G	$2 \times [60, 4]$
CS-07d0fcb6/C011	(6, 10)	[120, 4]	10	Exact	—	3.33	$1 + xy + x^5y^5$	$1 + y$	M	2	G	conn.
CS-d74f23a3/C001	(6, 10)	[120, 4]	4	Exact	—	0.53	$1 + y^3$	$1 + x + x^2$	Exh.	1	G	conn.
CS-2a6c3475/C006	(6, 10)	[120, 4]	5; $\widehat{10}$	Estimate	Tighter	$0.83; \widehat{3.33}$	$1 + x^3y^8$	$1 + x^2y^5 + x^4y^4$	L4+B	2	G	conn.
CS-681e1b89/C007	(6, 10)	[120, 4]	5; $\widehat{10}$	Estimate	Tighter	$0.83; \widehat{3.33}$	$1 + x^4y^9 + x^5y$	$1 + x^3y$	L4+B	1	O	conn.
CS-442a59e3/C003	(6, 10)	[120, 4]	5; $\widehat{10}$	Estimate	Tighter	$0.83; \widehat{3.33}$	$1 + y^3$	$1 + xy^3 + x^5y^2$	L4+B	13	G/O	conn.
CS-1eb1785f/C002	(6, 10)	[120, 4]	5; $\widehat{8}$	Estimate	Tighter	$0.83; \widehat{2.13}$	$1 + xy + x^2y^2$	$1 + x^3y^4$	L4+B	1	O	conn.
CS-21cfa1e9/C017	(6, 10)	[120, 4]	5; $\widehat{10}$	Estimate	—	$0.83; \widehat{3.33}$	$1 + xy^3 + x^5y^9$	$1 + x^3y^4$	L4+B	2	G	conn.
CS-23cbfbfd/C016	(6, 10)	[120, 4]	5; $\widehat{10}$	Estimate	—	$0.83; \widehat{3.33}$	$1 + xy^6 + x^5y$	$1 + y^3$	L4+B	1	G	conn.
CS-a583e335/C010	(6, 10)	[120, 4]	5; $\widehat{10}$	Estimate	—	$0.83; \widehat{3.33}$	$1 + xy^7 + x^5y^2$	$1 + y^3$	L4+B	2	G/O	conn.
CS-12c8b44e/C011	(6, 10)	[120, 4]	10	Estimate	Exact	3.33	$1 + x^2y^8 + x^4y^3$	$1 + x^3y^2$	L4+B	1	O	conn.
CS-06a00cb2/C012	(6, 10)	[120, 4]	5; $\widehat{10}$	Estimate	—	$0.83; \widehat{3.33}$	$1 + x^3y$	$1 + x^2y^7 + x^4y^7$	L4+B	1	G	conn.
CS-0a7f4a24/C011	(6, 10)	[120, 4]	10	Estimate	Exact	3.33	$1 + x^3y^3$	$1 + xy^2 + x^5$	L4+B	1	N	conn.
CS-85b452ec/C011	(6, 10)	[120, 4]	10	Estimate	Exact	3.33	$1 + x^3y^3$	$1 + x^2y^2 + x^4y^5$	L4+B	1	G	conn.
CS-e21fcdf6/C011	(6, 10)	[120, 4]	10	Estimate	Exact	3.33	$1 + x^3y^6$	$1 + x^4y^4 + x^5y^5$	L4+B	1	G	conn.

TABLE S5 (continued)

ID/class	(ℓ, m)	$\llbracket n, k \rrbracket$	d	Own evidence	Class effect	FOM	A	B	Evid.	Obs.	Attr.	Decomp.
CS-fa6ee2a0/C011	(6, 10)	$\llbracket 120, 4 \rrbracket$	10	Estimate	Exact	3.33	$1 + x^3 y^6$	$1 + x^4 y^4 + x^5 y^9$	L4+B	13	G/O	conn.
CS-289980e5/C004	(6, 10)	$\llbracket 120, 4 \rrbracket$	$5; \widehat{8}$	Estimate	Tighter	$0.83; \widehat{2.13}$	$1 + y^3$	$1 + xy^2 + x^5 y^2$	L4+B	7	G/O	conn.
CS-cd30a97d/C008	(6, 10)	$\llbracket 120, 4 \rrbracket$	$5; \widehat{8}$	Estimate	Tighter	$0.83; 2.13$	$1 + x + x^2 y^7$	$1 + x^3 y^7$	L4+B	2	G	conn.
CS-1b5f2c76/C014	(6, 10)	$\llbracket 120, 4 \rrbracket$	$5; \widehat{8}$	Estimate	Tighter	$0.83; 2.13$	$1 + xy + x^5 y^3$	$1 + x^3 y^4$	L4+B	1	G	conn.
CS-f9c94426/C005	(6, 10)	$\llbracket 120, 4 \rrbracket$	$5; \widehat{8}$	Estimate	Tighter	$0.83; \widehat{2.13}$	$1 + xy^7 + x^5 y^4$	$1 + x^3 y^6$	L4+B	3	G	conn.
CS-18b82d12/C013	(6, 10)	$\llbracket 120, 4 \rrbracket$	$5; \widehat{6}$	Estimate	Tighter	$0.83; \widehat{1.20}$	$1 + x^3 y^2$	$1 + x^4 y + x^5 y^4$	L4+B	15	G/O	conn.
CS-20bcc637/C015	(6, 10)	$\llbracket 120, 4 \rrbracket$	$5; \widehat{6}$	Estimate	—	$0.83; \widehat{1.20}$	$1 + x^3 y^3$	$1 + xy^5 + x^5$	L4+B	1	N	conn.
CS-06d9c8e4/C009	(6, 10)	$\llbracket 120, 4 \rrbracket$	$5; \widehat{6}$	Estimate	—	$0.83; \widehat{1.20}$	$1 + x^3 y^3$	$1 + x^2 y^2 + x^4 y$	L4+B	1	O	conn.
CS-5aa2af5d/C053	(9, 9)	$\llbracket 162, 12 \rrbracket$	6	Exact	—	2.67	$1 + y^7 + x^3 y^4$	$1 + x^3 y^5$	M	2	G	$3 \times \llbracket 54, 4 \rrbracket$
CS-0e4d0ef4/C056	(9, 9)	$\llbracket 162, 12 \rrbracket$	6	Exact	—	2.67	$1 + x^3 y$	$1 + x^3 y^3 + x^6$	M	1	O	$3 \times \llbracket 54, 4 \rrbracket$
CS-ed9ea89a/C053	(9, 9)	$\llbracket 162, 12 \rrbracket$	6	Exact	—	2.67	$1 + x^3 y^4 + x^6 y^4$	$1 + y^2$	M	28	G/O	$3 \times \llbracket 54, 4 \rrbracket$
CS-0c92ab37/C050	(9, 9)	$\llbracket 162, 12 \rrbracket$	6	Exact	—	2.67	$1 + x^4 y^6$	$1 + x^2 + x^2 y^6$	M	135	G/O/N	$3 \times \llbracket 54, 4 \rrbracket$
CS-7701034d/C055	(9, 9)	$\llbracket 162, 12 \rrbracket$	5	Exact	—	1.85	$1 + xy^6 + x^8 y^3$	$1 + x$	L4+W+B	1	O	$3 \times \llbracket 54, 4 \rrbracket$
CS-dc117df8/C055	(9, 9)	$\llbracket 162, 12 \rrbracket$	5	Exact	—	1.85	$1 + x^6 y^2$	$1 + x^3 y^5 + x^6 y^7$	L4+W+B	10	G/O	$3 \times \llbracket 54, 4 \rrbracket$
CS-df3ce049/C054	(9, 9)	$\llbracket 162, 12 \rrbracket$	4	Exact	—	1.19	$1 + x^2$	$1 + y^6 + x^2 y^3$	Exh.	147	G/O	$3 \times \llbracket 54, 4 \rrbracket$
CS-0b911359/C054	(9, 9)	$\llbracket 162, 12 \rrbracket$	4	Exact	—	1.19	$1 + x^5$	$1 + x^2 y^6 + x^3 y^3$	Exh.	1	G	$3 \times \llbracket 54, 4 \rrbracket$
CS-a8c58a3b/C052	(9, 9)	$\llbracket 162, 12 \rrbracket$	3	Exact	—	0.67	$1 + y^3$	$1 + x^4 y^4 + x^8 y^5$	Exh.	1	G	$3 \times \llbracket 54, 4 \rrbracket$
CS-fb456ca8/C048	(9, 9)	$\llbracket 162, 12 \rrbracket$	3	Exact	—	0.67	$1 + y^5 + x^3 y^7$	$1 + y^3$	Exh.	1	G	$3 \times \llbracket 54, 4 \rrbracket$
CS-52e7ba0e/C052	(9, 9)	$\llbracket 162, 12 \rrbracket$	3	Exact	—	0.67	$1 + xy^3 + x^2 y^3$	$1 + x^6 y^6$	Exh.	2	G	$3 \times \llbracket 54, 4 \rrbracket$
CS-c5681bf5/C049	(9, 9)	$\llbracket 162, 12 \rrbracket$	3	Exact	—	0.67	$1 + x^3 y^3$	$1 + xy^6 + x^5 y^3$	Exh.	1	G	$3 \times \llbracket 54, 4 \rrbracket$
CS-4dd38060/C051	(9, 9)	$\llbracket 162, 12 \rrbracket$	3	Exact	—	0.67	$1 + x^5 y^7 + x^8 y^7$	$1 + x^6 y^3$	Exh.	1	G	$3 \times \llbracket 54, 4 \rrbracket$
CS-47731b90/C051	(9, 9)	$\llbracket 162, 12 \rrbracket$	3	Exact	—	0.67	$1 + x^5 y^8 + x^8 y^8$	$1 + x^3 y^3$	Exh.	1	G	$3 \times \llbracket 54, 4 \rrbracket$
CS-f5cf4d10f/C052	(9, 9)	$\llbracket 162, 12 \rrbracket$	3	Exact	—	0.67	$1 + x^6$	$1 + xy^2 + x^5 y^7$	Exh.	18	G/O	$3 \times \llbracket 54, 4 \rrbracket$
CS-52200a16/C052	(9, 9)	$\llbracket 162, 12 \rrbracket$	3	Exact	—	0.67	$1 + x^6$	$1 + x^2 y^7 + x^7 y^8$	Exh.	1	G	$3 \times \llbracket 54, 4 \rrbracket$
CS-615a8fd3/C048	(9, 9)	$\llbracket 162, 12 \rrbracket$	3	Exact	—	0.67	$1 + x^6 y^2 + x^6 y^7$	$1 + y^6$	Exh.	2	G	$3 \times \llbracket 54, 4 \rrbracket$
CS-a6051954/C049	(9, 9)	$\llbracket 162, 12 \rrbracket$	3	Exact	—	0.67	$1 + x^6 y^6$	$1 + xy^8 + x^5 y^4$	Exh.	11	G/O	$3 \times \llbracket 54, 4 \rrbracket$
CS-18882b71/C050	(9, 9)	$\llbracket 162, 12 \rrbracket$	6	Estimate	Exact	2.67	$1 + x^2 y^4$	$1 + x^4 y^5 + x^7 y^8$	L4+B	1	G	$3 \times \llbracket 54, 4 \rrbracket$
CS-a86a3fa0/C053	(9, 9)	$\llbracket 162, 12 \rrbracket$	6	Estimate	Exact	2.67	$1 + x^3 y^6 + x^8 y^2$	$1 + xy^4$	L4+B	1	O	$3 \times \llbracket 54, 4 \rrbracket$
CS-78838dd5/C053	(9, 9)	$\llbracket 162, 12 \rrbracket$	6	Estimate	Exact	2.67	$1 + y$	$1 + x^3 + x^6 y^7$	L4+B	1	G	$3 \times \llbracket 54, 4 \rrbracket$
CS-85b036e7/C050	(9, 9)	$\llbracket 162, 12 \rrbracket$	6	Estimate	Exact	2.67	$1 + y + x^3 y$	$1 + x^6 y^7$	L4+B	1	G	$3 \times \llbracket 54, 4 \rrbracket$
CS-cd3a1f8b/C053	(9, 9)	$\llbracket 162, 12 \rrbracket$	6	Estimate	Exact	2.67	$1 + y^3 + xy^8$	$1 + x^5 y$	L4+B	7	G/O	$3 \times \llbracket 54, 4 \rrbracket$
CS-b21878a6/C053	(9, 9)	$\llbracket 162, 12 \rrbracket$	6	Estimate	Exact	2.67	$1 + y^4 + x^6 y^6$	$1 + x^3 y^5$	L4+B	1	O	$3 \times \llbracket 54, 4 \rrbracket$
CS-9b5d54bf/C053	(9, 9)	$\llbracket 162, 12 \rrbracket$	6	Estimate	Exact	2.67	$1 + y^7 + x^3 y^6$	$1 + x^6 y^2$	L4+B	1	G	$3 \times \llbracket 54, 4 \rrbracket$
CS-8367793c/C050	(9, 9)	$\llbracket 162, 12 \rrbracket$	6	Estimate	Exact	2.67	$1 + xy^2 + x^3$	$1 + xy^5$	L4+B	1	G	$3 \times \llbracket 54, 4 \rrbracket$
CS-bc57e3bb/C050	(9, 9)	$\llbracket 162, 12 \rrbracket$	6	Estimate	Exact	2.67	$1 + xy^3 + xy^6$	$1 + x^7$	L4+B	1	G	$3 \times \llbracket 54, 4 \rrbracket$
CS-6eaa2f94/C055	(9, 9)	$\llbracket 162, 12 \rrbracket$	5	Estimate	Exact	1.85	$1 + xy^6$	$1 + x^2 + x^7$	L4+B	1	G	$3 \times \llbracket 54, 4 \rrbracket$
CS-3e385a2f/C050	(9, 9)	$\llbracket 162, 12 \rrbracket$	6	Estimate	Exact	2.67	$1 + x^3 + x^5 y^4$	$1 + x^2 y$	L4+B	7	G	$3 \times \llbracket 54, 4 \rrbracket$
CS-98c3dcd5/C053	(9, 9)	$\llbracket 162, 12 \rrbracket$	6	Estimate	Exact	2.67	$1 + x^3 y$	$1 + x^3 + x^3 y^2$	L4+B	1	O	$3 \times \llbracket 54, 4 \rrbracket$
CS-e6c3d6f7/C050	(9, 9)	$\llbracket 162, 12 \rrbracket$	6	Estimate	Exact	2.67	$1 + x^4 y$	$1 + y^6 + x^2 y^8$	L4+B	149	G/O	$3 \times \llbracket 54, 4 \rrbracket$
CS-a99e1b35/C056	(9, 9)	$\llbracket 162, 12 \rrbracket$	6	Estimate	Exact	2.67	$1 + x^4 y^5$	$1 + x^6 + x^6 y^6$	L4+B	1	N	$3 \times \llbracket 54, 4 \rrbracket$
CS-766a79b8/C050	(9, 9)	$\llbracket 162, 12 \rrbracket$	6	Estimate	Exact	2.67	$1 + x^5 y^2$	$1 + xy + x^6$	L4+B	1	N	$3 \times \llbracket 54, 4 \rrbracket$
CS-ca862d04/C050	(9, 9)	$\llbracket 162, 12 \rrbracket$	6	Estimate	Exact	2.67	$1 + x^5 y^2 + x^6$	$1 + x^4 y^4$	L4+B	2	G	$3 \times \llbracket 54, 4 \rrbracket$
CS-4f82636c/C050	(9, 9)	$\llbracket 162, 12 \rrbracket$	6	Estimate	Exact	2.67	$1 + x^5 y^5 + x^6 y^3$	$1 + x^5 y^2$	L4+B	1	G	$3 \times \llbracket 54, 4 \rrbracket$
CS-929021ff/C050	(9, 9)	$\llbracket 162, 12 \rrbracket$	6	Estimate	Exact	2.67	$1 + x^7$	$1 + y^6 + x^8 y^3$	L4+B	4	G/O	$3 \times \llbracket 54, 4 \rrbracket$
CS-b50a397b/C046	(9, 9)	$\llbracket 162, 4 \rrbracket$	3	Exact	—	0.22	$1 + y^3$	$1 + y^4 + x^4 y^7$	Exh.	2	G	conn.
CS-8672e923/C042	(9, 9)	$\llbracket 162, 4 \rrbracket$	3	Exact	—	0.22	$1 + y^6$	$1 + xy^8 + x^5 y^8$	Exh.	1	G	conn.
CS-dee1f3c3/C046	(9, 9)	$\llbracket 162, 4 \rrbracket$	3	Exact	—	0.22	$1 + x^3 y^3$	$1 + x^4 y^2 + x^5 y^5$	Exh.	1	O	conn.
CS-83c45948/C046	(9, 9)	$\llbracket 162, 4 \rrbracket$	3	Exact	—	0.22	$1 + x^3 y^6$	$1 + x + x^2 y$	Exh.	1	N	conn.
CS-57124862/C046	(9, 9)	$\llbracket 162, 4 \rrbracket$	3	Exact	—	0.22	$1 + x^3 y^6$	$1 + xy^8 + x^5 y^6$	Exh.	1	G	conn.
CS-a4ab3427/C046	(9, 9)	$\llbracket 162, 4 \rrbracket$	3	Exact	—	0.22	$1 + x^4 y^2 + x^5 y^2$	$1 + x^3 y^6$	Exh.	3	G	conn.
CS-892b4e2d/C046	(9, 9)	$\llbracket 162, 4 \rrbracket$	3	Exact	—	0.22	$1 + x^6$	$1 + x + xy$	Exh.	1	N	conn.
CS-4f70cb6a/C046	(9, 9)	$\llbracket 162, 4 \rrbracket$	3	Exact	—	0.22	$1 + x^6 y^6$	$1 + xy + x^4$	Exh.	1	N	conn.
CS-acb810ff/C046	(9, 9)	$\llbracket 162, 4 \rrbracket$	3	Exact	—	0.22	$1 + x^7 y^2 + x^7 y^7$	$1 + x^3 y^3$	Exh.	1	G	conn.
CS-5cddb079/C045	(9, 9)	$\llbracket 162, 4 \rrbracket$	$5; \widehat{9}$	Estimate	Tighter	$0.62; \widehat{2.00}$	$1 + y$	$1 + x^4 y^2 + x^5 y^4$	L4+B	2	G	conn.
CS-dae6552/C047	(9, 9)	$\llbracket 162, 4 \rrbracket$	$5; \widehat{9}$	Estimate	Tighter	$0.62; 2.00$	$1 + x^4 y^4 + x^7 y^2$	$1 + x^2 y^6$	L4+B	1	G	conn.
CS-0A81f585/C041	(9, 9)	$\llbracket 162, 4 \rrbracket$	$[5, 9]$	Estimate	Supported	$[0.62, 2.00]$	$1 + y + xy^8$	$1 + x^2$	L4+B	1	G	conn.
CS-29a56c86/C043	(9, 9)	$\llbracket 162, 4 \rrbracket$	$5; \widehat{9}$	Estimate	—	$0.62; \widehat{2.00}$	$1 + xy^2$	$1 + x^3 y^5 + x^5 y^5$	L4+B	1	O	conn.
CS-000c990b/C044	(9, 9)	$\llbracket 162, 4 \rrbracket$	$5; \widehat{6}$	Estimate	Tighter	$0.62; \widehat{0.89}$	$1 + x^5$	$1 + x^4 y^4 + x^8 y^8$	L4+B	2	G	conn.
CS-6a1b1dfe/C164	(10, 9)	$\llbracket 180, 24 \rrbracket$	5	Exact	—	3.33	$1 + x^2 y^3 + x^2 y^6$	$1 + x^6$	M	158	G/O	$6 \times \llbracket 30, 4 \rrbracket$
CS-fb880f70/C163	(10, 9)	$\llbracket 180, 24 \rrbracket$	3	Exact	—	1.20	$1 + x^2$	$1 + x^2 y^6 + x^6 y^3$	Exh.	1	G	$6 \times \llbracket 30, 4 \rrbracket$
CS-1a073225/C163	(10, 9)	$\llbracket 180, 24 \rrbracket$	3	Exact	—	1.20	$1 + x^2 y^3 + x^8 y^6$	$1 + x^4$	Exh.	1	G	$6 \times \llbracket 30, 4 \rrbracket$
CS-31aff634/C162	(10, 9)	$\llbracket 180, 20 \rrbracket$	5	Exact	—	2.78	$1 + x^5 y^2 + x^5 y^4$	$1 + x^5 y^3$	M	15	G/O	$5 \times \llbracket 36, 4 \rrbracket$
CS-4496a7f0/C161	(10, 9)	$\llbracket 180, 16 \rrbracket$	3	Exact	—	0.80	$1 + y^3$	$1 + x^2 y^7 + x^4$	Exh.	12	G	$2 \times \llbracket 90, 8 \rrbracket$
CS-091bd1b5/C161	(10, 9)	$\llbracket 180, 16 \rrbracket$	3	Exact	—	0.80	$1 + x^4 y^5 + x^6 y^5$	$1 + y^3$	Exh.	18	G/O	$2 \times \llbracket 90, 8 \rrbracket$
CS-6d6a31f9/C156	(10, 9)	$\llbracket 180, 12 \rrbracket$	5	Exact	—	1.67	$1 + x^6$	$1 + xy^3 + x^5 y^6$	L4+W+B	16	G/O	$3 \times \llbracket 60, 4 \rrbracket$
CS-c4765355/C160	(10, 9)	$\llbracket 180, 12 \rrbracket$	5	Exact	—	1.67	$1 + x^6$	$1 + xy^6 + x^4 y^3$	M	38	G/O	$3 \times \llbracket 60, 4 \rrbracket$
CS-359b0899/C158	(10, 9)	$\llbracket 180, 12 \rrbracket$	4	Exact	—	1.07	$1 + y^3 + x^5 y^6$	$1 + x^2$	Exh.	167	G/O	$3 \times \llbracket 60, 4 \rrbracket$
CS-c25cf74/C155	(10, 9)	$\llbracket 180, 12 \rrbracket$	4	Exact	—	1.07	$1 + y^6 + x^3 y^3$	$1 + x^3$	Exh.	1	G	$3 \times \llbracket 60, 4 \rrbracket$
CS-265c73ea/C157	(10, 9)	$\llbracket 180, 12 \rrbracket$	$5; \widehat{8}$	Estimate	—	$1.67; \widehat{4.27}$	$1 + x^4 y^6 + x^5 y^3$	$1 + x^2$	L4+B	1	O	$3 \times \llbracket 60, 4 \rrbracket$

TABLE S5 (continued)

ID/class	(ℓ, m)	$\llbracket n, k \rrbracket$	d	Own evidence	Class effect	FOM	A	B	Evid.	Obs.	Attr.	Decomp.
CS-1eaea74d/C159	(10, 9)	$\llbracket 180, 12 \rrbracket$	5; $\widehat{8}$	Estimate	—	1.67; $\widehat{4.27}$	$1 + x^8 y^3 + x^8 y^6$	$1 + x^3$	L4+B	2	G	$3 \times \llbracket 60, 4 \rrbracket$
CS-d473da9b/C142	(10, 9)	$\llbracket 180, 8 \rrbracket$	9	Exact	—	3.60	$1 + x^4 y^8 + x^6 y^7$	$1 + x^4 y^6$	M	8	G/O	$2 \times \llbracket 90, 4 \rrbracket$
CS-770cc2a8/C138	(10, 9)	$\llbracket 180, 8 \rrbracket$	5	Exact	—	1.11	$1 + y^2 + x^8 y$	$1 + x^6$	L4+W+B 111	G/O	$2 \times \llbracket 90, 4 \rrbracket$	
CS-2d69d929/C152	(10, 9)	$\llbracket 180, 8 \rrbracket$	5	Exact	—	1.11	$1 + y^4 + x^6 y^2$	$1 + x^4$	L4+W+B 1	G	$2 \times \llbracket 90, 4 \rrbracket$	
CS-8ca0927a/C141	(10, 9)	$\llbracket 180, 8 \rrbracket$	5	Exact	—	1.11	$1 + x^2$	$1 + x^4 y^2 + x^4 y^4$	L4+W+B 8	G/O	$2 \times \llbracket 90, 4 \rrbracket$	
CS-de1d2921/C148	(10, 9)	$\llbracket 180, 8 \rrbracket$	5	Exact	—	1.11	$1 + x^2 y^2 + x^4 y^4$	$1 + x^2$	L4+W+B 1	G	$2 \times \llbracket 90, 4 \rrbracket$	
CS-a6d60d3d/C154	(10, 9)	$\llbracket 180, 8 \rrbracket$	5	Exact	—	1.11	$1 + x^4$	$1 + y^2 + y^4$	L4+W+B 1	N	$2 \times \llbracket 90, 4 \rrbracket$	
CS-c29564a1/C144	(10, 9)	$\llbracket 180, 8 \rrbracket$	5	Exact	—	1.11	$1 + x^4$	$1 + x^2 y^5 + x^8 y^4$	L4+W+B 1	G	$2 \times \llbracket 90, 4 \rrbracket$	
CS-0893ef17/C140	(10, 9)	$\llbracket 180, 8 \rrbracket$	5	Exact	—	1.11	$1 + x^8 y^6$	$1 + x^4 y^2 + x^4 y^4$	L4+W+B 2	G	$2 \times \llbracket 90, 4 \rrbracket$	
CS-accf4ce7/C140	(10, 9)	$\llbracket 180, 8 \rrbracket$	5	Exact	—	1.11	$1 + x^8 y^7 + x^8 y^8$	$1 + x^6 y^6$	L4+W+B 2	G	$2 \times \llbracket 90, 4 \rrbracket$	
CS-54e115ab/C146	(10, 9)	$\llbracket 180, 8 \rrbracket$	4	Exact	—	0.71	$1 + x^6 y + x^8 y^5$	$1 + x^2 y^3$	Exh.	4	G	$2 \times \llbracket 90, 4 \rrbracket$
CS-7223b27c/C139	(10, 9)	$\llbracket 180, 8 \rrbracket$	3	Exact	—	0.40	$1 + y^3$	$1 + x^6 y^2 + x^9 y^5$	Exh.	6	G	conn.
CS-69c61aee/C145	(10, 9)	$\llbracket 180, 8 \rrbracket$	3	Exact	—	0.40	$1 + x^4 y + x^6 y^8$	$1 + y^6$	Exh.	3	G	$2 \times \llbracket 90, 4 \rrbracket$
CS-ab23b32b/C147	(10, 9)	$\llbracket 180, 8 \rrbracket$	3	Exact	—	0.40	$1 + x^6 y^6$	$1 + x^6 y^5 + x^8 y^7$	Exh.	1	G	$2 \times \llbracket 90, 4 \rrbracket$
CS-a24f8c4e/C152	(10, 9)	$\llbracket 180, 8 \rrbracket$	5	Estimate	Exact	1.11	$1 + y^7 + x^4 y^8$	$1 + x^4$	L4+B	1	G	$2 \times \llbracket 90, 4 \rrbracket$
CS-b6830444/C143	(10, 9)	$\llbracket 180, 8 \rrbracket$	5; $\widehat{8}$	Estimate	—	1.11; $\widehat{2.84}$	$1 + x^4 y^5 + x^6 y$	$1 + x^2 y^3$	L4+B	2	G	$2 \times \llbracket 90, 4 \rrbracket$
CS-b7fe6960/C151	(10, 9)	$\llbracket 180, 8 \rrbracket$	5; $\widehat{7}$	Estimate	—	1.11; $\widehat{2.18}$	$1 + x^2 y^5 + x^4 y^7$	$1 + x^2 y^3$	L4+B	9	G/O	$2 \times \llbracket 90, 4 \rrbracket$
CS-36a5bff0/C153	(10, 9)	$\llbracket 180, 8 \rrbracket$	5; $\widehat{6}$	Estimate	—	1.11; $\widehat{1.60}$	$1 + y + y^2$	$1 + x^2 y^6$	L4+B	156	G/O	$2 \times \llbracket 90, 4 \rrbracket$
CS-2c0066a4/C140	(10, 9)	$\llbracket 180, 8 \rrbracket$	5	Estimate	Exact	1.11	$1 + x^2 y^3$	$1 + y^5 + x^6 y^4$	L4+B	1	G	$2 \times \llbracket 90, 4 \rrbracket$
CS-fe2ca145/C149	(10, 9)	$\llbracket 180, 8 \rrbracket$	5; $\widehat{6}$	Estimate	—	1.11; $\widehat{1.60}$	$1 + x^4 y^6$	$1 + x^2 y^5 + x^4 y$	L4+B	2	G	$2 \times \llbracket 90, 4 \rrbracket$
CS-0bd3fc2a/C150	(10, 9)	$\llbracket 180, 8 \rrbracket$	5; $\widehat{6}$	Estimate	—	1.11; $\widehat{1.60}$	$1 + x^5 y^3$	$1 + xy^4 + x^2$	L4+B	1	N	conn.
CS-77a004c8/C136	(10, 9)	$\llbracket 180, 8 \rrbracket$	5; $\widehat{6}$	Estimate	—	1.11; $\widehat{1.60}$	$1 + x^5 y^3$	$1 + x^2 + x^6 y$	L4+B	1	N	conn.
CS-1062c49e/C135	(10, 9)	$\llbracket 180, 8 \rrbracket$	5; $\widehat{6}$	Estimate	—	1.11; $\widehat{1.60}$	$1 + x^5 y^3$	$1 + x^3 + x^4 y^4$	L4+B	1	N	conn.
CS-ceda6f39/C137	(10, 9)	$\llbracket 180, 8 \rrbracket$	5; $\widehat{6}$	Estimate	—	1.11; $\widehat{1.60}$	$1 + x^5 y^6$	$1 + x^6 y^5 + x^7 y^3$	L4+B	112	G/O	conn.
CS-1530f6fa/C119	(10, 9)	$\llbracket 180, 4 \rrbracket$	14	Exact	—	4.356	$1 + x^2 y^2 + x^5 y$	$1 + x^3 y^3$	M	1	O	conn.
CS-6a4be7ad/C072	(10, 9)	$\llbracket 180, 4 \rrbracket$	13	Exact	—	3.76	$1 + x^2 y^3$	$1 + x^5 y + x^7 y^8$	M	3	G/O	conn.
CS-712a2f67/C129	(10, 9)	$\llbracket 180, 4 \rrbracket$	5	Exact	—	0.56	$1 + xy + x^2 y^2$	$1 + x^6$	L4+W+B 1	G	conn.	
CS-98bca4ef/C096	(10, 9)	$\llbracket 180, 4 \rrbracket$	5	Exact	—	0.56	$1 + xy^2 + xy^7$	$1 + x^4$	L4+W+B 2	G/O	conn.	
CS-6cecbcd3/C085	(10, 9)	$\llbracket 180, 4 \rrbracket$	5	Exact	—	0.56	$1 + x^2$	$1 + y^5 + x^3 y$	L4+W+B 5	G	conn.	
CS-02441504/C085	(10, 9)	$\llbracket 180, 4 \rrbracket$	5	Exact	—	0.56	$1 + x^2$	$1 + y^5 + x^7 y$	L4+W+B 1	G	conn.	
CS-c73a5314/C129	(10, 9)	$\llbracket 180, 4 \rrbracket$	5	Exact	—	0.56	$1 + x^2$	$1 + x^3 y^4 + x^6 y^8$	L4+W+B 9	G	conn.	
CS-4a07bbcc/C102	(10, 9)	$\llbracket 180, 4 \rrbracket$	5	Exact	—	0.56	$1 + x^3 y^5 + x^7 y$	$1 + x^6$	L4+W+B 3	G/O	conn.	
CS-fcc9d58d/C104	(10, 9)	$\llbracket 180, 4 \rrbracket$	5	Exact	—	0.56	$1 + x^3 y^8 + x^7 y^7$	$1 + x^8$	L4+W+B 1	G	conn.	
CS-21e1c012/C068	(10, 9)	$\llbracket 180, 4 \rrbracket$	5	Exact	—	0.56	$1 + x^4$	$1 + x^3 y^8 + x^5 y$	L4+W+B 1	G	conn.	
CS-0f220f7a/C122	(10, 9)	$\llbracket 180, 4 \rrbracket$	5	Exact	—	0.56	$1 + x^4 y^5 + x^5 y$	$1 + x^6$	L4+W+B 1	G	conn.	
CS-67685e6f/C129	(10, 9)	$\llbracket 180, 4 \rrbracket$	5	Exact	—	0.56	$1 + x^4 y^5 + x^7 y^7$	$1 + x^2$	L4+W+B 1	G	conn.	
CS-64d66132/C080	(10, 9)	$\llbracket 180, 4 \rrbracket$	5	Exact	—	0.56	$1 + x^6$	$1 + x^2 y^5 + x^5 y^4$	L4+W+B 1	G	conn.	
CS-d501d19e/C067	(10, 9)	$\llbracket 180, 4 \rrbracket$	3	Exact	—	0.20	$1 + y^2 + x^3 y^4$	$1 + y^6$	Exh.	5	G	conn.
CS-101586ac/C095	(10, 9)	$\llbracket 180, 4 \rrbracket$	3	Exact	—	0.20	$1 + y^3$	$1 + xy + x^2 y^5$	Exh.	1	G	conn.
CS-7934812b/C095	(10, 9)	$\llbracket 180, 4 \rrbracket$	3	Exact	—	0.20	$1 + y^3$	$1 + x^8 y^5 + x^9 y$	Exh.	1	G	conn.
CS-cc0df40f/C073	(10, 9)	$\llbracket 180, 4 \rrbracket$	3	Exact	—	0.20	$1 + xy^5 + x^2 y$	$1 + x^3 y^6$	Exh.	5	G/O	conn.
CS-4f746110/C111	(10, 9)	$\llbracket 180, 4 \rrbracket$	3	Exact	—	0.20	$1 + x^4 y + x^7 y^5$	$1 + y^6$	Exh.	1	G	conn.
CS-70b8f4ec/C060	(10, 9)	$\llbracket 180, 4 \rrbracket$	3	Exact	—	0.20	$1 + x^5 y^7 + x^6 y^2$	$1 + y^6$	Exh.	1	G	conn.
CS-0d018ae6/C123	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{14}$	Estimate	—	0.56; $\widehat{4.36}$	$1 + xy^3$	$1 + x^2 y^8 + x^3 y$	L4+B	7	G/O	conn.
CS-0bdaa7ef/C074	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{11}$	Estimate	Tighter	0.56; $\widehat{2.69}$	$1 + xy^6$	$1 + y^5 + xy$	L4+B	4	G	conn.
CS-e6cde6bd/C098	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{14}$	Estimate	—	0.56; $\widehat{4.36}$	$1 + xy^6$	$1 + xy^4 + x^2 y^2$	L4+B	152	G/O	conn.
CS-3c0c301f/C101	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{14}$	Estimate	—	0.56; $\widehat{4.36}$	$1 + x^2 y^3$	$1 + xy^5 + x^3 y^4$	L4+B	1	O	conn.
CS-9e96e9b1/C112	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{14}$	Estimate	—	0.56; $\widehat{4.36}$	$1 + x^2 y^7 + x^5 y^2$	$1 + x^2 y^6$	L4+B	2	G/O	conn.
CS-7a2dfee7/C109	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{14}$	Estimate	—	0.56; $\widehat{4.36}$	$1 + x^3 y^3$	$1 + xy^2 + x^5 y$	L4+B	1	O	conn.
CS-80fb270b/C078	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{14}$	Estimate	—	0.56; $\widehat{4.36}$	$1 + x^6 y^6$	$1 + x^5 y^7 + x^9 y^8$	L4+B	5	G	conn.
CS-8c50fa75/C126	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{12}$	Estimate	Tighter	0.56; $\widehat{3.20}$	$1 + x^8 y^6$	$1 + x^3 y^8 + x^6 y$	L4+B	1	G	conn.
CS-89a9b5af/C114	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{13}$	Estimate	—	0.56; $\widehat{3.76}$	$1 + y + x^8 y^2$	$1 + x^7 y^6$	L4+B	1	G	conn.
CS-ada32c2c/C092	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{13}$	Estimate	—	0.56; $\widehat{3.76}$	$1 + y^8 + xy^7$	$1 + x^2 y^3$	L4+B	1	G	conn.
CS-6d15be47/C099	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{13}$	Estimate	—	0.56; $\widehat{3.76}$	$1 + xy^2 + xy^7$	$1 + x^6 y^3$	L4+B	2	G	conn.
CS-4c09b2cb/C087	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{13}$	Estimate	—	0.56; $\widehat{3.76}$	$1 + xy^2 + x^4 y^4$	$1 + x^8 y^6$	L4+B	1	O	conn.
CS-d1864182/C058	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{13}$	Estimate	—	0.56; $\widehat{3.76}$	$1 + xy^6$	$1 + x^2 y + x^5 y^5$	L4+B	2	G	conn.
CS-56761fec/C082	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{13}$	Estimate	—	0.56; $\widehat{3.76}$	$1 + x^2 y^3$	$1 + x^4 y^7 + x^7 y^5$	L4+B	1	G	conn.
CS-7e3ffbde/C134	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{13}$	Estimate	—	0.56; $\widehat{3.76}$	$1 + x^3 y^5 + x^5 y^4$	$1 + x^8 y^6$	L4+B	1	G	conn.
CS-c71b9786/C113	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{13}$	Estimate	—	0.56; $\widehat{3.76}$	$1 + x^4 y^3$	$1 + x^5 y^7 + x^8 y^2$	L4+B	1	G	conn.
CS-7f731439/C132	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{10}$	Estimate	Tighter	0.56; $\widehat{2.22}$	$1 + x^6 y^3$	$1 + x^3 y^4 + x^3 y^5$	L4+B	1	G	conn.
CS-1646a2ea/C066	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{12}$	Estimate	—	0.56; $\widehat{3.20}$	$1 + y + xy^5$	$1 + x$	L4+B	1	G	conn.
CS-851ee57c/C077	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{12}$	Estimate	—	0.56; $\widehat{3.20}$	$1 + xy^3$	$1 + y^5 + x^5 y^4$	L4+B	1	O	conn.
CS-3ac0d487/C091	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{12}$	Estimate	—	0.56; $\widehat{3.20}$	$1 + xy^3$	$1 + x^2 y + x^4 y^5$	L4+B	1	O	conn.
CS-b1faded5/C116	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{12}$	Estimate	—	0.56; $\widehat{3.20}$	$1 + xy^3$	$1 + x^2 y^5 + x^6 y^4$	L4+B	1	O	conn.
CS-c4be49cb/C103	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{12}$	Estimate	—	0.56; $\widehat{3.20}$	$1 + xy^6$	$1 + x^4 y^8 + x^6 y$	L4+B	2	G	conn.
CS-598df9f0/C079	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{12}$	Estimate	—	0.56; $\widehat{3.20}$	$1 + x^2 y + x^2 y^8$	$1 + x^7 y^3$	L4+B	1	G	conn.
CS-abf6bdb6/C100												

TABLE S5 (continued)

ID/class	(ℓ, m)	$\llbracket n, k \rrbracket$	d	Own evidence	Class effect	FOM	A	B	Evid.	Obs. Attr.	Decomp.	
CS-1a8512d1/C061	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{12}$	Estimate	—	0.56; $\widehat{3.20}$	$1 + x^6 y^3$	$1 + x y^7 + x^8 y^2$	L4+B	1	G	conn.
CS-fc37a8ea/C075	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{12}$	Estimate	—	0.56; $\widehat{3.20}$	$1 + x^6 y^4 + x^7 y^2$	$1 + x^6 y^3$	L4+B	1	O	conn.
CS-ddbf7682/C070	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{12}$	Estimate	—	0.56; $\widehat{3.20}$	$1 + x^9 y^6$	$1 + x^7 y^5 + x^8 y$	L4+B	1	G	conn.
CS-00515ca4/C115	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{11}$	Estimate	—	0.56; $\widehat{2.69}$	$1 + y^4 + x^3 y^5$	$1 + x^6 y^6$	L4+B	1	G	conn.
CS-fa20271f/C128	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{11}$	Estimate	—	0.56; $\widehat{2.69}$	$1 + y^7 + x^3 y^2$	$1 + x^2 y^6$	L4+B	2	G/O	conn.
CS-b13425d8/C106	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{11}$	Estimate	—	0.56; $\widehat{2.69}$	$1 + x^3 y^3$	$1 + x y^5 + x^2 y^4$	L4+B	2	G/O	conn.
CS-b09acff6/C063	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{10}$	Estimate	Tighter	0.56; $\widehat{2.22}$	$1 + x^3 y^3$	$1 + x^3 y + x^4 y^2$	L4+B	3	G/O	conn.
CS-33b930a7/C108	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{11}$	Estimate	—	0.56; $\widehat{2.69}$	$1 + x^4 y^4 + x^5 y^2$	$1 + x^7 y^3$	L4+B	1	G	conn.
CS-17bc0f33/C118	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{10}$	Estimate	—	0.56; $\widehat{2.22}$	$1 + y + x y^5$	$1 + x^8$	L4+B	1	G	conn.
CS-3ff88cc4/C083	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{10}$	Estimate	—	0.56; $\widehat{2.22}$	$1 + y^8 + x^4 y^7$	$1 + x^3$	L4+B	1	G	conn.
CS-ba416885/C093	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{10}$	Estimate	—	0.56; $\widehat{2.22}$	$1 + x$	$1 + x y^5 + x^7 y$	L4+B	2	G/O	conn.
CS-1ce7d07e/C065	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{10}$	Estimate	—	0.56; $\widehat{2.22}$	$1 + x y^3$	$1 + y^4 + x^8 y^2$	L4+B	5	G	conn.
CS-07daf45c/C071	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{10}$	Estimate	—	0.56; $\widehat{2.22}$	$1 + x y^6$	$1 + x y^4 + x^7 y^8$	L4+B	4	G	conn.
CS-fdc311e1/C131	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{10}$	Estimate	—	0.56; $\widehat{2.22}$	$1 + x^3$	$1 + x^2 y^7 + x^9 y^8$	L4+B	5	G/O	conn.
CS-c1829d74/C110	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{10}$	Estimate	—	0.56; $\widehat{2.22}$	$1 + x^3$	$1 + x^3 y^5 + x^4 y$	L4+B	1	G	conn.
CS-74fea5e9/C088	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{10}$	Estimate	—	0.56; $\widehat{2.22}$	$1 + x^3$	$1 + x^3 y^5 + x^5 y$	L4+B	4	G	conn.
CS-10b07986/C094	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{10}$	Estimate	—	0.56; $\widehat{2.22}$	$1 + x^3 y + x^3 y^2$	$1 + x^9 y^6$	L4+B	1	G	conn.
CS-0bc90821/C124	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{10}$	Estimate	—	0.56; $\widehat{2.22}$	$1 + x^3 y + x^8 y^2$	$1 + x^3$	L4+B	143	G/O	conn.
CS-e6c23694/C089	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{10}$	Estimate	—	0.56; $\widehat{2.22}$	$1 + x^3 y^3$	$1 + x^5 y + x^8 y^2$	L4+B	1	G	conn.
CS-1d84a2a3/C107	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{10}$	Estimate	—	0.56; $\widehat{2.22}$	$1 + x^4 y^2 + x^7 y^7$	$1 + x$	L4+B	1	G	conn.
CS-5ee45526/C069	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{10}$	Estimate	—	0.56; $\widehat{2.22}$	$1 + x^6 y^5 + x^9 y$	$1 + x^3$	L4+B	27	G/O	conn.
CS-879fb436/C133	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{10}$	Estimate	—	0.56; $\widehat{2.22}$	$1 + x^7$	$1 + x^5 y + x^9 y^8$	L4+B	1	G	conn.
CS-202f136b/C090	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{10}$	Estimate	—	0.56; $\widehat{2.22}$	$1 + x^7 y^8 + x^9 y^7$	$1 + x^2 y^6$	L4+B	1	G	conn.
CS-93af0190/C105	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{9}$	Estimate	—	0.56; $\widehat{1.80}$	$1 + y^2 + x y^4$	$1 + x^3 y^6$	L4+B	1	O	conn.
CS-436f137b/C130	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{9}$	Estimate	—	0.56; $\widehat{1.80}$	$1 + x^3 y^3$	$1 + x y + x^2 y^5$	L4+B	1	O	conn.
CS-53609662/C117	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{9}$	Estimate	—	0.56; $\widehat{1.80}$	$1 + x^4 y^3$	$1 + x^5 y + x^5 y^8$	L4+B	17	G/O	conn.
CS-8590bd89/C125	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{9}$	Estimate	—	0.56; $\widehat{1.80}$	$1 + x^5 y^3$	$1 + x y^8 + x^7 y^7$	L4+B	1	G	conn.
CS-192b33e7/C064	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{8}$	Estimate	—	0.56; $\widehat{1.42}$	$1 + x^3 y^5 + x^6 y$	$1 + x^8 y^6$	L4+B	1	G	conn.
CS-7c4decca/C086	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{7}$	Estimate	—	0.56; $\widehat{1.09}$	$1 + x y^3$	$1 + x^5 y^2 + x^6 y$	L4+B	1	O	conn.
CS-4480c730/C121	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{6}$	Estimate	—	0.56; $\widehat{0.80}$	$1 + y^4 + x^9 y^8$	$1 + x^5 y^6$	L4+B	1	G	conn.
CS-652bb7cb/C120	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{6}$	Estimate	—	0.56; $\widehat{0.80}$	$1 + x$	$1 + y + y^2$	L4+B	1	N	conn.
CS-6c8c0afc/C084	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{6}$	Estimate	—	0.56; $\widehat{0.80}$	$1 + x y^2 + x^6 y^4$	$1 + x^5 y^6$	L4+B	1	G	conn.
CS-9e4d1df4/C081	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{6}$	Estimate	—	0.56; $\widehat{0.80}$	$1 + x^2 y^7 + x^3 y^5$	$1 + x^5 y^6$	L4+B	1	G	conn.
CS-2c2a34da/C057	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{6}$	Estimate	—	0.56; $\widehat{0.80}$	$1 + x^2 y^8 + x^8 y^7$	$1 + x^5 y^6$	L4+B	7	G/O	conn.
CS-6be91ab3/C062	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{6}$	Estimate	—	0.56; $\widehat{0.80}$	$1 + x^5 y^3$	$1 + x^8 y^7 + x^9 y^2$	L4+B	6	G/O	conn.
CS-237fb0ad/C097	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{6}$	Estimate	—	0.56; $\widehat{0.80}$	$1 + x^5 y^6$	$1 + x^3 y^2 + x^9 y^7$	L4+B	141	G/O	conn.
CS-2c036c5a/C076	(10, 9)	$\llbracket 180, 4 \rrbracket$	5; $\widehat{6}$	Estimate	—	0.56; $\widehat{0.80}$	$1 + x^5 y^6$	$1 + x^8 y^5 + x^9 y^7$	L4+B	2	G/O	conn.

TABLE S6: CSS-large compact catalogue (391 displayed of 547 retained proposals; 371 presentation classes). The table includes every proposal in an exact class and one deterministic representative of every nonexact class. Own evidence and Evid. describe the proposal before class merging; Class effect reports when merging makes the displayed d and FOM exact, tighter, or rigorously supported. Column abbreviations, filtering, and evidence conventions are defined above.

ID/class	(ℓ, m)	$\llbracket n, k \rrbracket$	d	Own evidence	Class effect	FOM	A	B	Evid.	Obs. Attr.	Decomp.	
CL-929c2a70/C248	(16, 12)	$\llbracket 384, 64 \rrbracket$	3	Exact	–	1.50	$1 + x^4 y^4 + x^{12} y^8$	$1 + x^4$	H _c	1	–	$16 \times \llbracket 24, 4 \rrbracket$
CL-a25b2baf/C247	(16, 12)	$\llbracket 384, 32 \rrbracket$	6	Exact	–	3.00	$1 + x^4 y^2 + x^4 y^{10}$	$1 + x^2 y^9$	H _c	1	–	$8 \times \llbracket 48, 4 \rrbracket$
CL-83dd7b5d/C245	(16, 12)	$\llbracket 384, 32 \rrbracket$	5	Exact	–	2.08	$1 + x^2 y^6$	$1 + x^2 y^2 + x^4 y^4$	M	2	G	$8 \times \llbracket 48, 4 \rrbracket$
CL-a94de289/C244	(16, 12)	$\llbracket 384, 32 \rrbracket$	4	Exact	–	1.33	$1 + x^2 y^5 + x^4 y^{10}$	$1 + x^4 y^6$	Exh.+I	4	G/O	$8 \times \llbracket 48, 4 \rrbracket$
CL-02bb1787/C243	(16, 12)	$\llbracket 384, 32 \rrbracket$	4	Exact	–	1.33	$1 + x^4$	$1 + y^2 + y^4$	Exh.	2	G/N	$8 \times \llbracket 48, 4 \rrbracket$
CL-d9f65a7e/C246	(16, 12)	$\llbracket 384, 32 \rrbracket$	3	Exact	–	0.75	$1 + x^4 y^2 + x^{10} y^7$	$1 + x^2 y^3$	H _c	1	–	$8 \times \llbracket 48, 4 \rrbracket$
CL-885dfa58/C233	(16, 12)	$\llbracket 384, 16 \rrbracket$	10	Exact	–	4.17	$1 + y^4 + x^2 y^8$	$1 + x^3 y^6$	M	1	O	$4 \times \llbracket 96, 4 \rrbracket$
CL-49103759/C232	(16, 12)	$\llbracket 384, 16 \rrbracket$	8	Exact	–	2.67	$1 + x$	$1 + x^3 y^4 + x^7 y^8$	M	3	G/O	$4 \times \llbracket 96, 4 \rrbracket$
CL-b71ff5d9/C223	(16, 12)	$\llbracket 384, 16 \rrbracket$	8	Exact	–	2.67	$1 + x^2$	$1 + y^8 + x^2 y^{10}$	M	1	G	$4 \times \llbracket 96, 4 \rrbracket$
CL-05841223/C232	(16, 12)	$\llbracket 384, 16 \rrbracket$	8	Exact	–	2.67	$1 + x^4 y^8 + x^7 y^{10}$	$1 + xy^6$	M	2	G/O	$4 \times \llbracket 96, 4 \rrbracket$
CL-8c40001b/C229	(16, 12)	$\llbracket 384, 16 \rrbracket$	8	Exact	–	2.67	$1 + x^6 y^3$	$1 + x^{10} y + x^{12} y^8$	M	2	G	$4 \times \llbracket 96, 4 \rrbracket$
CL-ebc1d7c4/C229	(16, 12)	$\llbracket 384, 16 \rrbracket$	8	Exact	–	2.67	$1 + x^6 y^7 + x^{10} y^{11}$	$1 + x^6 y^3$	M	1	G	$4 \times \llbracket 96, 4 \rrbracket$
CL-454c87bc/C228	(16, 12)	$\llbracket 384, 16 \rrbracket$	8	Exact	–	2.67	$1 + x^8 y^2 + x^{14} y^7$	$1 + x^6 y^3$	M	1	G	$4 \times \llbracket 96, 4 \rrbracket$
CL-9fc416af/C242	(16, 12)	$\llbracket 384, 16 \rrbracket$	7	Exact	–	2.04	$1 + x^2$	$1 + y^4 + x^3 y^8$	M	2	G	$4 \times \llbracket 96, 4 \rrbracket$
CL-Offd9a2d/C225	(16, 12)	$\llbracket 384, 16 \rrbracket$	7	Exact	–	2.04	$1 + x^2 y^{10} + x^{15} y^{11}$	$1 + xy^9$	M	1	G	$4 \times \llbracket 96, 4 \rrbracket$
CL-52dcc64d/C226	(16, 12)	$\llbracket 384, 16 \rrbracket$	6	Exact	–	1.50	$1 + x^2 y^9$	$1 + x^2 y^{11} + x^{14} y$	M	1	G	$4 \times \llbracket 96, 4 \rrbracket$
CL-9daeb04b/C235	(16, 12)	$\llbracket 384, 16 \rrbracket$	6	Exact	–	1.50	$1 + x^{10} y^6$	$1 + y^4 + x^5 y^5$	M	1	G	$4 \times \llbracket 96, 4 \rrbracket$
CL-52dc7734/C239	(16, 12)	$\llbracket 384, 16 \rrbracket$	4	Exact	–	0.67	$1 + y^4 + x^4 y^2$	$1 + x^4 y^3$	Exh.	1	O	$4 \times \llbracket 96, 4 \rrbracket$
CL-7aa28e28/C224	(16, 12)	$\llbracket 384, 16 \rrbracket$	4	Exact	–	0.67	$1 + x^2$	$1 + y^2 + y^4$	Exh.	6	G/O/N	$4 \times \llbracket 96, 4 \rrbracket$
CL-ff17aa92/C237	(16, 12)	$\llbracket 384, 16 \rrbracket$	4	Exact	–	0.67	$1 + x^4$	$1 + y + y^2$	Exh.	1	G	$4 \times \llbracket 96, 4 \rrbracket$
CL-f1c90545/C222	(16, 12)	$\llbracket 384, 16 \rrbracket$	4	Exact	–	0.67	$1 + x^4$	$1 + x^2 y + x^{15} y^5$	Exh.	1	G	conn.
CL-47fcea25/C236	(16, 12)	$\llbracket 384, 16 \rrbracket$	4	Exact	–	0.67	$1 + x^4 y^6$	$1 + x^6 y^8 + x^8 y^{10}$	Exh.	1	G	$4 \times \llbracket 96, 4 \rrbracket$
CL-da717524/C231	(16, 12)	$\llbracket 384, 16 \rrbracket$	4	Exact	–	0.67	$1 + x^4 y^{11} + x^8 y$	$1 + y^3$	Exh.	1	G	$4 \times \llbracket 96, 4 \rrbracket$

TABLE S6 (continued)

ID/class	(ℓ, m)	$[n, k]$	d	Own evidence	Class effect	FOM	A	B	Evid.	Obs. Attr.	Decomp.
CL-0fcae59b/C234	(16, 12)	[384, 16]	4	Exact	—	0.67	$1 + x^5 y^6$	$1 + x^8 y^4 + x^8 y^8$	Exh.	1 O	$4 \times [96, 4]$
CL-9e72e18f/C230	(16, 12)	[384, 16]	4	Exact	—	0.67	$1 + x^5 y^9$	$1 + y^4 + x^5 y^5$	Exh.	1 G	$4 \times [96, 4]$
CL-326e2b7f/C224	(16, 12)	[384, 16]	4	Exact	—	0.67	$1 + x^6 y^6$	$1 + y^2 + y^{10}$	Exh.	1 O	$4 \times [96, 4]$
CL-64686929/C240	(16, 12)	[384, 16]	3	Exact	—	0.38	$1 + x^5 y^4 + x^{11} y^8$	$1 + x$	Exh.	1 G	$4 \times [96, 4]$
CL-bda4c4b2/C238	(16, 12)	[384, 16]	$5; \widehat{6}$	Estimate	Tighter	$1.04; \widehat{1.50}$	$1 + x^5 y^9$	$1 + x^{10} y^2 + x^{10} y^{10}$	L4+B	1 G	$4 \times [96, 4]$
CL-a26ff6ea/C226	(16, 12)	[384, 16]	6	Estimate	Exact	1.50	$1 + x^2 y^3$	$1 + x^2 y + x^{14} y^{11}$	L4+B	1 G	$4 \times [96, 4]$
CL-a24fe148/C227	(16, 12)	[384, 16]	$5; \widehat{7}$	Estimate	—	$1.04; \widehat{2.04}$	$1 + x^2 y^2 + x^{11} y^7$	$1 + x^3 y^3$	L4+B	1 G	$4 \times [96, 4]$
CL-5bf06ff4/C241	(16, 12)	[384, 16]	$5; \widehat{6}$	Estimate	—	$1.04; \widehat{1.50}$	$1 + x^{15} y^3$	$1 + x y^5 + x^{15} y^7$	L4+B	1 G	$4 \times [96, 4]$
CL-cbea6bcb/C210	(16, 12)	[384, 8]	10	Exact	—	2.08	$1 + x y^3$	$1 + x^4 y^{10} + x^{11} y^5$	M	1 O	$2 \times [192, 4]$
CL-bea18d22/C207	(16, 12)	[384, 8]	4	Exact	—	0.33	$1 + x$	$1 + y^2 + y^4$	Exh.	2 G/N	$2 \times [192, 4]$
CL-526b1ee0/C207	(16, 12)	[384, 8]	4	Exact	—	0.33	$1 + x^3$	$1 + y^2 + y^4$	Exh.	1 G	$2 \times [192, 4]$
CL-e473f2e1/C212	(16, 12)	[384, 8]	4	Exact	—	0.33	$1 + x^4 y^3$	$1 + x^6 y^{10} + x^{12} y^8$	Exh.	1 G	$2 \times [192, 4]$
CL-537533d8/C204	(16, 12)	[384, 8]	4	Exact	—	0.33	$1 + x^4 y^9$	$1 + y^5 + x^{10} y^7$	Exh.	1 G	$2 \times [192, 4]$
CL-7929daca/C212	(16, 12)	[384, 8]	4	Exact	—	0.33	$1 + x^8 y^3$	$1 + x^{12} y^8 + x^{14} y^{10}$	Exh.	1 G	$2 \times [192, 4]$
CL-cca60bc4/C218	(16, 12)	[384, 8]	$5; \widehat{15}$	Estimate	—	$0.52; \widehat{4.69}$	$1 + x^5 y^3$	$1 + x^8 y^{10} + x^{15} y^5$	L4+B	1 G	$2 \times [192, 4]$
CL-13a07207/C209	(16, 12)	[384, 8]	$5; \widehat{13}$	Estimate	Tighter	$0.52; \widehat{3.52}$	$1 + x y^{11} + x^3 y^7$	$1 + x^5 y^9$	L4+B	1 G	$2 \times [192, 4]$
CL-59b650fb/C216	(16, 12)	[384, 8]	$5; \widehat{14}$	Estimate	—	$0.52; \widehat{4.08}$	$1 + x^3 y^6$	$1 + x y^4 + x^8 y^8$	L4+B	5 G/O	$2 \times [192, 4]$
CL-4e8f3f5e/C220	(16, 12)	[384, 8]	$5; \widehat{14}$	Estimate	—	$0.52; \widehat{4.08}$	$1 + x^7 y^{10} + x^{12} y^2$	$1 + x^3$	L4+B	1 G	$2 \times [192, 4]$
CL-b06bdc8c/C210	(16, 12)	[384, 8]	10	Estimate	Exact	2.08	$1 + x^3$	$1 + x^4 y^2 + x^5 y^{10}$	L4+B	1 O	$2 \times [192, 4]$
CL-0462d4da/C201	(16, 12)	[384, 8]	[5, 12]	Estimate	Supported	[0.52, 3.00]	$1 + x^5$	$1 + x^2 y^{10} + x^{10} y^8$	L4+B	1 G	$2 \times [192, 4]$
CL-ff471113/C219	(16, 12)	[384, 8]	$5; \widehat{12}$	Estimate	—	$0.52; \widehat{3.00}$	$1 + x^5$	$1 + x^5 y^2 + x^{15} y^4$	L4+B	1 G	$2 \times [192, 4]$
CL-c35bf5e5/C203	(16, 12)	[384, 8]	$5; \widehat{12}$	Estimate	—	$0.52; \widehat{3.00}$	$1 + x^5 y^4 + x^{10} y^2$	$1 + x^5 y^6$	L4+B	2 G	$2 \times [192, 4]$
CL-c034f0c7/C213	(16, 12)	[384, 8]	$5; \widehat{12}$	Estimate	—	$0.52; \widehat{3.00}$	$1 + x^{12} y^{10} + x^{12} y^{11}$	$1 + x^6 y^6$	L4+B	1 G	$2 \times [192, 4]$
CL-ef3d7a2a/C210	(16, 12)	[384, 8]	10	Estimate	Exact	2.08	$1 + x y^3$	$1 + x y^5 + x^{12} y^{10}$	L4+B	1 G	$2 \times [192, 4]$
CL-03df4877/C206	(16, 12)	[384, 8]	$5; \widehat{11}$	Estimate	—	$0.52; \widehat{2.52}$	$1 + x y^3$	$1 + x^{12} y^8 + x^{13} y$	L4+B	1 G	$2 \times [192, 4]$
CL-e3bbf85c/C214	(16, 12)	[384, 8]	$5; \widehat{8}$	Estimate	Tighter	$0.52; \widehat{1.33}$	$1 + x^5$	$1 + y^8 + x^5 y^{10}$	L4+B	1 G	$2 \times [192, 4]$
CL-262ec76e/C202	(16, 12)	[384, 8]	$5; \widehat{8}$	Estimate	Tighter	$0.52; \widehat{1.33}$	$1 + x^5 y^6$	$1 + x^3 y^4 + x^8 y^2$	L4+B	1 G	$2 \times [192, 4]$
CL-b1d747a7/C208	(16, 12)	[384, 8]	$5; \widehat{8}$	Estimate	—	$0.52; \widehat{1.33}$	$1 + y^2 + x^{15} y$	$1 + x^6$	L4+B	1 G	$2 \times [192, 4]$
CL-61cad729/C211	(16, 12)	[384, 8]	$5; \widehat{8}$	Estimate	—	$0.52; \widehat{1.33}$	$1 + x^2$	$1 + y + y^2$	L4+B	2 G/N	$2 \times [192, 4]$
CL-d410d5a1/C221	(16, 12)	[384, 8]	$5; \widehat{8}$	Estimate	—	$0.52; \widehat{1.33}$	$1 + x^2 y^5 + x^4 y^{10}$	$1 + x^2 y^6$	L4+B	2 G	$2 \times [192, 4]$
CL-2f9daa68/C200	(16, 12)	[384, 8]	$5; \widehat{8}$	Estimate	—	$0.52; \widehat{1.33}$	$1 + x^2 y^6$	$1 + x^5 y^{11} + x^{11} y^4$	L4+B	1 G	conn.
CL-157539a4/C217	(16, 12)	[384, 8]	$5; \widehat{8}$	Estimate	—	$0.52; \widehat{1.33}$	$1 + x^4 y + x^{14} y^{11}$	$1 + x^2$	L4+B	1 G	$2 \times [192, 4]$
CL-1ed36a91/C205	(16, 12)	[384, 8]	$5; \widehat{8}$	Estimate	—	$0.52; \widehat{1.33}$	$1 + x^6 y^5 + x^8 y^{10}$	$1 + x^2 y^6$	L4+B	1 G	$2 \times [192, 4]$
CL-f1196537/C199	(16, 12)	[384, 8]	$5; \widehat{8}$	Estimate	—	$0.52; \widehat{1.33}$	$1 + x^6 y^6$	$1 + x^3 y^2 + x^3 y^4$	L4+B	1 G	$2 \times [192, 4]$
CL-3edcdab9/C215	(16, 12)	[384, 8]	$5; \widehat{6}$	Estimate	—	$0.52; \widehat{0.75}$	$1 + x$	$1 + x^5 y^2 + x^{10} y^{10}$	L4+B	2 G/O	$2 \times [192, 4]$
CL-6e950816/C189	(16, 12)	[384, 4]	4	Exact	—	0.17	$1 + y^3$	$1 + x^3 y^{10} + x^{13} y^5$	Exh.	1 O	conn.
CL-3a2ea8c7/C176	(16, 12)	[384, 4]	4	Exact	—	0.17	$1 + x y^7 + x^{10} y^8$	$1 + x^4 y^3$	Exh.	1 G	conn.
CL-c63d3c22/C189	(16, 12)	[384, 4]	4	Exact	—	0.17	$1 + x^2 y + x^{15} y^{11}$	$1 + x^4 y^3$	Exh.	1 G	conn.
CL-604a1049/C189	(16, 12)	[384, 4]	4	Exact	—	0.17	$1 + x^4 y^3$	$1 + x y + x^{11} y^8$	Exh.	1 G	conn.
CL-3db67857/C192	(16, 12)	[384, 4]	4	Exact	—	0.17	$1 + x^4 y^3$	$1 + x y + x^{15} y^{11}$	Exh.	6 G/O	conn.
CL-96f673ac/C196	(16, 12)	[384, 4]	4	Exact	—	0.17	$1 + x^4 y^3$	$1 + x^2 y + x^3 y^{11}$	Exh.	1 G	conn.
CL-dbf2f6a0/C179	(16, 12)	[384, 4]	4	Exact	—	0.17	$1 + x^4 y^3$	$1 + x^3 y + x^5 y^{11}$	Exh.	1 G	conn.
CL-3bdd5f4a/C197	(16, 12)	[384, 4]	4	Exact	—	0.17	$1 + x^4 y^3$	$1 + x^3 y^4 + x^4 y^8$	Exh.	1 O	conn.
CL-eca6b561/C189	(16, 12)	[384, 4]	4	Exact	—	0.17	$1 + x^4 y^3$	$1 + x^3 y^8 + x^{10} y^7$	Exh.	1 G	conn.
CL-387ff10e/C189	(16, 12)	[384, 4]	4	Exact	—	0.17	$1 + x^4 y^3$	$1 + x^5 y^4 + x^{14} y^{11}$	Exh.	1 G	conn.
CL-8d3eab41/C189	(16, 12)	[384, 4]	4	Exact	—	0.17	$1 + x^4 y^3$	$1 + x^5 y^4 + x^{15} y^{11}$	Exh.	1 G	conn.
CL-c4b43b09/C195	(16, 12)	[384, 4]	$5; \widehat{16}$	Estimate	Tighter	$0.26; \widehat{2.67}$	$1 + x^2 y^5 + x^{15} y^7$	$1 + x^5$	L4+B	1 G	conn.
CL-c8819726/C178	(16, 12)	[384, 4]	$5; \widehat{16}$	Estimate	—	$0.26; \widehat{2.67}$	$1 + y + x^4 y^2$	$1 + x y^6$	L4+B	1 O	conn.
CL-4b2eb5d5/C171	(16, 12)	[384, 4]	$5; \widehat{16}$	Estimate	—	$0.26; \widehat{2.67}$	$1 + y + x^6 y^2$	$1 + x y^6$	L4+B	1 O	conn.
CL-7e7edaed/C186	(16, 12)	[384, 4]	$5; \widehat{16}$	Estimate	—	$0.26; \widehat{2.67}$	$1 + y + x^{15} y^5$	$1 + x y^9$	L4+B	1 G	conn.
CL-84dbe6c0/C198	(16, 12)	[384, 4]	$5; \widehat{16}$	Estimate	—	$0.26; \widehat{2.67}$	$1 + y^2 + x^2 y$	$1 + x^3 y^3$	L4+B	1 O	conn.
CL-9b6ebfad/C184	(16, 12)	[384, 4]	$5; \widehat{16}$	Estimate	—	$0.26; \widehat{2.67}$	$1 + x$	$1 + x y^{11} + x^{14} y^7$	L4+B	1 G	conn.
CL-015c0d4f/C183	(16, 12)	[384, 4]	$5; \widehat{16}$	Estimate	—	$0.26; \widehat{2.67}$	$1 + x y^2 + x^6 y^4$	$1 + x y^9$	L4+B	1 G	conn.
CL-762872ac/C170	(16, 12)	[384, 4]	$5; \widehat{16}$	Estimate	—	$0.26; \widehat{2.67}$	$1 + x y^6$	$1 + x^3 y^2 + x^{11} y$	L4+B	1 G	conn.
CL-5623f517/C191	(16, 12)	[384, 4]	$5; \widehat{16}$	Estimate	—	$0.26; \widehat{2.67}$	$1 + x y^6$	$1 + x^{11} y^2 + x^{12} y$	L4+B	1 G	conn.
CL-0823cb54/C190	(16, 12)	[384, 4]	$5; \widehat{16}$	Estimate	—	$0.26; \widehat{2.67}$	$1 + x^2 y^2 + x^2 y^7$	$1 + x y^6$	L4+B	1 G	conn.
CL-47fe1ca1/C166	(16, 12)	[384, 4]	$5; \widehat{16}$	Estimate	—	$0.26; \widehat{2.67}$	$1 + x^3$	$1 + x y^5 + x^{14} y$	L4+B	1 O	conn.
CL-2e330ac5/C187	(16, 12)	[384, 4]	$5; \widehat{16}$	Estimate	—	$0.26; \widehat{2.67}$	$1 + x^3 y^2 + x^8 y^{10}$	$1 + x y^3$	L4+B	3 G/O	conn.
CL-6f55f40d/C173	(16, 12)	[384, 4]	$5; \widehat{16}$	Estimate	—	$0.26; \widehat{2.67}$	$1 + x^3 y^6$	$1 + x y^7 + x^5 y^{11}$	L4+B	1 G	conn.
CL-43a4f306/C193	(16, 12)	[384, 4]	$5; \widehat{16}$	Estimate	—	$0.26; \widehat{2.67}$	$1 + x^5$	$1 + x^3 y^5 + x^{12} y^{10}$	L4+B	3 G	conn.
CL-a8aab347/C165	(16, 12)	[384, 4]	$5; \widehat{16}$	Estimate	—	$0.26; \widehat{2.67}$	$1 + x^5 y^3$	$1 + x^{14} y + x^{15} y^2$	L4+B	1 G	conn.
CL-764bc1ba/C168	(16, 12)	[384, 4]	$5; \widehat{16}$	Estimate	—	$0.26; \widehat{2.67}$	$1 + x^5 y^6$	$1 + y + x^6 y^5$	L4+B	1 O	conn.
CL-90020de8/C182	(16, 12)	[384, 4]	$5; \widehat{16}$	Estimate	—	$0.26; \widehat{2.67}$	$1 + x^5 y^6$	$1 + x y^7 + x^6 y^{11}$	L4+B	2 G	conn.
CL-78232732/C172	(16, 12)	[384, 4]	$5; \widehat{16}$	Estimate	—	$0.26; \widehat{2.67}$	$1 + x^5 y^6$	$1 + x^{12} y^{11} + x^{15} y^4$	L4+B	1 G	conn.
CL-4735294a/C188	(16, 12)	[384, 4]	$5; \widehat{16}$	Estimate	—	$0.26; \widehat{2.67}$	$1 + x^{13} y^3$	$1 + x^2 y^5 + x^{14} y^7$	L4+B	1 O	conn.
CL-8fe6ec97/C194	(16, 12)	[384, 4]	$5; \widehat{14}$	Estimate	Tighter	$0.26; \widehat{2.04}$	$1 + x^{13}$	$1 + x^2 y + x^7 y^2$	L4+B	1 O	conn.
CL-518180e8/C167	(16, 12)	[384, 4]	$5; \widehat{12}$	Estimate	Tighter	$0.26; \widehat{1.50}$	$1 + x^{15} y^9$	$1 + x y^2 + x^{15} y^{10}$	L4+B	1 O	conn.
CL-4b75f13a/C174	(16, 12)	[384, 4]	$5; \widehat{8}$	Estimate	Tighter	$0.26; \widehat{0.67}$	$1 + y^7 + x^9 y^2$	$1 + x^2 y^9$	L4+B	2 G/O	conn.
CL-9b17db49/C181	(16, 12)	[384, 4]	$5; \widehat{8}$	Estimate	—	$0.26; \widehat{0.67}$	$1 + y^4 + y^5$	$1 + x^5 y^3$	L4+B	1 O	conn.

TABLE S6 (continued)

ID/class	(ℓ, m)	$\llbracket n, k \rrbracket$	d	Own evidence	Class effect	FOM	A	B	Evid.	Obs. Attr.	Decomp.
CL-02dbb137/C175	(16, 12)	$\llbracket 384, 4 \rrbracket$	$5; \widehat{8}$	Estimate	—	$0.26; \widehat{0.67}$	$1 + x$	$1 + y + y^2$	L4+B	2	G/N conn.
CL-3230671f/C177	(16, 12)	$\llbracket 384, 4 \rrbracket$	$5; \widehat{8}$	Estimate	—	$0.26; \widehat{0.67}$	$1 + x^2 y^3$	$1 + xy^8 + x^{14} y^{10}$	L4+B	1	G conn.
CL-45c62c8a/C180	(16, 12)	$\llbracket 384, 4 \rrbracket$	$5; \widehat{8}$	Estimate	—	$0.26; \widehat{0.67}$	$1 + x^2 y^9$	$1 + x^6 y^2 + x^{15} y$	L4+B	1	G conn.
CL-2102bcd/C185	(16, 12)	$\llbracket 384, 4 \rrbracket$	$5; \widehat{8}$	Estimate	—	$0.26; \widehat{0.67}$	$1 + x^5 y + x^9 y^2$	$1 + x^6 y^3$	L4+B	1	G conn.
CL-1b76c9a2/C169	(16, 12)	$\llbracket 384, 4 \rrbracket$	$5; \widehat{8}$	Estimate	—	$0.26; \widehat{0.67}$	$1 + x^7 y^{10} + x^9 y^2$	$1 + x^6 y^9$	L4+B	1	G conn.
CL-4b943aa2/C423	(15, 14)	$\llbracket 420, 60 \rrbracket$	3	Exact	—	1.29	$1 + x^5 y^{12} + x^{10} y^8$	$1 + x^{10}$	H _c	1	— $10 \times \llbracket 42, 6 \rrbracket$
CL-437b6e5f/C422	(15, 14)	$\llbracket 420, 40 \rrbracket$	5	Exact	—	2.38	$1 + x^5 y^4 + x^{10} y^6$	$1 + y^6$	H _c	7	— $10 \times \llbracket 42, 4 \rrbracket$
CL-abc3d2fe/C421	(15, 14)	$\llbracket 420, 30 \rrbracket$	3	Exact	—	0.64	$1 + x^5 y^2 + x^{10} y^6$	$1 + y^2 + x^5 y^3$	Exh.	1	G $5 \times \llbracket 84, 6 \rrbracket$
CL-1830493f/C420	(15, 14)	$\llbracket 420, 20 \rrbracket$	8	Exact	—	3.05	$1 + x^5 y^2 + x^{10} y^6$	$1 + y^3$	M	3	G/O $5 \times \llbracket 84, 4 \rrbracket$
CL-0eaa9430/C416	(15, 14)	$\llbracket 420, 20 \rrbracket$	7	Exact	—	2.33	$1 + y^4$	$1 + x^5 y^{10} + x^{10} y^5$	M	1	G $5 \times \llbracket 84, 4 \rrbracket$
CL-e2ab51c4/C418	(15, 14)	$\llbracket 420, 20 \rrbracket$	6	Exact	—	1.71	$1 + y^2$	$1 + x^5 y^3 + x^{10} y^{13}$	M	5	G/O $5 \times \llbracket 84, 4 \rrbracket$
CL-e891e16d/C419	(15, 14)	$\llbracket 420, 20 \rrbracket$	6	Exact	—	1.71	$1 + x^5 y^4 + x^{10} y^{11}$	$1 + y$	M	1	O $5 \times \llbracket 84, 4 \rrbracket$
CL-253c2c53/C417	(15, 14)	$\llbracket 420, 20 \rrbracket$	5	Exact	—	1.19	$1 + x^6$	$1 + x^7 y^2 + x^{14} y^6$	M	1	O $2 \times \llbracket 210, 10 \rrbracket$
CL-f8ef7ae2/C414	(15, 14)	$\llbracket 420, 18 \rrbracket$	10	Exact	—	4.29	$1 + y^{13} + x^{12} y^{11}$	$1 + x^3 y^7$	M	1	G $3 \times \llbracket 140, 6 \rrbracket$
CL-0286b883/C415	(15, 14)	$\llbracket 420, 18 \rrbracket$	9	Exact	—	3.47	$1 + x^3 y^7$	$1 + x^6 y^5 + x^9 y$	M	2	G $3 \times \llbracket 140, 6 \rrbracket$
CL-e6216eab/C413	(15, 14)	$\llbracket 420, 16 \rrbracket$	7	Exact	—	1.87	$1 + y^2$	$1 + x^3 + x^4 y^4$	M	1	G $2 \times \llbracket 210, 8 \rrbracket$
CL-1553fac6/C410	(15, 14)	$\llbracket 420, 16 \rrbracket$	7	Exact	—	1.87	$1 + y^4$	$1 + xy^4 + x^4 y^8$	M	1	G $2 \times \llbracket 210, 8 \rrbracket$
CL-10274ff6/C412	(15, 14)	$\llbracket 420, 16 \rrbracket$	7	Exact	—	1.87	$1 + y^4$	$1 + x^3 y^{12} + x^4 y^8$	M	1	G $2 \times \llbracket 210, 8 \rrbracket$
CL-7e34d25c/C411	(15, 14)	$\llbracket 420, 16 \rrbracket$	$5; \widehat{8}$	Estimate	—	$0.95; \widehat{2.44}$	$1 + y^2$	$1 + x^3 + x^4$	L4+B	1	G $2 \times \llbracket 210, 8 \rrbracket$
CL-97118c06/C406	(15, 14)	$\llbracket 420, 12 \rrbracket$	10	Exact	—	2.86	$1 + x^4$	$1 + y^4 + x^9 y^6$	M	1	G $2 \times \llbracket 210, 6 \rrbracket$
CL-84c084df/C405	(15, 14)	$\llbracket 420, 12 \rrbracket$	9	Exact	—	2.31	$1 + x$	$1 + x^2 y^2 + x^8 y^6$	M	1	G $2 \times \llbracket 210, 6 \rrbracket$
CL-7a5b9b2b/C408	(15, 14)	$\llbracket 420, 12 \rrbracket$	3	Exact	—	0.26	$1 + x^5$	$1 + xy^4 + x^8 y^{12}$	Exh.	1	G $2 \times \llbracket 210, 6 \rrbracket$
CL-28abb736/C407	(15, 14)	$\llbracket 420, 12 \rrbracket$	$5; \widehat{12}$	Estimate	—	$0.71; \widehat{4.11}$	$1 + xy^2 + xy^{10}$	$1 + x^8$	L4+B	1	G $2 \times \llbracket 210, 6 \rrbracket$
CL-2075fcbcc/C404	(15, 14)	$\llbracket 420, 12 \rrbracket$	$5; \widehat{10}$	Estimate	—	$0.71; \widehat{2.86}$	$1 + x^6$	$1 + x^3 y^8 + x^4 y^{10}$	L4+B	1	G $2 \times \llbracket 210, 6 \rrbracket$
CL-4efbd3b7/C403	(15, 14)	$\llbracket 420, 12 \rrbracket$	$5; \widehat{9}$	Estimate	—	$0.71; \widehat{2.31}$	$1 + x^2 y^8 + x^2 y^{12}$	$1 + x^2$	L4+B	1	G $2 \times \llbracket 210, 6 \rrbracket$
CL-94086b72/C409	(15, 14)	$\llbracket 420, 10 \rrbracket$	5	Exact	—	0.60	$1 + x^2 y + x^{10} y^5$	$1 + x^6$	L4+W+B	1	G conn.
CL-18ce8ea0/C402	(15, 14)	$\llbracket 420, 10 \rrbracket$	$5; \widehat{10}$	Estimate	—	$0.60; \widehat{2.38}$	$1 + x^5 y^2 + x^{13} y^3$	$1 + x^3 y^7$	L4+B	1	G conn.
CL-37987300/C395	(15, 14)	$\llbracket 420, 8 \rrbracket$	5	Exact	—	0.48	$1 + xy^6 + x^{11} y^2$	$1 + x^3 y^6$	L4+W+B	1	G $2 \times \llbracket 210, 4 \rrbracket$
CL-ae04e166/C401	(15, 14)	$\llbracket 420, 8 \rrbracket$	4	Exact	—	0.30	$1 + xy^2 + x^2 y^4$	$1 + x^6 y^4$	Exh.	3	G/O $2 \times \llbracket 210, 4 \rrbracket$
CL-f1fe99b8/C391	(15, 14)	$\llbracket 420, 8 \rrbracket$	$5; \widehat{16}$	Estimate	—	$0.48; \widehat{4.88}$	$1 + x^3 y^2$	$1 + x^5 y^{12} + x^{13} y^{10}$	L4+B	1	G $2 \times \llbracket 210, 4 \rrbracket$
CL-86fa16fa/C396	(15, 14)	$\llbracket 420, 8 \rrbracket$	$5; \widehat{15}$	Estimate	—	$0.48; \widehat{4.29}$	$1 + xy^6 + x^{14} y^8$	$1 + x^3 y^2$	L4+B	1	G $2 \times \llbracket 210, 4 \rrbracket$
CL-53812e7c/C372	(15, 14)	$\llbracket 420, 8 \rrbracket$	$\llbracket 5, 15 \rrbracket$	Interval	—	$\llbracket 0.48, 4.29 \rrbracket$	$1 + x^6 y^4$	$1 + x^8 y^{10} + x^{10} y^{12}$	L4+I+B	2	G $2 \times \llbracket 210, 4 \rrbracket$
CL-eb460c33/C380	(15, 14)	$\llbracket 420, 8 \rrbracket$	$5; \widehat{14}$	Estimate	—	$0.48; \widehat{3.73}$	$1 + y$	$1 + xy^5 + x^{12} y^3$	L4+B	1	G conn.
CL-e2918556/C384	(15, 14)	$\llbracket 420, 8 \rrbracket$	$5; \widehat{14}$	Estimate	—	$0.48; \widehat{3.73}$	$1 + y$	$1 + x^3 y^4 + x^4 y^9$	L4+B	3	G/O conn.
CL-576cc4f9/C385	(15, 14)	$\llbracket 420, 8 \rrbracket$	$5; \widehat{14}$	Estimate	—	$0.48; \widehat{3.73}$	$1 + y^3$	$1 + xy^3 + x^4 y^8$	L4+B	1	G conn.
CL-5e7d0343/C373	(15, 14)	$\llbracket 420, 8 \rrbracket$	$5; \widehat{14}$	Estimate	—	$0.48; \widehat{3.73}$	$1 + y^{11}$	$1 + x^{11} y^{11} + x^{14} y^{13}$	L4+B	1	G conn.
CL-4a95cc39/C389	(15, 14)	$\llbracket 420, 8 \rrbracket$	$\llbracket 5, 13 \rrbracket$	Interval	—	$\llbracket 0.48, 3.22 \rrbracket$	$1 + y^3$	$1 + xy^3 + x^4$	L4+W+B	2	G/O conn.
CL-d192c3d1/C375	(15, 14)	$\llbracket 420, 8 \rrbracket$	$\llbracket 5, 13 \rrbracket$	Interval	—	$\llbracket 0.48, 3.22 \rrbracket$	$1 + y^5$	$1 + x^6 + x^8 y^5$	L4+W+B	3	G/O conn.
CL-c822290f/C394	(15, 14)	$\llbracket 420, 8 \rrbracket$	$5; \widehat{13}$	Estimate	—	$0.48; \widehat{3.22}$	$1 + x^{12} y^2$	$1 + x^4 y^{12} + x^{14} y^2$	L4+B	1	G $2 \times \llbracket 210, 4 \rrbracket$
CL-0e057d70/C386	(15, 14)	$\llbracket 420, 8 \rrbracket$	$5; \widehat{13}$	Estimate	—	$0.48; \widehat{3.22}$	$1 + x^{12} y^4$	$1 + x^4 + x^5 y^6$	L4+B	1	G $2 \times \llbracket 210, 4 \rrbracket$
CL-d377ce58/C399	(15, 14)	$\llbracket 420, 8 \rrbracket$	$5; \widehat{12}$	Estimate	—	$0.48; \widehat{2.74}$	$1 + xy^6 + x^8 y^2$	$1 + x^6 y^6$	L4+B	1	G $2 \times \llbracket 210, 4 \rrbracket$
CL-0533da8a/C376	(15, 14)	$\llbracket 420, 8 \rrbracket$	$5; \widehat{12}$	Estimate	—	$0.48; \widehat{2.74}$	$1 + x^{12} y^4$	$1 + x^2 y^2 + x^4$	L4+B	1	G $2 \times \llbracket 210, 4 \rrbracket$
CL-f742ab37/C374	(15, 14)	$\llbracket 420, 8 \rrbracket$	$5; \widehat{11}$	Estimate	—	$0.48; \widehat{2.30}$	$1 + x^3 y^{10}$	$1 + x^5 y^6 + x^{13} y^2$	L4+B	1	G $2 \times \llbracket 210, 4 \rrbracket$
CL-f7222fc1/C379	(15, 14)	$\llbracket 420, 8 \rrbracket$	$5; \widehat{10}$	Estimate	—	$0.48; \widehat{1.90}$	$1 + x^3 y^4$	$1 + x + x^{14} y^{10}$	L4+B	1	G $2 \times \llbracket 210, 4 \rrbracket$
CL-84deda58/C383	(15, 14)	$\llbracket 420, 8 \rrbracket$	$5; \widehat{10}$	Estimate	—	$0.48; \widehat{1.90}$	$1 + x^6 y^4$	$1 + x^2 + x^4$	L4+B	1	G $2 \times \llbracket 210, 4 \rrbracket$
CL-9de7ed68/C378	(15, 14)	$\llbracket 420, 8 \rrbracket$	$5; \widehat{9}$	Estimate	—	$0.48; \widehat{1.54}$	$1 + x^3 y^4$	$1 + xy^4 + x^2$	L4+B	1	N $2 \times \llbracket 210, 4 \rrbracket$
CL-845845df/C397	(15, 14)	$\llbracket 420, 8 \rrbracket$	$5; \widehat{9}$	Estimate	—	$0.48; \widehat{1.54}$	$1 + x^4 y^{10} + x^{14} y^{12}$	$1 + x^3 y^{12}$	L4+B	1	G $2 \times \llbracket 210, 4 \rrbracket$
CL-568676b0/C400	(15, 14)	$\llbracket 420, 8 \rrbracket$	$5; \widehat{9}$	Estimate	—	$0.48; \widehat{1.54}$	$1 + x^6 y^4$	$1 + x + x^2$	L4+B	1	G $2 \times \llbracket 210, 4 \rrbracket$
CL-19f71a7e/C377	(15, 14)	$\llbracket 420, 8 \rrbracket$	$5; \widehat{7}$	Estimate	—	$0.48; \widehat{0.93}$	$1 + y^4$	$1 + x + x^2$	L4+B	1	G $2 \times \llbracket 210, 4 \rrbracket$
CL-263c4ed6/C388	(15, 14)	$\llbracket 420, 8 \rrbracket$	$5; \widehat{7}$	Estimate	—	$0.48; \widehat{0.93}$	$1 + y^4$	$1 + x^2 y^{13} + x^8 y^6$	L4+B	1	G conn.
CL-e437a721/C381	(15, 14)	$\llbracket 420, 8 \rrbracket$	$5; \widehat{7}$	Estimate	—	$0.48; \widehat{0.93}$	$1 + y^4$	$1 + x^4 y^2 + x^{11} y^8$	L4+B	1	O $2 \times \llbracket 210, 4 \rrbracket$
CL-ec5f2a26/C387	(15, 14)	$\llbracket 420, 8 \rrbracket$	$5; \widehat{7}$	Estimate	—	$0.48; \widehat{0.93}$	$1 + y^6$	$1 + xy^6 + x^2 y^2$	L4+B	1	G $2 \times \llbracket 210, 4 \rrbracket$
CL-9bb8225d/C390	(15, 14)	$\llbracket 420, 8 \rrbracket$	$5; \widehat{7}$	Estimate	—	$0.48; \widehat{0.93}$	$1 + y^6$	$1 + x^{13} y^{10} + x^{14} y^6$	L4+B	1	G $2 \times \llbracket 210, 4 \rrbracket$
CL-8311524f/C393	(15, 14)	$\llbracket 420, 8 \rrbracket$	$5; \widehat{7}$	Estimate	—	$0.48; \widehat{0.93}$	$1 + x + x^2 y^2$	$1 + x^6 y^6$	L4+B	1	O $2 \times \llbracket 210, 4 \rrbracket$
CL-f30dec8e/C392	(15, 14)	$\llbracket 420, 8 \rrbracket$	$5; \widehat{7}$	Estimate	—	$0.48; \widehat{0.93}$	$1 + x^2 y^4 + x^4 y^{10}$	$1 + y^4$	L4+B	1	O $2 \times \llbracket 210, 4 \rrbracket$
CL-722d606c/C398	(15, 14)	$\llbracket 420, 8 \rrbracket$	$5; \widehat{7}$	Estimate	—	$0.48; \widehat{0.93}$	$1 + x^2 y^{12} + x^{10} y^{12}$	$1 + y^6$	L4+B	1	O $2 \times \llbracket 210, 4 \rrbracket$
CL-373036fd/C382	(15, 14)	$\llbracket 420, 8 \rrbracket$	$5; \widehat{7}$	Estimate	—	$0.48; \widehat{0.93}$	$1 + x^4 + x^8 y^{12}$	$1 + y^2$	L4+B	1	G $2 \times \llbracket 210, 4 \rrbracket$
CL-32806d84/C395	(15, 14)	$\llbracket 420, 8 \rrbracket$	5	Estimate	Exact	0.48	$1 + xy^{10} + x^5 y^8$	$1 + x^3 y^2$	L4+B	3	G/O $2 \times \llbracket 210, 4 \rrbracket$
CL-c065aeaf/C357	(15, 14)	$\llbracket 420, 6 \rrbracket$	5	Exact	—	0.36	$1 + xy^7$	$1 + x^9 y^9 + x^{13} y^3$	L4+W+B	1	G conn.
CL-01ad3134/C362	(15, 14)	$\llbracket 420, 6 \rrbracket$	3	Exact	—	0.13	$1 + x^2 y^2 + x^3 y^3$	$1 + x^5$	Exh.	1	G conn.
CL-dcca76b0/C364	(15, 14)	$\llbracket 420, 6 \rrbracket$	$\llbracket 5, 16 \rrbracket$	Interval	—	$\llbracket 0.36, 3.66 \rrbracket$	$1 + x^4 y^7$	$1 + x^{14} y^2 + x^{14} y^6$	L4+I+B	1	G conn.
CL-acd29413/C353	(15, 14)	$\llbracket 420, 6 \rrbracket$	$\llbracket 5, 15 \rrbracket$	Interval	—	$\llbracket 0.36, 3.21 \rrbracket$	$1 + y^5 + x^2 y$	$1 + xy^7$	L4+I+B	1	G conn.
CL-bc646161/C371	(15, 14)	$\llbracket 420, 6 \rrbracket$	$5; \widehat{15}$	Estimate	—	$0.36; \widehat{3.21}$	$1 + x$	$1 + y^2 + xy^3$	L4+B	1	G conn.
CL-2530c11b/C354	(15, 14)	$\llbracket 420, 6 \rrbracket$	$5; \widehat{15}$	Estimate	—	$0.36; \widehat{3.21}$	$1 + x$	$1 + xy + x^6 y^3$	L4+B	1	G conn.
CL-69bf71ea/C361	(15, 14)	$\llbracket 420, 6 \rrbracket$	$5; \widehat{15}$	Estimate	—	$0.36; \widehat{3.21}$	$1 + x$	$1 + x^5 y + x^{13} y^3$	L4+B	1	G conn.
CL-f7a6bc43/C363	(15, 14)	$\llbracket 420, 6 \rrbracket$	$5; \widehat{15}$	Estimate	—	$0.36; \widehat{3.21}$	$1 + x^2$	$1 + x^6 y^{12} + x^{11} y^{11}$	L4+B	2	G/O conn.
CL-735d8776/C365	(15, 14)	$\llbracket 420, 6 \rrbracket$	$5; \widehat{15}$	Estimate	—	$0.36; \widehat{3.21}$	$1 + x^4 y + x^5 y^3$	$1 + x$	L4+B	1	G conn.
CL-85245770/C356	(15, 14)	$\llbracket 420, 6 \rrbracket$	$\llbracket 5, 14 \rrbracket$	Interval	—	$\llbracket 0.36, 2.80 \rrbracket$	$1 + x^4 y^7$	$1 + x^5 y^8 + x^{11} y^3$	L4+W+B	1	O conn.
CL-8ba9617f/C368	(15, 14)	$\llbracket 420, 6 \rrbracket$	$5; \widehat{13}$	Estimate	—	$0.36; \widehat{2.41}$	$1 + x^2 y^{11} + x^{12} y^{12}$	$1 + x^4 y^7$	L4+B	1	G conn.

TABLE S6 (continued)

ID/class	(ℓ, m)	$\llbracket n, k \rrbracket$	d	Own evidence	Class	effect	FOM	A	B	Evid.	Obs.	Attr.	Decomp.
CL-4f88e824/C366	(15, 14)	$\llbracket 420, 6 \rrbracket$	$5; \widehat{13}$	Estimate	—	0.36; $\widehat{2.41}$	$1 + x^4 y^7$	$1 + x^3 y^6 + x^{10} y^4$	L4+B	1	G	conn.	
CL-54d0129e/C355	(15, 14)	$\llbracket 420, 6 \rrbracket$	$[5, 13]$	Interval	—	$[0.36, 2.41]$	$1 + x^7 y^7$	$1 + x^4 y + x^{14} y^{10}$	L4+W+B	1	O	conn.	
CL-0ec9877d/C370	(15, 14)	$\llbracket 420, 6 \rrbracket$	$5; \widehat{10}$	Estimate	—	0.36; $\widehat{1.43}$	$1 + x^3 y^7$	$1 + x y^4 + x^{12} y^{13}$	L4+B	1	G	conn.	
CL-aa1f235d/C367	(15, 14)	$\llbracket 420, 6 \rrbracket$	$5; \widehat{10}$	Estimate	—	0.36; $\widehat{1.43}$	$1 + x^5 y^2 + x^{12} y^6$	$1 + x^5 y^7$	L4+B	1	G	conn.	
CL-1c3813aa/C358	(15, 14)	$\llbracket 420, 6 \rrbracket$	$5; \widehat{8}$	Estimate	—	0.36; $\widehat{0.91}$	$1 + y^2 + y^3$	$1 + x y^7$	L4+B	3	G	conn.	
CL-21aeb4fa/C369	(15, 14)	$\llbracket 420, 6 \rrbracket$	$5; \widehat{8}$	Estimate	—	0.36; $\widehat{0.91}$	$1 + x^4$	$1 + y + y^3$	L4+B	5	G/N	conn.	
CL-d42149ac/C360	(15, 14)	$\llbracket 420, 6 \rrbracket$	$5; \widehat{6}$	Estimate	—	0.36; $\widehat{0.51}$	$1 + x^5 y^7$	$1 + x^2 y^5 + x^{10} y^{11}$	L4+B	1	G	conn.	
CL-1cc5db1e/C359	(15, 14)	$\llbracket 420, 6 \rrbracket$	$5; \widehat{6}$	Estimate	—	0.36; $\widehat{0.51}$	$1 + x^5 y^7$	$1 + x^7 y^2 + x^8 y^6$	L4+B	1	G	conn.	
CL-89dd6c68/C321	(15, 14)	$\llbracket 420, 4 \rrbracket$	5	Exact	—	0.24	$1 + x^2 y^{10} + x^{10} y^5$	$1 + x^6$	L4+W+B	1	O	conn.	
CL-e72cc9d8/C334	(15, 14)	$\llbracket 420, 4 \rrbracket$	5	Exact	—	0.24	$1 + x^6$	$1 + x^7 y + x^{11} y^2$	L4+W+B	2	G/O	conn.	
CL-03413624/C328	(15, 14)	$\llbracket 420, 4 \rrbracket$	5	Exact	—	0.24	$1 + x^6 y^4$	$1 + x y^3 + x^2 y^6$	L4+W+B	3	G/O	conn.	
CL-9fbce668/C290	(15, 14)	$\llbracket 420, 4 \rrbracket$	$[5, 22]$	Interval	—	$[0.24, 4.61]$	$1 + x^3 y$	$1 + x^{10} y^6 + x^{11} y^2$	L4+I+B	1	G	conn.	
CL-c271b632/C279	(15, 14)	$\llbracket 420, 4 \rrbracket$	$[5, 22]$	Interval	—	$[0.24, 4.61]$	$1 + x^4 y^3 + x^8 y^{11}$	$1 + x^3 y^3$	L4+I+B	1	O	conn.	
CL-f1e29899/C298	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{21}$	Estimate	—	0.24; $\widehat{4.20}$	$1 + x + x^2 y$	$1 + x^6 y^2$	L4+B	1	G	conn.	
CL-3e5c8799/C313	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{21}$	Estimate	—	0.24; $\widehat{4.20}$	$1 + x^2 y + x^{10} y^{11}$	$1 + x^3 y^2$	L4+B	1	G	conn.	
CL-bdfa840a/C327	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{21}$	Estimate	—	0.24; $\widehat{4.20}$	$1 + x^3 y$	$1 + x y^3 + x^2 y^6$	L4+B	1	G	conn.	
CL-170eab7b/C340	(15, 14)	$\llbracket 420, 4 \rrbracket$	$[5, 21]$	Interval	—	$[0.24, 4.20]$	$1 + x^3 y^3$	$1 + x^2 + x^{13} y^{13}$	L4+I+B	1	G	conn.	
CL-35afb936/C278	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{21}$	Estimate	—	0.24; $\widehat{4.20}$	$1 + x^3 y^6$	$1 + x^4 y^2 + x^5 y$	L4+B	1	G	conn.	
CL-de17c9d3/C303	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{21}$	Estimate	—	0.24; $\widehat{4.20}$	$1 + x^6 y^8$	$1 + x^8 y^7 + x^{13} y$	L4+B	1	G	conn.	
CL-bbe854ab/C286	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{20}$	Estimate	—	0.24; $\widehat{3.81}$	$1 + x y^4 + x^8 y^{11}$	$1 + x^{12} y^{12}$	L4+B	1	G	conn.	
CL-34564918/C293	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{20}$	Estimate	—	0.24; $\widehat{3.81}$	$1 + x^3 y$	$1 + x y^{11} + x^{14} y^7$	L4+B	1	G	conn.	
CL-035a9186/C306	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{20}$	Estimate	—	0.24; $\widehat{3.81}$	$1 + x^3 y$	$1 + x^4 y^5 + x^{14} y^{10}$	L4+B	1	G	conn.	
CL-921146b8/C273	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{20}$	Estimate	—	0.24; $\widehat{3.81}$	$1 + x^3 y^2$	$1 + x y + x^2 y$	L4+B	1	G	conn.	
CL-47b94d92/C308	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{20}$	Estimate	—	0.24; $\widehat{3.81}$	$1 + x^4 y^{13} + x^{14} y^3$	$1 + x^6 y^5$	L4+B	2	G/O	conn.	
CL-84e4fd49/C263	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{20}$	Estimate	—	0.24; $\widehat{3.81}$	$1 + x^5 y + x^{10} y^{12}$	$1 + x^{12} y^{13}$	L4+B	1	G	conn.	
CL-d0e1c43e/C339	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{20}$	Estimate	—	0.24; $\widehat{3.81}$	$1 + x^6 y^3$	$1 + x y + x^5$	L4+B	1	N	conn.	
CL-a72299e5/C312	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{18}$	Estimate	Tighter	0.24; $\widehat{3.09}$	$1 + x^6 y^3$	$1 + x^2 y^{13} + x^4 y^4$	L4+B	6	G/O	conn.	
CL-dc41e42a/C288	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{20}$	Estimate	—	0.24; $\widehat{3.81}$	$1 + x^6 y^3$	$1 + x^5 y^{11} + x^{10} y^{13}$	L4+B	1	G	conn.	
CL-b30c14d0/C287	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{20}$	Estimate	—	0.24; $\widehat{3.81}$	$1 + x^6 y^5$	$1 + x^2 y^8 + x^{10} y^6$	L4+B	1	G	conn.	
CL-fb0e1722/C259	(15, 14)	$\llbracket 420, 4 \rrbracket$	$[5, 20]$	Interval	—	$[0.24, 3.81]$	$1 + x^6 y^5$	$1 + x^4 + x^5 y^2$	L4+I+B	1	N	conn.	
CL-68eefaad/C302	(15, 14)	$\llbracket 420, 4 \rrbracket$	$[5, 20]$	Interval	—	$[0.24, 3.81]$	$1 + x^6 y^5$	$1 + x^8 y^7 + x^{10} y^2$	L4+I+B	1	G	conn.	
CL-cfe9c60e/C342	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{20}$	Estimate	—	0.24; $\widehat{3.81}$	$1 + x^6 y^9$	$1 + x y^6 + x^5 y^3$	L4+B	1	G	conn.	
CL-3ba7a33d/C268	(15, 14)	$\llbracket 420, 4 \rrbracket$	$[5, 20]$	Interval	—	$[0.24, 3.81]$	$1 + x^6 y^9$	$1 + x^2 + x^7 y^9$	L4+I+B	2	G	conn.	
CL-859e9a93/C322	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{20}$	Estimate	—	0.24; $\widehat{3.81}$	$1 + x^6 y^{13}$	$1 + x^2 y^2 + x^7$	L4+B	1	G	conn.	
CL-959df55a/C315	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{20}$	Estimate	—	0.24; $\widehat{3.81}$	$1 + x^7 + x^{11} y^7$	$1 + x^6 y^4$	L4+B	1	G	conn.	
CL-3dc9e172/C260	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{19}$	Estimate	—	0.24; $\widehat{3.44}$	$1 + x y^3 + x^5 y$	$1 + x^3 y$	L4+B	1	O	conn.	
CL-79baf488/C305	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{19}$	Estimate	—	0.24; $\widehat{3.44}$	$1 + x^2 y^3 + x^7 y^{11}$	$1 + x^3 y^5$	L4+B	1	G	conn.	
CL-58bc09e7/C284	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{19}$	Estimate	—	0.24; $\widehat{3.44}$	$1 + x^2 y^{13} + x^{10} y^{11}$	$1 + x^6 y^4$	L4+B	1	O	conn.	
CL-efd28323/C329	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{19}$	Estimate	—	0.24; $\widehat{3.44}$	$1 + x^3 y^3$	$1 + x^2 y^{12} + x^4 y^2$	L4+B	1	G	conn.	
CL-56365497/C350	(15, 14)	$\llbracket 420, 4 \rrbracket$	$[5, 19]$	Interval	—	$[0.24, 3.44]$	$1 + x^3 y^4$	$1 + x^5 y + x^{13} y^2$	L4+I+B	1	G	conn.	
CL-aabd155e/C299	(15, 14)	$\llbracket 420, 4 \rrbracket$	$[5, 19]$	Interval	—	$[0.24, 3.44]$	$1 + x^3 y^5$	$1 + x y^2 + x^{14} y^3$	L4+I+B	1	G	conn.	
CL-d18d0097/C291	(15, 14)	$\llbracket 420, 4 \rrbracket$	$[5, 19]$	Interval	—	$[0.24, 3.44]$	$1 + x^3 y^5$	$1 + x^2 y^5 + x^4 y^{12}$	L4+I+B	1	O	conn.	
CL-16ddab0d/C285	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{16}$	Estimate	Tighter	0.24; $\widehat{2.44}$	$1 + x^4 + x^5 y^5$	$1 + x^3 y^9$	L4+B	2	G/O	conn.	
CL-6f3c91dc/C352	(15, 14)	$\llbracket 420, 4 \rrbracket$	$[5, 19]$	Interval	—	$[0.24, 3.44]$	$1 + x^6 y^5$	$1 + x^4 y^5 + x^5 y^6$	L4+I+B	2	G	conn.	
CL-92173649/C269	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{19}$	Estimate	—	0.24; $\widehat{3.44}$	$1 + x^6 y^5$	$1 + x^8 y^9 + x^{13} y^5$	L4+B	1	G	conn.	
CL-d75eef3d/C330	(15, 14)	$\llbracket 420, 4 \rrbracket$	$[5, 18]$	Interval	—	$[0.24, 3.09]$	$1 + x y + x^2$	$1 + x^3 y^4$	L4+I+B	1	O	conn.	
CL-3517f047/C307	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{18}$	Estimate	—	0.24; $\widehat{3.09}$	$1 + x y + x^{14}$	$1 + x^3 y$	L4+B	1	G	conn.	
CL-1c01f1af/C326	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{18}$	Estimate	—	0.24; $\widehat{3.09}$	$1 + x^2 y^3 + x^{10} y$	$1 + x^6 y^5$	L4+B	1	O	conn.	
CL-0f0f7823/C300	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{18}$	Estimate	—	0.24; $\widehat{3.09}$	$1 + x^2 y^{10} + x^7 y^{11}$	$1 + x^6 y^{13}$	L4+B	2	G	conn.	
CL-a68b54eb/C295	(15, 14)	$\llbracket 420, 4 \rrbracket$	$[5, 18]$	Interval	—	$[0.24, 3.09]$	$1 + x^3 y$	$1 + x^2 y^{10} + x^7 y^6$	L4+I+B	1	O	conn.	
CL-cd1a95aa/C280	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{18}$	Estimate	—	0.24; $\widehat{3.09}$	$1 + x^3 y^5$	$1 + x^5 y^6 + x^{13} y^{13}$	L4+B	4	G/O	conn.	
CL-243c1ce8/C276	(15, 14)	$\llbracket 420, 4 \rrbracket$	$[5, 18]$	Interval	—	$[0.24, 3.09]$	$1 + x^3 y^6$	$1 + x y^8 + x^5 y^7$	L4+I+B	1	G	conn.	
CL-a01dbf6f/C346	(15, 14)	$\llbracket 420, 4 \rrbracket$	$[5, 18]$	Interval	—	$[0.24, 3.09]$	$1 + x^4 y + x^{11} y^5$	$1 + x^6 y^3$	L4+I+B	1	O	conn.	
CL-52ea7c29/C320	(15, 14)	$\llbracket 420, 4 \rrbracket$	$[5, 18]$	Interval	—	$[0.24, 3.09]$	$1 + x^6 y^4$	$1 + x + x^5 y^3$	L4+I+B	1	N	conn.	
CL-97ce5a71/C274	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{18}$	Estimate	—	0.24; $\widehat{3.09}$	$1 + x^6 y^6$	$1 + x y^5 + x^5 y$	L4+B	4	G/O	conn.	
CL-f4f0dd58/C348	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{18}$	Estimate	—	0.24; $\widehat{3.09}$	$1 + x^7 y^9 + x^{14} y^7$	$1 + x^6 y^2$	L4+B	1	O	conn.	
CL-39516f7a/C311	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{18}$	Estimate	—	0.24; $\widehat{3.09}$	$1 + x^8 y^{10} + x^{10} y^{13}$	$1 + x^3 y^{12}$	L4+B	2	G/O	conn.	
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TABLE S6 (continued)

ID/class	(ℓ, m)	$\llbracket n, k \rrbracket$	d	Own evidence	Class	effect	FOM	A	B	Evid.	Obs. Attr.	Decomp.
CL-012b5093/C261	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{16}$	Estimate	—	0.24; $\widehat{2.44}$	$1 + x^9 y^{13}$	$1 + x^7 y^{13} + x^8 y$	L4+B	2	G/O	conn.
CL-18c67b02/C317	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{16}$	Estimate	—	0.24; $\widehat{2.44}$	$1 + x^{10} y^2 + x^{14} y^{11}$	$1 + x^3 y^2$	L4+B	1	G	conn.
CL-45dbbd81/C301	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{15}$	Estimate	—	0.24; $\widehat{2.14}$	$1 + x^2 y^7 + x^{13} y^{11}$	$1 + x^3 y^{13}$	L4+B	1	G	conn.
CL-00f55eb4/C347	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{14}$	Estimate	—	0.24; $\widehat{1.87}$	$1 + y$	$1 + x^4 y + x^{11} y$	L4+B	3	G/O	conn.
CL-ee1a67cc/C331	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{14}$	Estimate	—	0.24; $\widehat{1.87}$	$1 + y$	$1 + x^{13} y^4 + x^{14} y$	L4+B	2	G	conn.
CL-1d74ed97/C255	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{14}$	Estimate	—	0.24; $\widehat{1.87}$	$1 + y^3$	$1 + x^2 y + x^{13} y^7$	L4+B	3	G/O	conn.
CL-4218c470/C297	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{14}$	Estimate	—	0.24; $\widehat{1.87}$	$1 + y^5$	$1 + x y^4 + x^{14}$	L4+B	1	G	conn.
CL-20a74088/C332	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{14}$	Estimate	—	0.24; $\widehat{1.87}$	$1 + y^5$	$1 + x^4 y^5 + x^8 y$	L4+B	1	G	conn.
CL-63d37244/C252	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{14}$	Estimate	—	0.24; $\widehat{1.87}$	$1 + y^5$	$1 + x^4 y^6 + x^8 y^8$	L4+B	1	O	conn.
CL-39202e68/C343	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{14}$	Estimate	—	0.24; $\widehat{1.87}$	$1 + x y + x^{14}$	$1 + y$	L4+B	1	G	conn.
CL-e0159989/C296	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{14}$	Estimate	—	0.24; $\widehat{1.87}$	$1 + x^2 + x^{13} y$	$1 + y^5$	L4+B	1	G	conn.
CL-1a333605/C349	(15, 14)	$\llbracket 420, 4 \rrbracket$	$\llbracket 5, 14 \rrbracket$	Interval	—	$[0.24, 1.87]$	$1 + x^3 y^4$	$1 + x y^2 + x^2 y^3$	L4+W+B	1	G	conn.
CL-96e74570/C282	(15, 14)	$\llbracket 420, 4 \rrbracket$	$\llbracket 5, 14 \rrbracket$	Interval	—	$[0.24, 1.87]$	$1 + x^3 y^5$	$1 + x^2 + x^{13} y^4$	L4+I+B	2	G	conn.
CL-a517ace2/C281	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{14}$	Estimate	—	0.24; $\widehat{1.87}$	$1 + x^3 y^{13}$	$1 + x^5 y^2 + x^{10} y^{12}$	L4+B	2	G/O	conn.
CL-23177205/C341	(15, 14)	$\llbracket 420, 4 \rrbracket$	$\llbracket 5, 14 \rrbracket$	Interval	—	$[0.24, 1.87]$	$1 + x^4 y^2 + x^{11} y^{12}$	$1 + x^3 y^3$	L4+W+B	2	G/O	conn.
CL-0bcb2cf/C265	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{14}$	Estimate	—	0.24; $\widehat{1.87}$	$1 + x^6 y^5$	$1 + x^5 y^8 + x^{10} y^2$	L4+B	1	G	conn.
CL-d0bba1bf/C351	(15, 14)	$\llbracket 420, 4 \rrbracket$	$\llbracket 5, 14 \rrbracket$	Interval	—	$[0.24, 1.87]$	$1 + x^6 y^5$	$1 + x^{10} y^3 + x^{14} y^8$	L4+I+B	1	G	conn.
CL-bd9385b1/C262	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{14}$	Estimate	—	0.24; $\widehat{1.87}$	$1 + x^7 y + x^{14} y^4$	$1 + y^2$	L4+B	2	G	conn.
CL-cabac8af/C283	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{13}$	Estimate	—	0.24; $\widehat{1.61}$	$1 + x^4 y + x^5 y^7$	$1 + x^3 y^3$	L4+B	1	G	conn.
CL-6245f632/C270	(15, 14)	$\llbracket 420, 4 \rrbracket$	$\llbracket 5, 13 \rrbracket$	Interval	—	$[0.24, 1.61]$	$1 + x^{12} y^{12}$	$1 + x^7 + x^{14} y^{11}$	L4+I+B	1	G	conn.
CL-27163520/C294	(15, 14)	$\llbracket 420, 4 \rrbracket$	$\llbracket 5, 12 \rrbracket$	Interval	—	$[0.24, 1.37]$	$1 + x^3 y^3$	$1 + x^4 y^2 + x^5$	L4+I+B	1	N	conn.
CL-7f05804a/C335	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{10}$	Estimate	Tighter	0.24; $\widehat{0.95}$	$1 + x^4 y^5 + x^8 y^{10}$	$1 + x^6 y^5$	L4+B	1	O	conn.
CL-3e07faef/C250	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{12}$	Estimate	—	0.24; $\widehat{1.37}$	$1 + x^9 y^{13}$	$1 + x y^7 + x^2$	L4+B	1	O	conn.
CL-013659b7/C292	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{11}$	Estimate	—	0.24; $\widehat{1.15}$	$1 + y^3$	$1 + x^4 y^3 + x^8 y^6$	L4+B	1	G	conn.
CL-c5751133/C267	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{11}$	Estimate	—	0.24; $\widehat{1.15}$	$1 + x^2 y^6 + x^7 y^5$	$1 + x^6 y^3$	L4+B	1	G	conn.
CL-0892ddc0/C324	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{11}$	Estimate	—	0.24; $\widehat{1.15}$	$1 + x^3 y^5$	$1 + x y^2 + x^{14} y^{12}$	L4+B	1	G	conn.
CL-22178d24/C289	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{11}$	Estimate	—	0.24; $\widehat{1.15}$	$1 + x^4 y^{12} + x^{14} y$	$1 + x^3 y^{13}$	L4+B	1	G	conn.
CL-52590f51/C304	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{10}$	Estimate	—	0.24; $\widehat{0.95}$	$1 + y$	$1 + x + x^2$	L4+B	1	G	conn.
CL-30f2d7e8/C318	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{10}$	Estimate	—	0.24; $\widehat{0.95}$	$1 + x y^3 + x^5 y$	$1 + x^6 y$	L4+B	1	G	conn.
CL-c1097279/C249	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{10}$	Estimate	—	0.24; $\widehat{0.95}$	$1 + x^3 y$	$1 + x + x^2$	L4+B	1	G	conn.
CL-969d5906/C333	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{10}$	Estimate	—	0.24; $\widehat{0.95}$	$1 + x^3 y^3$	$1 + x^8 y^2 + x^{13} y^2$	L4+B	1	O	conn.
CL-7e2d38ae/C336	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{10}$	Estimate	—	0.24; $\widehat{0.95}$	$1 + x^6 y^3$	$1 + x + x^2$	L4+B	1	G	conn.
CL-c087367c/C254	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{9}$	Estimate	—	0.24; $\widehat{0.77}$	$1 + x^3 y^2$	$1 + x y + x^{14} y^{13}$	L4+B	1	G	conn.
CL-64c71487/C309	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{9}$	Estimate	—	0.24; $\widehat{0.77}$	$1 + x^3 y^4$	$1 + x^2 y^3 + x^7 y^{10}$	L4+B	1	G	conn.
CL-3cadff79/C314	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{9}$	Estimate	—	0.24; $\widehat{0.77}$	$1 + x^6 y$	$1 + x y^8 + x^{11} y$	L4+B	1	G	conn.
CL-224f3680/C310	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{9}$	Estimate	—	0.24; $\widehat{0.77}$	$1 + x^6 y^{13}$	$1 + x^7 y^3 + x^{11} y^{13}$	L4+B	1	G	conn.
CL-b18bd481/C271	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{8}$	Estimate	—	0.24; $\widehat{0.61}$	$1 + x^6 y^{13}$	$1 + x y^3 + x^{14} y^{11}$	L4+B	1	O	conn.
CL-a467cd49/C272	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{7}$	Estimate	—	0.24; $\widehat{0.47}$	$1 + x y + x^2 y^8$	$1 + y^2$	L4+B	4	G/O	conn.
CL-e8dd03a4/C319	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{7}$	Estimate	—	0.24; $\widehat{0.47}$	$1 + x y^3 + x^{14} y^{11}$	$1 + y^2$	L4+B	5	G/O	conn.
CL-71175309/C345	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{7}$	Estimate	—	0.24; $\widehat{0.47}$	$1 + x^3 y$	$1 + x^4 y^2 + x^{11} y^5$	L4+B	1	G	conn.
CL-229f68f/C256	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{7}$	Estimate	—	0.24; $\widehat{0.47}$	$1 + x^4 y^3 + x^{11} y$	$1 + y^2$	L4+B	2	G/O	conn.
CL-da88d8de/C275	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{6}$	Estimate	—	0.24; $\widehat{0.34}$	$1 + x^3 y^5$	$1 + x y^{12} + x^2 y^{10}$	L4+B	1	G	conn.
CL-a8b29562/C258	(15, 14)	$\llbracket 420, 4 \rrbracket$	$5; \widehat{6}$	Estimate	—	0.24; $\widehat{0.34}$	$1 + x^4 y^2 + x^{14} y^2$	$1 + x^{12} y$	L4+B	1	G	conn.
CL-b9275593/C535	(18, 12)	$\llbracket 432, 48 \rrbracket$	3	Exact	—	1.00	$1 + y^4$	$1 + x^6 y^2 + x^{12} y^4$	H _c	1	—	$12 \times \llbracket 36, 4 \rrbracket$
CL-21ad24a2/C533	(18, 12)	$\llbracket 432, 32 \rrbracket$	6	Exact	—	2.67	$1 + x^4 + x^{12} y^4$	$1 + x^4 y^4$	M	2	G	$8 \times \llbracket 54, 4 \rrbracket$
CL-5e6f356a/C534	(18, 12)	$\llbracket 432, 32 \rrbracket$	3	Exact	—	0.67	$1 + y^4$	$1 + x^2 + x^4$	Exh.	1	G	$8 \times \llbracket 54, 4 \rrbracket$
CL-f93732aa/C527	(18, 12)	$\llbracket 432, 24 \rrbracket$	7	Exact	—	2.72	$1 + x^3 y^3$	$1 + y^{10} + x^3 y^5$	M	3	G/O	$6 \times \llbracket 72, 4 \rrbracket$
CL-63ba351e/C529	(18, 12)	$\llbracket 432, 24 \rrbracket$	6	Exact	—	2.00	$1 + y + x^{12} y^{11}$	$1 + x^6 y^6$	M	2	G/O	$6 \times \llbracket 72, 4 \rrbracket$
CL-7827219c/C526	(18, 12)	$\llbracket 432, 24 \rrbracket$	5	Exact	—	1.39	$1 + x^6 y^6$	$1 + x y + x^2 y^2$	M	1	G	$6 \times \llbracket 72, 4 \rrbracket$
CL-2c508b86/C530	(18, 12)	$\llbracket 432, 24 \rrbracket$	4	Exact	—	0.89	$1 + y^3$	$1 + x^2 + x^4$	Exh.	5	G/O	$6 \times \llbracket 72, 4 \rrbracket$
CL-86e25310/C528	(18, 12)	$\llbracket 432, 24 \rrbracket$	4	Exact	—	0.89	$1 + x^3$	$1 + y^2 + y^4$	Exh.	1	G	$6 \times \llbracket 72, 4 \rrbracket$
CL-18750838/C532	(18, 12)	$\llbracket 432, 24 \rrbracket$	4	Exact	—	0.89	$1 + x^3 y^9$	$1 + y^4 + x^3 y^5$	Exh.+I	2	G	$6 \times \llbracket 72, 4 \rrbracket$
CL-fedc4d95/C531	(18, 12)	$\llbracket 432, 24 \rrbracket$	3	Exact	—	0.50	$1 + x^3 y^{11} + x^{12} y$	$1 + x^6$	Exh.+I	1	G	$3 \times \llbracket 144, 8 \rrbracket$
CL-8fe7fff8/C523	(18, 12)	$\llbracket 432, 16 \rrbracket$	9	Exact	—	3.00	$1 + x^5 y^4$	$1 + x^5 + x^{17} y^8$	M	1	O	$4 \times \llbracket 108, 4 \rrbracket$
CL-c5ca6b9a/C525	(18, 12)	$\llbracket 432, 16 \rrbracket$	8	Exact	—	2.37	$1 + x y^6$	$1 + x^{11} y^{10} + x^{15} y^2$	M	1	G	$4 \times \llbracket 108, 4 \rrbracket$
CL-a1a678e0/C519	(18, 12)	$\llbracket 432, 16 \rrbracket$	8	Exact	—	2.37	$1 + x^{16} y^4$	$1 + x^3 y^4 + x^5 y^4$	M	1	G	$4 \times \llbracket 108, 4 \rrbracket$
CL-d72254a8/C517	(18, 12)	$\llbracket 432, 16 \rrbracket$	7	Exact	—	1.81	$1 + x^2 + x^3 y^8$	$1 + x y^4$	M	4	G/O	$4 \times \llbracket 108, 4 \rrbracket$
CL-8e614a5f/C517	(18, 12)	$\llbracket 432, 16 \rrbracket$	7	Exact	—	1.81	$1 + x^2 y^4 + x^{17} y^2$	$1 + x y^6$	M	1	G	$4 \times \llbracket 108, 4 \rrbracket$
CL-c9d42fcb/C518	(18, 12)	$\llbracket 432, 16 \rrbracket$	6	Exact	—	1.33	$1 + y^2$	$1 + x^2 + x^4$	M	1	G	$4 \times \llbracket 108, 4 \rrbracket$
CL-dca40892/C521	(18, 12)	$\llbracket 432, 16 \rrbracket$	6	Exact	—	1.33	$1 + x^3 y^{10}$	$1 + x^4 + x^{17} y^{10}$	M	1	G	$4 \times \llbracket 108, 4 \rrbracket$
CL-8157fe5b/C522	(18, 12)	$\llbracket 432, 16 \rrbracket$	4	Exact	—	0.59	$1 + x^4$	$1 + y^2 + y^4$	Exh.	1	G	$4 \times \llbracket 108, 4 \rrbracket$
CL-207f9524/C524	(18, 12)	$\llbracket 432, 16 \rrbracket$	4	Exact	—	0.59	$1 + x^4 y^6$	$1 + x^6 y^8 + x^8 y^{10}$	Exh.	1	G	$4 \times \llbracket 108, 4 \rrbracket$
CL-5c6fb0dc/C524	(18, 12)	$\llbracket 432, 16 \rrbracket$	4	Exact	—	0.59	$1 + x^5$	$1 + y^8 + x^5 y^4$	Exh.	2	G	$4 \times \llbracket 108, 4 \rrbracket$
CL-aa2abeb9/C520	(18, 12)	$\llbracket 432, 16 \rrbracket$	3	Exact	—	0.33	$1 + x^{12} y^8$	$1 + x y^4 + x^5$	Exh.	1	G	$4 \times \llbracket 108, 4 \rrbracket$
CL-f85559a3/C513	(18, 12)	$\llbracket 432, 12 \rrbracket$	10	Exact	—	2.78	$1 + x^3 y^3$	$1 + x^5 y^7 + x^{16} y^{11}$	M	2	G	$3 \times \llbracket 144, 4 \rrbracket$
CL-db6df8c9/C516	(18, 12)	$\llbracket 432, 12 \rrbracket$	4	Exact	—	0.44	$1 + y^3$	$1 + x + x^2$	Exh.	1	G	$3 \times \llbracket 144, 4 \rrbracket$
CL-3f55460c/C512	(18, 12)	$\llbracket 432, 12 \rrbracket$	$5; \widehat{12}$	Estimate	—	0.69; $\widehat{4.00}$	$1 + x^2 + x^4 y^9$	$1 + x^3 y^3$	L4+B	1	G	$3 \times \llbracket 144, 4 \rrbracket$
CL-771b1149/C515	(18, 12)	$\llbracket 432, 12 \rrbracket$	$5; \widehat{6}$	Estimate	—	0.69; $\widehat{1.00}$	$1 + x^3$	$1 + y + y^2$	L4+B	1	G	$3 \times \llbracket 144, 4 \rrbracket$
CL-800f5ab6/C514	(18, 12)	$\llbracket 432, 12 \rrbracket$	$5; \widehat{6}$	Estimate	—	0.69; $\widehat{1.00}$	$1 + x^3$	$1 + x^4 y^{11} + x^5 y^{10}$	L4+B	1	G	$3 \times \llbracket 144, 4 \rrbracket$

TABLE S6 (continued)

ID/class	(ℓ, m)	$[n, k]$	d	Own evidence	Class effect	FOM	A	B	Evid.	Obs. Attr.	Decomp.
CL-7ae78520/C497	(18, 12)	[432, 8]	4	Exact	–	0.30	$1 + x$	$1 + y^2 + y^4$	Exh.	1 G	$2 \times [216, 4]$
CL-24835ac8/C511	(18, 12)	[432, 8]	5; $\widehat{14}$	Estimate	Tighter	0.46; $\widehat{3.63}$	$1 + x^3y^5 + x^8y^4$	$1 + xy^3$	L4+B	1 G	$2 \times [216, 4]$
CL-793d3a34/C503	(18, 12)	[432, 8]	5; $\widehat{13}$	Estimate	Tighter	0.46; $\widehat{3.13}$	$1 + xy^9$	$1 + xy^{11} + x^{17}y$	L4+B	1 G	$2 \times [216, 4]$
CL-ab07a1d1/C495	(18, 12)	[432, 8]	[5, 15]	Interval	–	[0.46, 4.17]	$1 + x^4y^6$	$1 + xy^5 + x^5y^{10}$	L4+I+B	1 G	conn.
CL-74327c2b/C486	(18, 12)	[432, 8]	5; $\widehat{14}$	Estimate	–	0.46; $\widehat{3.63}$	$1 + x^2y^2$	$1 + x^2y^9 + x^6y^{11}$	L4+B	1 G	$2 \times [216, 4]$
CL-a4b8c21d/C474	(18, 12)	[432, 8]	[5, 14]	Interval	–	[0.46, 3.63]	$1 + x^2y^2$	$1 + x^3y + x^7$	L4+I+B	1 G	conn.
CL-1056d548/C487	(18, 12)	[432, 8]	5; $\widehat{12}$	Estimate	Tighter	0.46; $\widehat{2.67}$	$1 + x^4y^2$	$1 + x^5 + x^5y^2$	L4+B	2 O/N	$2 \times [216, 4]$
CL-a845ee15/C472	(18, 12)	[432, 8]	5; $\widehat{14}$	Estimate	–	0.46; $\widehat{3.63}$	$1 + x^4y^6 + x^5$	$1 + xy^2$	L4+B	1 O	$2 \times [216, 4]$
CL-09a77c34/C473	(18, 12)	[432, 8]	5; $\widehat{14}$	Estimate	–	0.46; $\widehat{3.63}$	$1 + x^5y^7$	$1 + x^9y^5 + x^{16}$	L4+B	1 G	$2 \times [216, 4]$
CL-a1456a39/C485	(18, 12)	[432, 8]	5; $\widehat{13}$	Estimate	–	0.46; $\widehat{3.13}$	$1 + xy^3 + x^{14}$	$1 + xy^5$	L4+B	1 O	$2 \times [216, 4]$
CL-60b043d7/C477	(18, 12)	[432, 8]	5; $\widehat{12}$	Estimate	Tighter	0.46; $\widehat{2.67}$	$1 + x^2y^2$	$1 + x + x^5y^6$	L4+B	1 N	$2 \times [216, 4]$
CL-9adf9c82/C492	(18, 12)	[432, 8]	5; $\widehat{13}$	Estimate	–	0.46; $\widehat{3.13}$	$1 + x^2y^2$	$1 + x^{11}y^7 + x^{14}y^9$	L4+B	1 G	conn.
CL-6b338767/C484	(18, 12)	[432, 8]	5; $\widehat{13}$	Estimate	–	0.46; $\widehat{3.13}$	$1 + x^2y^3$	$1 + x^4y^{11} + x^{16}y^7$	L4+B	2 O	$2 \times [216, 4]$
CL-e9ccd5a7/C493	(18, 12)	[432, 8]	5; $\widehat{12}$	Estimate	Tighter	0.46; $\widehat{2.67}$	$1 + x^2y^6$	$1 + x^2y^8 + x^{17}y^4$	L4+B	1 G	$2 \times [216, 4]$
CL-4ec75a5c/C488	(18, 12)	[432, 8]	5; $\widehat{13}$	Estimate	–	0.46; $\widehat{3.13}$	$1 + x^{10}y^8 + x^{16}y^3$	$1 + x^4y$	L4+B	2 G/O	$2 \times [216, 4]$
CL-d3cd6d5d/C505	(18, 12)	[432, 8]	5; $\widehat{12}$	Estimate	–	0.46; $\widehat{2.67}$	$1 + x^2y^6$	$1 + xy^5 + x^{15}y$	L4+B	2 G	$2 \times [216, 4]$
CL-802642c3/C483	(18, 12)	[432, 8]	5; $\widehat{12}$	Estimate	–	0.46; $\widehat{2.67}$	$1 + x^2y^6 + x^4y^9$	$1 + y^5$	L4+B	1 O	$2 \times [216, 4]$
CL-47f1818f/C501	(18, 12)	[432, 8]	5; $\widehat{12}$	Estimate	–	0.46; $\widehat{2.67}$	$1 + x^4y + x^{14}y^4$	$1 + y^5$	L4+B	1 O	$2 \times [216, 4]$
CL-05d7189f/C506	(18, 12)	[432, 8]	5; $\widehat{12}$	Estimate	–	0.46; $\widehat{2.67}$	$1 + x^4y^3$	$1 + y^4 + x^8y^2$	L4+B	1 G	$2 \times [216, 4]$
CL-2329ffcc/C489	(18, 12)	[432, 8]	5; $\widehat{10}$	Estimate	Tighter	0.46; $\widehat{1.85}$	$1 + x^5$	$1 + xy^2 + x^{17}y^{10}$	L4+B	5 G/O	$2 \times [216, 4]$
CL-6109dbbe/C478	(18, 12)	[432, 8]	5; $\widehat{12}$	Estimate	–	0.46; $\widehat{2.67}$	$1 + x^5y^2$	$1 + x + x^3y^4$	L4+B	2 O/N	$2 \times [216, 4]$
CL-401c648e/C475	(18, 12)	[432, 8]	[5, 12]	Interval	–	[0.46, 2.67]	$1 + x^5y^3$	$1 + x^2y^{10} + x^6y^2$	L4+I+B	3 G	$2 \times [216, 4]$
CL-eb4e5df6/C482	(18, 12)	[432, 8]	5; $\widehat{12}$	Estimate	–	0.46; $\widehat{2.67}$	$1 + x^5y^5$	$1 + x^{11}y^7 + x^{16}y^8$	L4+B	3 G/O	$2 \times [216, 4]$
CL-deb9e5a7/C500	(18, 12)	[432, 8]	5; $\widehat{12}$	Estimate	–	0.46; $\widehat{2.67}$	$1 + x^6y$	$1 + x^{14}y^6 + x^{16}y^9$	L4+B	1 G	$2 \times [216, 4]$
CL-cde49a87/C471	(18, 12)	[432, 8]	5; $\widehat{12}$	Estimate	–	0.46; $\widehat{2.67}$	$1 + x^6y^5$	$1 + x^2y^8 + x^4y$	L4+B	1 O	$2 \times [216, 4]$
CL-53bfe094/C480	(18, 12)	[432, 8]	5; $\widehat{12}$	Estimate	–	0.46; $\widehat{2.67}$	$1 + x^6y^7$	$1 + x^2y^5 + x^{10}y^8$	L4+B	1 G	$2 \times [216, 4]$
CL-56e2dbbb/C507	(18, 12)	[432, 8]	5; $\widehat{8}$	Estimate	Tighter	0.46; $\widehat{1.19}$	$1 + x^{10}y^8 + x^{14}y^{10}$	$1 + x^5y^5$	L4+B	1 O	$2 \times [216, 4]$
CL-070a0e56/C481	(18, 12)	[432, 8]	5; $\widehat{12}$	Estimate	–	0.46; $\widehat{2.67}$	$1 + x^{11}y^9 + x^{15}y^{11}$	$1 + x^5y^{11}$	L4+B	2 G	$2 \times [216, 4]$
CL-a110e8aa/C502	(18, 12)	[432, 8]	5; $\widehat{9}$	Estimate	Tighter	0.46; $\widehat{1.50}$	$1 + x^3y$	$1 + x^4y^6 + x^5y^{11}$	L4+B	2 G/O	$2 \times [216, 4]$
CL-07986a64/C479	(18, 12)	[432, 8]	5; $\widehat{10}$	Estimate	–	0.46; $\widehat{1.85}$	$1 + x^2y^8 + x^4y^4$	$1 + x^3y$	L4+B	1 G	$2 \times [216, 4]$
CL-2e6fe66f/C499	(18, 12)	[432, 8]	5; $\widehat{9}$	Estimate	–	0.46; $\widehat{1.50}$	$1 + x^2$	$1 + y^{10} + x^2y^5$	L4+B	1 G	$2 \times [216, 4]$
CL-1aaa038c/C476	(18, 12)	[432, 8]	5; $\widehat{9}$	Estimate	–	0.46; $\widehat{1.50}$	$1 + x^2y^3 + x^3y^5$	$1 + x^{10}y^4$	L4+B	1 G	conn.
CL-2af009b4/C496	(18, 12)	[432, 8]	5; $\widehat{9}$	Estimate	–	0.46; $\widehat{1.50}$	$1 + x^2y^5$	$1 + x^6y^{10} + x^{14}y$	L4+B	2 G/O	$2 \times [216, 4]$
CL-0f7f26a5/C504	(18, 12)	[432, 8]	5; $\widehat{9}$	Estimate	–	0.46; $\widehat{1.50}$	$1 + x^4y^4$	$1 + xy^3 + x^3y$	L4+B	1 G	$2 \times [216, 4]$
CL-93e856dd/C498	(18, 12)	[432, 8]	5; $\widehat{9}$	Estimate	–	0.46; $\widehat{1.50}$	$1 + x^4y^4$	$1 + x^3y^7 + x^{14}y^{10}$	L4+B	1 G	$2 \times [216, 4]$
CL-58125eba/C480	(18, 12)	[432, 8]	5; $\widehat{8}$	Estimate	–	0.46; $\widehat{1.19}$	$1 + x^2y^4$	$1 + y + y^2$	L4+B	1 G	$2 \times [216, 4]$
CL-f9fcbea8/C509	(18, 12)	[432, 8]	5; $\widehat{8}$	Estimate	–	0.46; $\widehat{1.19}$	$1 + x^4y^2 + x^{11}y^7$	$1 + x^{17}y^{11}$	L4+B	1 G	$2 \times [216, 4]$
CL-303631c7/C510	(18, 12)	[432, 8]	5; $\widehat{6}$	Estimate	–	0.46; $\widehat{0.67}$	$1 + x^2 + x^4$	$1 + xy$	L4+B	1 G	$2 \times [216, 4]$
CL-3901c61d/C494	(18, 12)	[432, 8]	5; $\widehat{6}$	Estimate	–	0.46; $\widehat{0.67}$	$1 + x^2y^6$	$1 + x^4y^5 + x^{16}y$	L4+B	2 G	$2 \times [216, 4]$
CL-2925a0eb/C491	(18, 12)	[432, 8]	5; $\widehat{6}$	Estimate	–	0.46; $\widehat{0.67}$	$1 + x^3y^2$	$1 + x + x^{17}y^{10}$	L4+B	2 G	$2 \times [216, 4]$
CL-fd352b42/C508	(18, 12)	[432, 8]	5; $\widehat{6}$	Estimate	–	0.46; $\widehat{0.67}$	$1 + x^6y^2$	$1 + x^{10}y^4 + x^{17}y^6$	L4+B	1 G	$2 \times [216, 4]$
CL-afd67a34/C434	(18, 12)	[432, 4]	10	Exact	–	0.93	$1 + x^{14}y^5$	$1 + xy^9 + x^{17}y^3$	M	1 O	conn.
CL-2de98d42/C438	(18, 12)	[432, 4]	4	Exact	–	0.15	$1 + x^4y^3$	$1 + x^3y^4 + x^6y^8$	Exh.	1 G	conn.
CL-ac793cd4/C426	(18, 12)	[432, 4]	5; $\widehat{22}$	Estimate	Tighter	0.23; $\widehat{4.48}$	$1 + xy^3 + x^8y^4$	$1 + x^6y$	L4+B	3 G	conn.
CL-5be73e9c/C434	(18, 12)	[432, 4]	10	Estimate	Exact	0.93	$1 + x^4y^3$	$1 + xy^{11} + x^2y^{10}$	L4+B	2 G	conn.
CL-f5754da6/C453	(18, 12)	[432, 4]	[5, 23]	Interval	–	[0.23, 4.90]	$1 + x^4y^3$	$1 + x^5y^{11} + x^{10}y$	L4+I+W+B	1 G	conn.
CL-b58f52c9/C459	(18, 12)	[432, 4]	5; $\widehat{22}$	Estimate	–	0.23; $\widehat{4.48}$	$1 + xy^9$	$1 + x^6y^2 + x^{13}y^{10}$	L4+B	1 G	conn.
CL-2cf3ba31/C437	(18, 12)	[432, 4]	5; $\widehat{22}$	Estimate	–	0.23; $\widehat{4.48}$	$1 + xy^9 + x^{17}y^6$	$1 + x^{17}y^5$	L4+B	1 G	conn.
CL-a78749f2/C434	(18, 12)	[432, 4]	10	Estimate	Exact	0.93	$1 + xy^{11}$	$1 + x^4y^9 + x^{14}y^3$	L4+B	1 O	conn.
CL-960fac4a/C444	(18, 12)	[432, 4]	5; $\widehat{22}$	Estimate	–	0.23; $\widehat{4.48}$	$1 + x^4y^5$	$1 + x^{17}y^3 + x^{17}y^5$	L4+B	1 G	conn.
CL-2c48572e/C467	(18, 12)	[432, 4]	5; $\widehat{18}$	Estimate	Tighter	0.23; $\widehat{3.00}$	$1 + x^5y^{11} + x^{12}y^{10}$	$1 + x^{17}y^6$	L4+B	1 G	conn.
CL-99b02885/C434	(18, 12)	[432, 4]	10	Estimate	Exact	0.93	$1 + x^5y^5$	$1 + x^2y + x^{16}y^{11}$	L4+B	3 G/O	conn.
CL-9b2cfe69/C458	(18, 12)	[432, 4]	5; $\widehat{18}$	Estimate	Tighter	0.23; $\widehat{3.00}$	$1 + xy^3$	$1 + y + xy^5$	L4+B	1 G	conn.
CL-2e0cf084/C469	(18, 12)	[432, 4]	5; $\widehat{18}$	Estimate	–	0.23; $\widehat{3.00}$	$1 + xy^2$	$1 + x^3y^8 + x^4y^9$	L4+B	2 G/O	conn.
CL-c562025d/C461	(18, 12)	[432, 4]	5; $\widehat{16}$	Estimate	Tighter	0.23; $\widehat{2.37}$	$1 + xy^2 + x^3y^{11}$	$1 + x^5y^8$	L4+B	2 G/O	conn.
CL-7b37e7a6/C445	(18, 12)	[432, 4]	5; $\widehat{18}$	Estimate	–	0.23; $\widehat{3.00}$	$1 + xy^4$	$1 + x^4 + x^6y$	L4+B	1 N	conn.
CL-888cfcaa/C462	(18, 12)	[432, 4]	5; $\widehat{18}$	Estimate	–	0.23; $\widehat{3.00}$	$1 + xy^4$	$1 + x^4y^{11} + x^{16}y^3$	L4+B	1 G	conn.
CL-0f1284f8/C464	(18, 12)	[432, 4]	5; $\widehat{18}$	Estimate	–	0.23; $\widehat{3.00}$	$1 + xy^4$	$1 + x^{13}y^9 + x^{14}y^3$	L4+B	1 G	conn.
CL-70597c08/C446	(18, 12)	[432, 4]	5; $\widehat{18}$	Estimate	–	0.23; $\widehat{3.00}$	$1 + xy^{11} + x^{17}y^4$	$1 + xy^{10}$	L4+B	2 G	conn.
CL-e206740e/C465	(18, 12)	[432, 4]	5; $\widehat{18}$	Estimate	–	0.23; $\widehat{3.00}$	$1 + x^2y^3$	$1 + xy^5 + x^{11}y^{10}$	L4+B	2 G/O	conn.
CL-935f70e2/C452	(18, 12)	[432, 4]	5; $\widehat{18}$	Estimate	–	0.23; $\widehat{3.00}$	$1 + x^3y^{10} + x^4y$	$1 + xy^2$	L4+B	1 O	conn.
CL-754eef8b/C449	(18, 12)	[432, 4]	5; $\widehat{16}$	Estimate	Tighter	0.23; $\widehat{2.37}$	$1 + x^5$	$1 + x^2y^7 + x^4y^2$	L4+B	1 G	conn.
CL-4961ba0d/C440	(18, 12)	[432, 4]	5; $\widehat{18}$	Estimate	–	0.23; $\widehat{3.00}$	$1 + x^5$	$1 + x^4y^5 + x^6y$	L4+B	1 O	conn.
CL-582ece9e/C428	(18, 12)	[432, 4]	5; $\widehat{18}$	Estimate	–	0.23; $\widehat{3.00}$	$1 + x^5y^3$	$1 + x^{11}y^4 + x^{12}y^8$	L4+B	1 O	conn.
CL-64d76631/C432	(18, 12)	[432, 4]	5; $\widehat{18}$	Estimate	–	0.23; $\widehat{3.00}$	$1 + x^5y^4$	$1 + y^5 + x^{17}y^{11}$	L4+B	1 G	conn.
CL-c459a3a7/C457	(18, 12)	[432, 4]	5; $\widehat{18}$	Estimate	–	0.23; $\widehat{3.00}$	$1 + x^7y^5 + x^{16}y^3$	$1 + x^5y^2$	L4+B	1 O	conn.
CL-9687b770/C455	(18, 12)	[432, 4]	5; $\widehat{18}$	Estimate	–	0.23; $\widehat{3.00}$	$1 + x^8y^6 + x^{14}y$	$1 + xy^4$	L4+B	1 G	conn.
CL-d78337f6/C436	(18, 12)	[432, 4]	5; $\widehat{16}$	Estimate	–	0.23; $\widehat{2.37}$	$1 + y^4 + xy^{10}$	$1 + x^4y^5$	L4+B	1 O	conn.
CL-7a6583b3/C447	(18, 12)	[432, 4]	5; $\widehat{15}$	Estimate	Tighter	0.23; $\widehat{2.08}$	$1 + xy^6 + x^3y^5$	$1 + xy^5$	L4+B	3 G	conn.

TABLE S6 (continued)

ID/class	(ℓ, m)	$[n, k]$	d	Own evidence Class effect FOM			A	B	Evid.	Obs. Attr.		Decomp.
CL-5e1d4e3b/C450	(18, 12)	$[432, 4]$	$[5, 16]$	Interval	–	$[0.23, 2.37]$	$1 + x^5 y^9$	$1 + x^{11} y^8 + x^{15} y^7$	L4+I+B	1	G	conn.
CL-6e6cae4b/C466	(18, 12)	$[432, 4]$	$[5, 12]$	Estimate	Tighter	$[0.23, 1.33]$	$1 + x^{10} y^7 + x^{13} y^5$	$1 + x^5$	L4+B	1	O	conn.
CL-7d5cd2d4/C454	(18, 12)	$[432, 4]$	$5; \widehat{12}$	Estimate	–	$0.23; \widehat{1.33}$	$1 + y$	$1 + x + x^2$	L4+B	1	G	conn.
CL-ba633f94/C442	(18, 12)	$[432, 4]$	$5; \widehat{12}$	Estimate	–	$0.23; \widehat{1.33}$	$1 + x^2 y^5$	$1 + xy^2 + x^2 y^4$	L4+B	1	G	conn.
CL-4065bf35/C435	(18, 12)	$[432, 4]$	$5; \widehat{12}$	Estimate	–	$0.23; \widehat{1.33}$	$1 + x^3 y^3$	$1 + x + x^3 y^4$	L4+B	3	O/N	conn.
CL-28f3c8be/C441	(18, 12)	$[432, 4]$	$5; \widehat{12}$	Estimate	–	$0.23; \widehat{1.33}$	$1 + x^3 y^3$	$1 + x^4 y^4 + x^5$	L4+B	1	N	conn.
CL-2c380f36/C425	(18, 12)	$[432, 4]$	$5; \widehat{12}$	Estimate	–	$0.23; \widehat{1.33}$	$1 + x^3 y^5$	$1 + xy^{10} + x^{17} y$	L4+B	1	G	conn.
CL-89827bd2/C429	(18, 12)	$[432, 4]$	$[5, 12]$	Interval	–	$[0.23, 1.33]$	$1 + x^4 y^{11}$	$1 + x^2 y^2 + x^5$	L4+I+B	1	G	conn.
CL-05950d9c/C430	(18, 12)	$[432, 4]$	$5; \widehat{12}$	Estimate	–	$0.23; \widehat{1.33}$	$1 + x^4 y^{11} + x^{11} y^{11}$	$1 + y^{11}$	L4+B	1	G	conn.
CL-6c6932a1/C456	(18, 12)	$[432, 4]$	$5; \widehat{12}$	Estimate	–	$0.23; \widehat{1.33}$	$1 + x^6 y$	$1 + xy^4 + x^2$	L4+B	2	O/N	conn.
CL-495ff19e/C460	(18, 12)	$[432, 4]$	$5; \widehat{12}$	Estimate	–	$0.23; \widehat{1.33}$	$1 + x^6 y^5$	$1 + x^4 y^{11} + x^5 y^5$	L4+B	1	G	conn.
CL-2faf8b1a/C433	(18, 12)	$[432, 4]$	$5; \widehat{12}$	Estimate	–	$0.23; \widehat{1.33}$	$1 + x^6 y^7$	$1 + x^5 y^{11} + x^{10} y^7$	L4+B	1	G	conn.
CL-373c7ffa/C470	(18, 12)	$[432, 4]$	$5; \widehat{12}$	Estimate	–	$0.23; \widehat{1.33}$	$1 + x^{10} y^2 + x^{11} y^2$	$1 + x^6 y$	L4+B	2	G/O	conn.
CL-f334fecb/C434	(18, 12)	$[432, 4]$	10	Estimate	Exact	0.93	$1 + xy^{11} + x^2 y^{10}$	$1 + x^{16} y^9$	L4+B	2	G	conn.
CL-4dc3dd1c/C448	(18, 12)	$[432, 4]$	$5; \widehat{6}$	Estimate	Tighter	$0.23; \widehat{0.33}$	$1 + x^5 + x^{16} y^3$	$1 + x^3 y^2$	L4+B	2	G/O	conn.
CL-ed2078cc/C451	(18, 12)	$[432, 4]$	$5; \widehat{9}$	Estimate	–	$0.23; \widehat{0.75}$	$1 + x^2 y^5$	$1 + xy^3 + x^{13} y^{11}$	L4+B	3	G/O	conn.
CL-11f56764/C439	(18, 12)	$[432, 4]$	$5; \widehat{8}$	Estimate	Tighter	$0.23; \widehat{0.59}$	$1 + x^3 y^2 + x^{16} y^{11}$	$1 + xy$	L4+B	1	G	conn.
CL-0ad5b376/C463	(18, 12)	$[432, 4]$	$5; \widehat{8}$	Estimate	–	$0.23; \widehat{0.59}$	$1 + x$	$1 + y + y^2$	L4+B	1	G	conn.
CL-3bee8f1a/C424	(18, 12)	$[432, 4]$	$5; \widehat{8}$	Estimate	–	$0.23; \widehat{0.59}$	$1 + x^4 y^5$	$1 + x^9 y + x^9 y^{11}$	L4+B	1	O	conn.
CL-5a1f8e7d/C431	(18, 12)	$[432, 4]$	$5; \widehat{6}$	Estimate	–	$0.23; \widehat{0.33}$	$1 + xy^4 + x^5 y^3$	$1 + x^3 y^4$	L4+B	1	O	conn.
CL-0266a837/C443	(18, 12)	$[432, 4]$	$5; \widehat{6}$	Estimate	–	$0.23; \widehat{0.33}$	$1 + x^3 y^2$	$1 + x^7 y^3 + x^{17} y$	L4+B	1	G	conn.
CL-9074474c/C427	(18, 12)	$[432, 4]$	$5; \widehat{6}$	Estimate	–	$0.23; \widehat{0.33}$	$1 + x^3 y^4$	$1 + x + x^2 y^3$	L4+B	2	O/N	conn.
CL-481d459f/C468	(18, 12)	$[432, 4]$	$5; \widehat{6}$	Estimate	–	$0.23; \widehat{0.33}$	$1 + x^{15} y^4$	$1 + x^2 y^{11} + x^{13} y^5$	L4+B	1	G	conn.

TABLE S7: PBB-small compact catalogue (151 displayed of 151 retained proposals; 39 presentation classes). The table includes every proposal in an exact class and one deterministic representative of every nonexact class. Own evidence and Evid. describe the proposal before class merging; Class effect reports when merging makes the displayed d and FOM exact, tighter, or rigorously supported. Column abbreviations, filtering, and evidence conventions are defined above.

ID/class	(ℓ, m)	$[n, k]$	d	Own evidence	Class effect	FOM	A	B	C	D	Evid.	Obs.	Attr.	Decomp.
PS-5feb6ebb/P013	(6, 6)	[72, 12]	3	Exact	—	1.50	$1 + x^3 y^3$	$1 + xy^3 + x^2$	0	$1 + x^2$	Exh.	1	G	$6 \times [12, 2]$
PS-64d64fd9/P012	(6, 6)	[72, 12]	3	Exact	—	1.50	$1 + x^3 y^3$	$1 + xy^3 + x^2$	0	xy^3	Exh.	8	G/O	$6 \times [12, 2]$
PS-106b8669/P014	(6, 6)	[72, 12]	3	Exact	—	1.50	$1 + x^3 y^3$	$1 + xy^3 + x^2$	$1 + x^3 y^3$	$1 + x^2$	Exh.	2	G/N	$6 \times [12, 2]$
PS-f4956d3c/P011	(6, 6)	[72, 12]	3	Exact	—	1.50	$1 + x^3 y^3$	$1 + xy^3 + x^2$	$1 + x^3 y^3$	xy^3	Exh.	1	G	$6 \times [12, 2]$
PS-edb2da66/P013	(6, 6)	[72, 12]	3	Exact	—	1.50	$1 + x^3 y^3$	$1 + x^4 + x^5 y^3$	0	$1 + x^4$	Exh.	1	G	$6 \times [12, 2]$
PS-1a20f6c1/P012	(6, 6)	[72, 12]	3	Exact	—	1.50	$1 + x^3 y^3$	$1 + x^4 + x^5 y^3$	0	$x^5 y^3$	Exh.	3	G	$6 \times [12, 2]$
PS-686df635/P014	(6, 6)	[72, 12]	3	Exact	—	1.50	$1 + x^3 y^3$	$1 + x^4 + x^5 y^3$	$1 + x^3 y^3$	$1 + x^4$	Exh.	1	G	$6 \times [12, 2]$
PS-66471642/P011	(6, 6)	[72, 12]	3	Exact	—	1.50	$1 + x^3 y^3$	$1 + x^4 + x^5 y^3$	$1 + x^3 y^3$	$x^5 y^3$	Exh.	1	G	$6 \times [12, 2]$
PS-bb41fc22/P004	(6, 6)	[72, 4]	4	Exact	—	0.89	$1 + y$	$1 + x + x^2$	$1 + y$	x	Exh.	49	G/O/F	conn.
PS-dd65d6ed/P004	(6, 6)	[72, 4]	4	Exact	—	0.89	$1 + x + x^2$	$1 + y$	x	$1 + y$	Exh.	37	G/O/F	conn.
PS-08fe6a9e/P001	(6, 6)	[72, 4]	4	Exact	—	0.89	$1 + xy$	$1 + xy^3 + x^2$	0	xy^3	Exh.	14	G/O/N	$2 \times [36, 2]$
PS-db456c42/P003	(6, 6)	[72, 4]	4	Exact	—	0.89	$1 + xy$	$1 + x^2 + x^4 y^3$	0	$1 + x^2$	Exh.	1	G	conn.
PS-327fdbe9/P003	(6, 6)	[72, 4]	4	Exact	—	0.89	$1 + xy$	$1 + x^2 y^3 + x^4$	0	$1 + x^4$	Exh.	16	G	conn.
PS-a18803a6/P006	(6, 6)	[72, 4]	4	Exact	—	0.89	$1 + xy$	$1 + x^2 y^3 + x^4$	0	$x^2 y^3$	Exh.	1	G	conn.
PS-ceb60d35/P003	(6, 6)	[72, 4]	4	Exact	—	0.89	$1 + xy^2$	$1 + xy^3 + x^2$	0	$1 + x^2$	Exh.	1	G	conn.
PS-6d4940be/P006	(6, 6)	[72, 4]	4	Exact	—	0.89	$1 + xy^2$	$1 + xy^3 + x^2$	0	xy^3	Exh.	5	G	conn.
PS-a72ff06a/P004	(6, 6)	[72, 4]	4	Exact	—	0.89	$1 + xy^2$	$1 + xy^3 + x^2$	$1 + xy^2$	xy^3	Exh.	17	G	conn.
PS-44aa1f41/P006	(6, 6)	[72, 4]	4	Exact	—	0.89	$1 + xy^2$	$1 + x^2 + x^4 y^3$	0	$x^4 y^3$	Exh.	1	G	conn.
PS-34d654cf/P003	(6, 6)	[72, 4]	4	Exact	—	0.89	$1 + xy^2$	$1 + x^2 y^3 + x^4$	0	$1 + x^4$	Exh.	4	G	conn.
PS-34a1c7a7/P006	(6, 6)	[72, 4]	4	Exact	—	0.89	$1 + xy^2$	$1 + x^2 y^3 + x^4$	0	$x^2 y^3$	Exh.	9	G	conn.
PS-5132ad68/P006	(6, 6)	[72, 4]	4	Exact	—	0.89	$1 + xy^2$	$1 + x^4 + x^5 y^3$	0	$x^5 y^3$	Exh.	27	G/O	conn.
PS-0d465075/P004	(6, 6)	[72, 4]	4	Exact	—	0.89	$1 + xy^4$	$1 + x^2 + x^4 y^3$	$1 + xy^4$	$x^4 y^3$	Exh.	11	G	conn.
PS-6612b0d7/P006	(6, 6)	[72, 4]	4	Exact	—	0.89	$1 + xy^4$	$1 + x^2 y^3 + x^4$	0	$x^2 y^3$	Exh.	4	G	conn.
PS-1db1d8fa/P002	(6, 6)	[72, 4]	4	Exact	—	0.89	$1 + xy^4$	$1 + x^4 + x^5 y^3$	$1 + xy^4$	$1 + x^4$	Exh.	71	G/O	conn.
PS-231b48e4/P004	(6, 6)	[72, 4]	4	Exact	—	0.89	$1 + xy^4$	$1 + x^4 + x^5 y^3$	$1 + xy^4$	$x^5 y^3$	Exh.	12	G	conn.
PS-264133e8/P003	(6, 6)	[72, 4]	4	Exact	—	0.89	$1 + x^2 y$	$1 + xy^3 + x^2$	0	$1 + x^2$	Exh.	1	G	conn.
PS-d9bb8009/P001	(6, 6)	[72, 4]	4	Exact	—	0.89	$1 + x^2 y$	$1 + x^2 y^3 + x^4$	0	$x^2 y^3$	Exh.	47	G/O	$2 \times [36, 2]$
PS-8095943f/P006	(6, 6)	[72, 4]	4	Exact	—	0.89	$1 + x^2 y$	$1 + x^4 + x^5 y^3$	0	$x^5 y^3$	Exh.	5	G	conn.
PS-ab75ae2d/P009	(6, 6)	[72, 4]	4	Exact	—	0.89	$1 + x^2 y^5$	$1 + x^2 + x^4 y^3$	$1 + x^2 y^5$	$1 + x^2$	Exh.	2	G	$2 \times [36, 2]$
PS-9b4e5941/P001	(6, 6)	[72, 4]	4	Exact	—	0.89	$1 + x^3 y$	$1 + xy^3 + x^2$	0	xy^3	Exh.	11	G/O	$2 \times [36, 2]$
PS-e3c4d761/P003	(6, 6)	[72, 4]	4	Exact	—	0.89	$1 + x^3 y$	$1 + x^2 + x^4 y^3$	0	$1 + x^2$	Exh.	1	G	conn.
PS-99d2a217/P002	(6, 6)	[72, 4]	4	Exact	—	0.89	$1 + x^3 y$	$1 + x^2 + x^4 y^3$	$1 + x^3 y$	$1 + x^2$	Exh.	31	G	conn.
PS-3a934cc1/P003	(6, 6)	[72, 4]	4	Exact	—	0.89	$1 + x^3 y^2$	$1 + xy^3 + x^2$	0	$1 + x^2$	Exh.	78	G/O	conn.
PS-a39ab3d8/P006	(6, 6)	[72, 4]	4	Exact	—	0.89	$1 + x^3 y^2$	$1 + xy^3 + x^2$	0	xy^3	Exh.	98	G/O	conn.
PS-9d8b733c/P004	(6, 6)	[72, 4]	4	Exact	—	0.89	$1 + x^3 y^2$	$1 + x^2 y^3 + x^4$	$1 + x^3 y^2$	$x^2 y^3$	Exh.	2	G	conn.
PS-e70df808/P003	(6, 6)	[72, 4]	4	Exact	—	0.89	$1 + x^3 y^4$	$1 + x^2 + x^4 y^3$	0	$1 + x^2$	Exh.	2	G	conn.
PS-09833b6a/P002	(6, 6)	[72, 4]	4	Exact	—	0.89	$1 + x^3 y^4$	$1 + x^2 + x^4 y^3$	$1 + x^3 y^4$	$1 + x^2$	Exh.	2	G	conn.

TABLE S7 (continued)

ID/class	(ℓ, m)	$[n, k]$	d	Own	evidence	Class	effect	FOM	A	B	C	D	Evid.	Obs.	Attr.	Decomp.
PS-350fe455/P002	(6, 6)	[72, 4]	4	Exact	—	0.89	$1 + x^3 y^4$	$1 + x^2 y^3 + x^4$	$1 + x^3 y^4$	$1 + x^4$	Exh.	2	G	conn.		
PS-29b7edf5/P001	(6, 6)	[72, 4]	4	Exact	—	0.89	$1 + x^3 y^5$	$1 + xy^3 + x^2$	0	xy^3	Exh.	1	G	$2 \times [36, 2]$		
PS-9bf6af1e/P009	(6, 6)	[72, 4]	4	Exact	—	0.89	$1 + x^3 y^5$	$1 + xy^3 + x^2$	$1 + x^3 y^5$	$1 + x^2$	Exh.	33	G	$2 \times [36, 2]$		
PS-e53b0829/P006	(6, 6)	[72, 4]	4	Exact	—	0.89	$1 + x^4 y$	$1 + xy^3 + x^2$	0	xy^3	Exh.	3	G/O	conn.		
PS-ecc218b0/P002	(6, 6)	[72, 4]	4	Exact	—	0.89	$1 + x^4 y$	$1 + xy^3 + x^2$	$1 + x^4 y$	$1 + x^2$	Exh.	14	G	conn.		
PS-a149b558/P006	(6, 6)	[72, 4]	4	Exact	—	0.89	$1 + x^4 y^5$	$1 + xy^3 + x^2$	0	xy^3	Exh.	2	G	conn.		
PS-d181cf11/P005	(6, 6)	[72, 4]	4	Exact	—	0.89	$1 + x^5 y$	$1 + xy^3 + x^2$	0	$1 + x^2$	Exh.	3	G	$2 \times [36, 2]$		
PS-f2f4d5a2/P001	(6, 6)	[72, 4]	4	Exact	—	0.89	$1 + x^5 y$	$1 + x^4 + x^5 y^3$	0	$x^5 y^3$	Exh.	3	G	$2 \times [36, 2]$		
PS-a1dab64c/P006	(6, 6)	[72, 4]	4	Exact	—	0.89	$1 + x^5 y^2$	$1 + x^2 + x^4 y^3$	0	$x^4 y^3$	Exh.	5	G	conn.		
PS-337b23a8/P002	(6, 6)	[72, 4]	4	Exact	—	0.89	$1 + x^5 y^2$	$1 + x^2 + x^4 y^3$	$1 + x^5 y^2$	$1 + x^2$	Exh.	2	G	conn.		
PS-c4a4cc2d/P004	(6, 6)	[72, 4]	4	Exact	—	0.89	$1 + x^5 y^2$	$1 + x^2 + x^4 y^3$	$1 + x^5 y^2$	$x^4 y^3$	Exh.	2	G	conn.		
PS-1434d4e7/P006	(6, 6)	[72, 4]	4	Exact	—	0.89	$1 + x^5 y^2$	$1 + x^4 + x^5 y^3$	0	$x^5 y^3$	Exh.	2	G	conn.		
PS-46a007cb/P007	(6, 6)	[72, 4]	3	Exact	—	0.50	$1 + x^2 y^2$	$1 + xy^3 + x^2$	$1 + x^2 y^2$	$1 + x^2$	Exh.	26	G/O	$2 \times [36, 2]$		
PS-9703d323/P010	(6, 6)	[72, 4]	3	Exact	—	0.50	$1 + x^2 y^2$	$1 + x^2 + x^4 y^3$	0	$x^4 y^3$	Exh.	17	G/O	$2 \times [36, 2]$		
PS-ec4998ee/P008	(6, 6)	[72, 4]	3	Exact	—	0.50	$1 + x^2 y^2$	$1 + x^2 y^3 + x^4$	$1 + x^2 y^2$	$x^2 y^3$	Exh.	23	G/O	$2 \times [36, 2]$		
PS-f872aa5b/P010	(6, 6)	[72, 4]	3	Exact	—	0.50	$1 + x^2 y^4$	$1 + xy^3 + x^2$	0	xy^3	Exh.	1	G	$2 \times [36, 2]$		
PS-3db41c4a/P007	(6, 6)	[72, 4]	3	Exact	—	0.50	$1 + x^2 y^4$	$1 + x^2 y^3 + x^4$	$1 + x^2 y^4$	$1 + x^4$	Exh.	1	G	$2 \times [36, 2]$		
PS-25a74b10/P010	(6, 6)	[72, 4]	3	Exact	—	0.50	$1 + x^4 y^4$	$1 + x^2 y^3 + x^4$	0	$x^2 y^3$	Exh.	11	G	$2 \times [36, 2]$		
PS-01443595/P008	(6, 6)	[72, 4]	3	Exact	—	0.50	$1 + x^4 y^4$	$1 + x^2 y^3 + x^4$	$1 + x^4 y^4$	$x^2 y^3$	Exh.	5	G	$2 \times [36, 2]$		
PS-dcdb4997/P015	(6, 9)	[108, 2]	4	Exact	—	0.30	$1 + y$	$1 + x + x^2$	$1 + y$	x	Exh.	162	G/O/N/F	conn.		
PS-02124a53/P015	(6, 9)	[108, 2]	4	Exact	—	0.30	$1 + x + x^2$	$1 + y$	x	$1 + y$	Exh.	162	G/O/N/F	conn.		
PS-194f5d29/P024	(6, 10)	[120, 20]	3	Exact	—	1.50	$1 + x^3 y^5$	$1 + xy^5 + x^2$	0	$1 + x^2$	Exh.	1	G	$10 \times [12, 2]$		
PS-3f8d6ded/P026	(6, 10)	[120, 20]	3	Exact	—	1.50	$1 + x^3 y^5$	$1 + xy^5 + x^2$	0	xy^5	Exh.	11	G/O	$10 \times [12, 2]$		
PS-5259dec3/P025	(6, 10)	[120, 20]	3	Exact	—	1.50	$1 + x^3 y^5$	$1 + xy^5 + x^2$	$1 + x^3 y^5$	$1 + x^2$	Exh.	7	G/O	$10 \times [12, 2]$		
PS-78481db9/P023	(6, 10)	[120, 20]	3	Exact	—	1.50	$1 + x^3 y^5$	$1 + xy^5 + x^2$	$1 + x^3 y^5$	xy^5	Exh.	1	G	$10 \times [12, 2]$		
PS-2e83cd7f/P024	(6, 10)	[120, 20]	3	Exact	—	1.50	$1 + x^3 y^5$	$1 + x^4 + x^5 y^5$	0	$1 + x^4$	Exh.	1	G	$10 \times [12, 2]$		
PS-4ab34380/P026	(6, 10)	[120, 20]	3	Exact	—	1.50	$1 + x^3 y^5$	$1 + x^4 + x^5 y^5$	0	$x^5 y^5$	Exh.	79	G/O	$10 \times [12, 2]$		
PS-94315715/P025	(6, 10)	[120, 20]	3	Exact	—	1.50	$1 + x^3 y^5$	$1 + x^4 + x^5 y^5$	$1 + x^3 y^5$	$1 + x^4$	Exh.	1	G	$10 \times [12, 2]$		
PS-5ff69f0e/P023	(6, 10)	[120, 20]	3	Exact	—	1.50	$1 + x^3 y^5$	$1 + x^4 + x^5 y^5$	$1 + x^3 y^5$	$x^5 y^5$	Exh.	1	G	$10 \times [12, 2]$		
PS-9828cd17/P022	(6, 10)	[120, 4]	4	Exact	—	0.53	$1 + y$	$1 + x + x^2$	$1 + y$	x	Exh.	37	G/N/F	conn.		
PS-af58ae1f/P022	(6, 10)	[120, 4]	4	Exact	—	0.53	$1 + x + x^2$	$1 + y$	x	$1 + y$	Exh.	70	G/O/N/F	conn.		
PS-240fa602/P019	(6, 10)	[120, 4]	4	Exact	—	0.53	$1 + x^3 y$	$1 + xy^5 + x^2$	0	xy^5	Exh.	59	G/O	$2 \times [60, 2]$		
PS-0c54bc34/P017	(6, 10)	[120, 4]	4	Exact	—	0.53	$1 + x^3 y$	$1 + x^2 + x^4 y^5$	0	$x^4 y^5$	Exh.	6	G	conn.		
PS-0279cf6d/P020	(6, 10)	[120, 4]	4	Exact	—	0.53	$1 + x^3 y$	$1 + x^2 + x^4 y^5$	$1 + x^3 y$	$1 + x^2$	Exh.	2	G	conn.		
PS-935a9444/P017	(6, 10)	[120, 4]	4	Exact	—	0.53	$1 + x^3 y$	$1 + x^2 y^5 + x^4$	0	$x^2 y^5$	Exh.	2	G	conn.		
PS-1eac6e27/P022	(6, 10)	[120, 4]	4	Exact	—	0.53	$1 + x^3 y$	$1 + x^2 y^5 + x^4$	$1 + x^3 y$	$x^2 y^5$	Exh.	65	G/O	conn.		
PS-d3184db2/P019	(6, 10)	[120, 4]	4	Exact	—	0.53	$1 + x^3 y$	$1 + x^4 + x^5 y^5$	0	$x^5 y^5$	Exh.	12	G	$2 \times [60, 2]$		
PS-6d57d019/P021	(6, 10)	[120, 4]	4	Exact	—	0.53	$1 + x^3 y^2$	$1 + xy^5 + x^2$	0	$1 + x^2$	Exh.	3	G	conn.		
PS-3700f610/P017	(6, 10)	[120, 4]	4	Exact	—	0.53	$1 + x^3 y^2$	$1 + xy^5 + x^2$	0	xy^5	Exh.	6	G	conn.		
PS-bfe0e15d/P020	(6, 10)	[120, 4]	4	Exact	—	0.53	$1 + x^3 y^2$	$1 + xy^5 + x^2$	$1 + x^3 y^2$	$1 + x^2$	Exh.	2	G	conn.		
PS-5bb63551/P022	(6, 10)	[120, 4]	4	Exact	—	0.53	$1 + x^3 y^2$	$1 + xy^5 + x^2$	$1 + x^3 y^2$	xy^5	Exh.	2	G	conn.		
PS-50d05cf8/P017	(6, 10)	[120, 4]	4	Exact	—	0.53	$1 + x^3 y^2$	$1 + x^2 + x^4 y^5$	0	$x^4 y^5$	Exh.	5	G	conn.		
PS-76c46f14/P020	(6, 10)	[120, 4]	4	Exact	—	0.53	$1 + x^3 y^2$	$1 + x^2 + x^4 y^5$	$1 + x^3 y^2$	$1 + x^2$	Exh.	5	G	conn.		
PS-4d498276/P022	(6, 10)	[120, 4]	4	Exact	—	0.53	$1 + x^3 y^2$	$1 + x^2 + x^4 y^5$	$1 + x^3 y^2$	$x^4 y^5$	Exh.	2	G	conn.		
PS-2ba743ef/P021	(6, 10)	[120, 4]	4	Exact	—	0.53	$1 + x^3 y^2$	$1 + x^2 y^5 + x^4$	0	$1 + x^4$	Exh.	21	G	conn.		
PS-a695189e/P020	(6, 10)	[120, 4]	4	Exact	—	0.53	$1 + x^3 y^2$	$1 + x^2 y^5 + x^4$	$1 + x^3 y^2$	$1 + x^4$	Exh.	2	G	conn.		
PS-08b59088/P022	(6, 10)	[120, 4]	4	Exact	—	0.53	$1 + x^3 y^2$	$1 + x^2 y^5 + x^4$	$1 + x^3 y^2$	$x^2 y^5$	Exh.	2	G	conn.		
PS-c9a7824f/P021	(6, 10)	[120, 4]	4	Exact	—	0.53	$1 + x^3 y^2$	$1 + x^4 + x^5 y^5$	0	$1 + x^4$	Exh.	1	G	conn.		
PS-d288ab92/P017	(6, 10)	[120, 4]	4	Exact	—	0.53	$1 + x^3 y^2$	$1 + x^4 + x^5 y^5$	0	$x^5 y^5$	Exh.	2	G	conn.		
PS-4592e602/P020	(6, 10)	[120, 4]	4	Exact	—	0.53	$1 + x^3 y^2$	$1 + x^4 + x^5 y^5$	$1 + x^3 y^2$	$1 + x^4$	Exh.	15	G	conn.		
PS-030a6803/P019	(6, 10)	[120, 4]	4	Exact	—	0.53	$1 + x^3 y^3$	$1 + xy^5 + x^2$	0	xy^5	Exh.	73	G/O	$2 \times [60, 2]$		
PS-a4a03513/P018	(6, 10)	[120, 4]	4	Exact	—	0.53	$1 + x^3 y^3$	$1 + xy^5 + x^2$	$1 + x^3 y^3$	xy^5	Exh.	27	G	$2 \times [60, 2]$		
PS-06e1c26a/P022	(6, 10)	[120, 4]	4	Exact	—	0.53	$1 + x^3 y^3$	$1 + x^2 + x^4 y^5$	$1 + x^3 y^3$	$x^4 y^5$	Exh.	9	G	conn.		
PS-0149b587/P021	(6, 10)	[120, 4]	4	Exact	—	0.53	$1 + x^3 y^3$	$1 + x^2 y^5 + x^4$	0	$1 + x^4$	Exh.	14	G	conn.		
PS-a5976097/P017	(6, 10)	[120, 4]	4	Exact	—	0.53	$1 + x^3 y^3$	$1 + x^2 y^5 + x^4$	0	$x^2 y^5$	Exh.	2	G	conn.		
PS-c64d05e8/P016	(6, 10)	[120, 4]	4	Exact	—	0.53	$1 + x^3 y^3$	$1 + x^4 + x^5 y^5$	0	$1 + x^4$	Exh.	2	G	$2 \times [60, 2]$		
PS-27a10a50/P017	(6, 10)	[120, 4]	4	Exact	—	0.53	$1 + x^3 y^4$	$1 + xy^5 + x^2$	0	xy^5	Exh.	4	G/O	conn.		
PS-965d4a57/P017	(6, 10)	[120, 4]	4	Exact	—	0.53	$1 + x^3 y^4$	$1 + x^2 y^5 + x^4$	0	$x^2 y^5$	Exh.	82	G/O	conn.		
PS-5f4ec5c0/P022	(6, 10)	[120, 4]	4	Exact	—	0.53	$1 + x^3 y^4$	$1 + x^2 y^5 + x^4$	$1 + x^3 y^4$	$x^2 y^5$	Exh.	2	G	conn.		
PS-2f58d2c0/P017	(6, 10)	[120, 4]	4	Exact	—	0.53	$1 + x^3 y^4$	$1 + x^4 + x^5 y^5$	0	$x^5 y^5$	Exh.	2	G	conn.		
PS-5898a489/P020	(6, 10)	[120, 4]	4	Exact	—	0.53	$1 + x^3 y^4$	$1 + x^4 + x^5 y^5$	$1 + x^3 y^4$	$1 + x^4$	Exh.	10	G	conn.		
PS-2ab9e3da/P022	(6, 10)	[120, 4]	4	Exact	—	0.53	$1 + x^3 y^4$	$1 + x^4 + x^5 y^5$	$1 + x^3 y^4$	$x^5 y^5$	Exh.	16	G	conn.		
PS-1d83fe20/P022	(6, 10)	[120, 4]	4	Exact	—	0.53	$1 + x^3 y^6$	$1 + x^2 + x^4 y^5$	$1 + x^3 y^6$	$x^4 y^5$	Exh.	44	G/O	conn.		
PS-49bdead8/P020	(6, 10)	[120, 4]	4	Exact	—	0.53	$1 + x^3 y^6$	$1 + x^2 y^5 + x^4$	$1 + x^3 y^6$	$1 + x^4$	Exh.	11	G	conn.		
PS-c1a685eb/P022	(6, 10)	[120, 4]	4	Exact	—	0.53	$1 + x^3 y^6$	$1 + x^2 y^5 + x^4$	$1 + x^3 y^6$	$x^2 y^5$	Exh.	2	G	conn.		
PS-85429f03/P038	(9, 8)	[144, 8]	4	Exact	—	0.89	$1 + x^3 y^2$	$1 + x + x^5 y^4$	0	$1 + x$	Exh.	3	G/O	$2 \times [72, 4]$		
PS-0b761a73/P032</																

TABLE S7 (continued)

ID/class	(ℓ, m)	$[n, k]$	d	Own evidence	Class effect	FOM	A	B	C	D	Evid.	Obs.	Attr.	Decomp.
PS-86a549d8/P032	(9, 8)	[144, 8]	4	Exact	—	0.89	$1 + x^3 y^2$	$1 + x^2 y^4 + x^4$	0	$x^2 y^4$	Exh.	7	G	$2 \times [72, 4]$
PS-24559419/P038	(9, 8)	[144, 8]	4	Exact	—	0.89	$1 + x^3 y^6$	$1 + x + x^5 y^4$	0	$1 + x$	Exh.	71	G/O	$2 \times [72, 4]$
PS-ad4a6ab8/P034	(9, 8)	[144, 8]	4	Exact	—	0.89	$1 + x^3 y^6$	$1 + xy^4 + x^2$	$1 + x^3 y^6$	$1 + x^2$	Exh.	2	G	$2 \times [72, 4]$
PS-01bb55ce1/P033	(9, 8)	[144, 8]	4	Exact	—	0.89	$1 + x^3 y^6$	$1 + xy^4 + x^2$	$1 + x^3 y^6$	xy^4	Exh.	5	G	$2 \times [72, 4]$
PS-b23256e8/P032	(9, 8)	[144, 8]	4	Exact	—	0.89	$1 + x^6 y^2$	$1 + x + x^5 y^4$	0	$x^5 y^4$	Exh.	3	G	$2 \times [72, 4]$
PS-ec306f78/P032	(9, 8)	[144, 8]	4	Exact	—	0.89	$1 + x^6 y^2$	$1 + xy^4 + x^2$	0	xy^4	Exh.	78	G/O	$2 \times [72, 4]$
PS-db11ff5e/P032	(9, 8)	[144, 8]	4	Exact	—	0.89	$1 + x^6 y^6$	$1 + x + x^5 y^4$	0	$x^5 y^4$	Exh.	83	G/O	$2 \times [72, 4]$
PS-c2455d37/P037	(9, 8)	[144, 8]	3	Exact	—	0.50	$1 + x^3 y^4$	$1 + x + x^5 y^4$	0	$1 + x$	Exh.	3	G/O	$4 \times [36, 2]$
PS-c94e786d/P036	(9, 8)	[144, 8]	3	Exact	—	0.50	$1 + x^3 y^4$	$1 + x + x^5 y^4$	0	$x^5 y^4$	Exh.	13	G/O	$4 \times [36, 2]$
PS-88f1815a/P036	(9, 8)	[144, 8]	3	Exact	—	0.50	$1 + x^3 y^4$	$1 + xy^4 + x^2$	0	xy^4	Exh.	13	G/O	$4 \times [36, 2]$
PS-ad3f6b0a/P035	(9, 8)	[144, 8]	3	Exact	—	0.50	$1 + x^3 y^4$	$1 + xy^4 + x^2$	$1 + x^3 y^4$	$1 + x^2$	Exh.	66	G/O	$4 \times [36, 2]$
PS-986012bb/P036	(9, 8)	[144, 8]	3	Exact	—	0.50	$1 + x^3 y^4$	$1 + x^2 y^4 + x^4$	0	$x^2 y^4$	Exh.	1	G	$4 \times [36, 2]$
PS-9a14d4a6/P036	(9, 8)	[144, 8]	3	Exact	—	0.50	$1 + x^6 y^4$	$1 + xy^4 + x^2$	0	xy^4	Exh.	9	G	$4 \times [36, 2]$
PS-24a01f0d/P037	(9, 8)	[144, 8]	3	Exact	—	0.50	$1 + x^6 y^4$	$1 + x^2 y^4 + x^4$	0	$1 + x^4$	Exh.	64	G/O	$4 \times [36, 2]$
PS-7bbac3f/P035	(9, 8)	[144, 8]	3	Exact	—	0.50	$1 + x^6 y^4$	$1 + x^2 y^4 + x^4$	$1 + x^6 y^4$	$1 + x^4$	Exh.	4	G	$4 \times [36, 2]$
PS-b9bd7fa4/P027	(9, 8)	[144, 4]	6	Exact	—	1.00	$1 + y$	$1 + x + x^2$	$1 + y$	x	Exh.+M 30	G/O/N/F conn.		
PS-22083bd2/P027	(9, 8)	[144, 4]	6	Exact	—	1.00	$1 + x + x^2$	$1 + y$	x	$1 + y$	Exh.+M 31	G/O/N/F conn.		
PS-383aa96d/P028	(9, 8)	[144, 4]	6	Exact	—	1.00	$1 + x^3 y$	$1 + x + x^5 y^4$	0	$1 + x$	Exh.+I 6	G	conn.	
PS-b3421fc2/P030	(9, 8)	[144, 4]	6	Exact	—	1.00	$1 + x^3 y$	$1 + x + x^5 y^4$	0	$x^5 y^4$	Exh.+M 33	G/O conn.		
PS-83570021/P029	(9, 8)	[144, 4]	6	Exact	—	1.00	$1 + x^3 y$	$1 + x + x^5 y^4$	$1 + x^3 y$	$1 + x$	Exh.+M 6	G	conn.	
PS-3e908b54/P031	(9, 8)	[144, 4]	6	Exact	—	1.00	$1 + x^3 y$	$1 + x + x^5 y^4$	$1 + x^3 y$	$x^5 y^4$	Exh.+M 8	G/O	conn.	
PS-7e09500e/P028	(9, 8)	[144, 4]	6	Exact	—	1.00	$1 + x^3 y$	$1 + xy^4 + x^2$	0	$1 + x^2$	Exh.+M 9	G/O	conn.	
PS-0d3127d2/P030	(9, 8)	[144, 4]	6	Exact	—	1.00	$1 + x^3 y$	$1 + xy^4 + x^2$	0	xy^4	Exh.+M 24	G/O	conn.	
PS-171308ed/P029	(9, 8)	[144, 4]	6	Exact	—	1.00	$1 + x^3 y$	$1 + xy^4 + x^2$	$1 + x^3 y$	$1 + x^2$	Exh.+M 6	G	conn.	
PS-7d33c1ec/P031	(9, 8)	[144, 4]	6	Exact	—	1.00	$1 + x^3 y$	$1 + xy^4 + x^2$	$1 + x^3 y$	xy^4	Exh.+M 8	G/O	conn.	
PS-9d4543d4/P028	(9, 8)	[144, 4]	6	Exact	—	1.00	$1 + x^3 y$	$1 + x^2 y^4 + x^4$	0	$1 + x^4$	Exh.+M 39	G	conn.	
PS-7a0360a4/P030	(9, 8)	[144, 4]	6	Exact	—	1.00	$1 + x^3 y$	$1 + x^2 y^4 + x^4$	0	$x^2 y^4$	Exh.+M 53	G/O	conn.	
PS-aa99a4dc/P030	(9, 8)	[144, 4]	6	Exact	—	1.00	$1 + x^3 y^3$	$1 + xy^4 + x^2$	0	xy^4	Exh.+M 15	G/O	conn.	
PS-9b58aa22/P030	(9, 8)	[144, 4]	6	Exact	—	1.00	$1 + x^3 y^3$	$1 + x^2 y^4 + x^4$	0	$x^2 y^4$	Exh.+M 6	G/O	conn.	
PS-beff910c/P030	(9, 8)	[144, 4]	6	Exact	—	1.00	$1 + x^3 y^5$	$1 + xy^4 + x^2$	0	xy^4	Exh.+M 129	G/O	conn.	
PS-f320ede8/P030	(9, 8)	[144, 4]	6	Exact	—	1.00	$1 + x^3 y^5$	$1 + x^2 y^4 + x^4$	0	$x^2 y^4$	Exh.+M 123	G/O	conn.	
PS-325fa0b8/P028	(9, 8)	[144, 4]	6	Exact	—	1.00	$1 + x^6 y$	$1 + x + x^5 y^4$	0	$1 + x$	Exh.+M 54	G/O	conn.	
PS-c0d3be0c/P030	(9, 8)	[144, 4]	6	Exact	—	1.00	$1 + x^6 y$	$1 + x + x^5 y^4$	0	$x^5 y^4$	Exh.+M 6	G/O	conn.	
PS-8679612c/P029	(9, 8)	[144, 4]	6	Exact	—	1.00	$1 + x^6 y$	$1 + x + x^5 y^4$	$1 + x^6 y$	$1 + x$	Exh.+M 1	G	conn.	
PS-05b1316c/P030	(9, 8)	[144, 4]	6	Exact	—	1.00	$1 + x^6 y$	$1 + x^2 y^4 + x^4$	0	$x^2 y^4$	Exh.+M 128	G/O	conn.	
PS-d8d3a477/P029	(9, 8)	[144, 4]	6	Exact	—	1.00	$1 + x^6 y$	$1 + x^2 y^4 + x^4$	$1 + x^6 y$	$1 + x^4$	Exh.+M 1	G	conn.	
PS-d7d8567d/P029	(9, 8)	[144, 4]	6	Exact	—	1.00	$1 + x^6 y^3$	$1 + x + x^5 y^4$	$1 + x^6 y^3$	$1 + x$	Exh.+M 115	G/O	conn.	
PS-2db30326/P030	(9, 8)	[144, 4]	6	Exact	—	1.00	$1 + x^6 y^3$	$1 + xy^4 + x^2$	0	xy^4	Exh.+M 8	G/O	conn.	
PS-9d2c2f66/P029	(9, 8)	[144, 4]	6	Exact	—	1.00	$1 + x^6 y^3$	$1 + x^2 y^4 + x^4$	$1 + x^6 y^3$	$1 + x^4$	Exh.+M 1	G	conn.	
PS-a475dd05/P028	(9, 8)	[144, 4]	6	Exact	—	1.00	$1 + x^6 y^5$	$1 + x + x^5 y^4$	0	$1 + x$	Exh.+M 115	G/O	conn.	
PS-3717b4f6/P030	(9, 8)	[144, 4]	6	Exact	—	1.00	$1 + x^6 y^5$	$1 + x + x^5 y^4$	0	$x^5 y^4$	Exh.+M 7	G/O	conn.	
PS-3031a34f/P028	(9, 8)	[144, 4]	6	Exact	—	1.00	$1 + x^6 y^5$	$1 + xy^4 + x^2$	0	$1 + x^2$	Exh.+M 39	G	conn.	
PS-de6ef659/P028	(9, 8)	[144, 4]	6	Exact	—	1.00	$1 + x^6 y^5$	$1 + x^2 y^4 + x^4$	0	$1 + x^4$	Exh.+M 43	G	conn.	
PS-aacbee20/P039	(9, 9)	[162, 2]	6	Exact	—	0.44	$1 + y$	$1 + x + x^2$	$1 + y$	x	Exh.+M 162	G/O/N/F conn.		
PS-514d488f/P039	(9, 9)	[162, 2]	6	Exact	—	0.44	$1 + x + x^2$	$1 + y$	x	$1 + y$	Exh.+M 162	G/O/N/F conn.		

TABLE S8: PBB-large compact catalogue (73 displayed of 79 retained proposals; 56 presentation classes). The table includes every proposal in an exact class and one deterministic representative of every nonexact class. Own evidence and Evid. describe the proposal before class merging; Class effect reports when merging makes the displayed d and FOM exact, tighter, or rigorously supported. Column abbreviations, filtering, and evidence conventions are defined above.

ID/class	(ℓ, m)	$[n, k]$	d	Own evidence	Class effect	FOM	A	B	C	D	Evid.	Obs. Attr.	Decomp.
PL-ba29a9df/P040 (12, 9)	[216, 2]	8	Exact	—	0.59	$1 + y$	$1 + x + x^2$	$1 + y$	x	M	126	G/O/N/F conn.	
PL-39082ddc/P040 (12, 9)	[216, 2]	8	Exact	—	0.59	$1 + x + x^2$	$1 + y$	x	$1 + y$	M	126	G/O/N/F conn.	
PL-9b97a02b/P053 (12, 12)	[288, 48]	3	Exact	—	1.50	$1 + x^6 y^6$	$1 + x^2 y^6 + x^4$	$1 + x^6 y^6$	$1 + x^4$	Exh.	20	G/O	$24 \times [12, 2]$
PL-78c3e7bf/P052 (12, 12)	[288, 24]	4	Exact	—	1.33	$1 + x^3 y^6$	$1 + x^2 y^6 + x^4$	0	$1 + x^4$	Exh.	3	G/O	$6 \times [48, 4]$
PL-ad1cc828/P049 (12, 12)	[288, 16]	4	Exact	—	0.89	$1 + xy$	$1 + x^2 y^6 + x^4$	0	$1 + x^4$	Exh.	28	G/O	$4 \times [72, 4]$
PL-0bbae5c3/P048 (12, 12)	[288, 16]	4	Exact	—	0.89	$1 + xy^5$	$1 + x^2 y^6 + x^4$	$1 + xy^5$	$x^2 y^6$	Exh.	13	G/O	$4 \times [72, 4]$
PL-d3deda3d/P048 (12, 12)	[288, 16]	4	Exact	—	0.89	$1 + x^3 y$	$1 + x^2 y^6 + x^4$	$1 + x^3 y$	$x^2 y^6$	Exh.	15	G/O	$4 \times [72, 4]$
PL-aecf3364/P050 (12, 12)	[288, 16]	4	Exact	—	0.89	$1 + x^4 y^2$	$1 + x^2 y^6 + x^4$	$1 + x^4 y^2$	$x^2 y^6$	Exh.	1	G	$4 \times [72, 4]$
PL-a1c1ac59/P048 (12, 12)	[288, 16]	4	Exact	—	0.89	$1 + x^5 y^5$	$1 + x^2 y^6 + x^4$	$1 + x^5 y^5$	$x^2 y^6$	Exh.	1	O	$4 \times [72, 4]$
PL-f2462ed9/P051 (12, 12)	[288, 16]	4	Exact	—	0.89	$1 + x^6 y^2$	$1 + x^2 y^6 + x^4$	0	$1 + x^4$	Exh.	1	G	$8 \times [36, 2]$
PL-adcbb3fc/P045 (12, 12)	[288, 8]	6	Exact	—	1.00	$1 + x^4 y^2$	$1 + xy^6 + x^2$	$1 + x^4 y^2$	$1 + x^2$	M	27	G/O	$2 \times [144, 4]$
PL-605688ff/P046 (12, 12)	[288, 8]	4	Exact	—	0.44	$1 + x^2 y$	$1 + x^2 y^6 + x^4$	0	$1 + x^4$	Exh.	9	G/O	$2 \times [144, 4]$
PL-592276b3/P047 (12, 12)	[288, 8]	4	Exact	—	0.44	$1 + x^2 y$	$1 + x^2 y^6 + x^4$	0	$x^2 y^6$	Exh.	5	G/O	$2 \times [144, 4]$
PL-b977d26c/P046 (12, 12)	[288, 8]	4	Exact	—	0.44	$1 + x^2 y^5$	$1 + x^2 y^6 + x^4$	0	$1 + x^4$	Exh.	1	O	$2 \times [144, 4]$
PL-40e78b8c/P046 (12, 12)	[288, 8]	4	Exact	—	0.44	$1 + x^3 y^2$	$1 + x^2 y^6 + x^4$	0	$1 + x^4$	Exh.	26	G/O	$2 \times [144, 4]$

TABLE S8 (continued)

ID/class	(ℓ, m)	$\llbracket n, k \rrbracket$	d	Own evidence	Class	effect	FOM	A	B	C	D	Evid.	Obs.	Attr.	Decomp.
PL-bff58260/P047	(12, 12)	$\llbracket 288, 8 \rrbracket$	4	Exact	—	0.44		$1 + x^4 y$	$1 + x^2 y^6 + x^4$	0	$x^2 y^6$	Exh.	27	G/O	$2 \times \llbracket 144, 4 \rrbracket$
PL-a608553f/P046	(12, 12)	$\llbracket 288, 8 \rrbracket$	4	Exact	—	0.44		$1 + x^6 y^5$	$1 + x^2 y^6 + x^4$	0	$1 + x^4$	Exh.	24	G/O	$2 \times \llbracket 144, 4 \rrbracket$
PL-592409d9/P044	(12, 12)	$\llbracket 288, 4 \rrbracket$	8	Exact	—	0.89		$1 + x + x^2$	$1 + y$	x	$1 + y$	M	100	G/O/N/F	conn.
PL-3bed6781/P041	(12, 12)	$\llbracket 288, 4 \rrbracket$	8	Exact	—	0.89		$1 + x^2 y$	$1 + x y^6 + x^2$	0	$1 + x^2$	M	16	G/O	conn.
PL-80889114/P042	(12, 12)	$\llbracket 288, 4 \rrbracket$	8	Exact	—	0.89		$1 + x^5 y$	$1 + x y^6 + x^2$	0	$x y^6$	M	12	G/O	conn.
PL-1df1bcd4/P044	(12, 12)	$\llbracket 288, 4 \rrbracket$	8	Interval	Exact	0.89		$1 + y$	$1 + x + x^2$	$1 + y$	x	L4+I+B 98	G/O/N/F	conn.	
PL-3f25e19f/P043	(12, 12)	$\llbracket 288, 4 \rrbracket$	[5, 8]	Interval	—	[0.35, 0.89]		$1 + x^3 y$	$1 + x y^6 + x^2$	$1 + x^3 y$	$1 + x^2$	L4+I+B 14	G/O	conn.	
PL-2da51468/P044	(12, 12)	$\llbracket 288, 4 \rrbracket$	8	Interval	Exact	0.89		$1 + x^4 y$	$1 + x y^6 + x^2$	$1 + x^4 y$	$x y^6$	L4+I+B 28	G/O	conn.	
PL-2b75ed37/P044	(12, 12)	$\llbracket 288, 4 \rrbracket$	8	Interval	Exact	0.89		$1 + x^5 y$	$1 + x y^6 + x^2$	$1 + x^5 y$	$x y^6$	L4+I+B 11	G/O	conn.	
PL-93be42eb/P068	(15, 12)	$\llbracket 360, 12 \rrbracket$	4	Exact	—	0.53		$1 + x^3 y^3$	$1 + x^2 y^6 + x^4$	$1 + x^3 y^3$	$x^2 y^6$	Exh.	28	G/O	$3 \times \llbracket 120, 4 \rrbracket$
PL-f49f60d8/P069	(15, 12)	$\llbracket 360, 12 \rrbracket$	4	Exact	—	0.53		$1 + x^6 y^3$	$1 + x y^6 + x^2$	0	$1 + x^2$	Exh.	12	G/O	$3 \times \llbracket 120, 4 \rrbracket$
PL-8ee754a5/P062	(15, 12)	$\llbracket 360, 4 \rrbracket$	11	Exact	—	1.34		$1 + x^2 y^4$	$1 + x^2 y^6 + x^4$	$1 + x^2 y^4$	$1 + x^4$	M	23	G/O	$2 \times \llbracket 180, 2 \rrbracket$
PL-4602a8fa/P059	(15, 12)	$\llbracket 360, 4 \rrbracket$	10	Exact	—	1.11		$1 + x + x^2$	$1 + y$	x	$1 + y$	M	101	G/O/N/F	conn.
PL-93708752/P060	(15, 12)	$\llbracket 360, 4 \rrbracket$	8	Exact	—	0.71		$1 + x y$	$1 + x^2 y^6 + x^4$	$1 + x y$	$1 + x^4$	M	3	G/O	conn.
PL-6eca49d3/P065	(15, 12)	$\llbracket 360, 4 \rrbracket$	8	Exact	—	0.71		$1 + x^3 y$	$1 + x y^6 + x^2$	0	$1 + x^2$	M	14	G/O	conn.
PL-689c272b/P060	(15, 12)	$\llbracket 360, 4 \rrbracket$	8	Exact	—	0.71		$1 + x^3 y$	$1 + x y^6 + x^2$	$1 + x^3 y$	$1 + x^2$	M	14	G/O	conn.
PL-48f3faa4/P058	(15, 12)	$\llbracket 360, 4 \rrbracket$	7	Exact	—	0.54		$1 + x^3 y^4$	$1 + x y^6 + x^2$	$1 + x^3 y^4$	$x y^6$	M	3	G	$2 \times \llbracket 180, 2 \rrbracket$
PL-6b21188d/P061	(15, 12)	$\llbracket 360, 4 \rrbracket$	5	Exact	—	0.28		$1 + x^2 y^2$	$1 + x^2 y^6 + x^4$	0	$x^2 y^6$	M	30	G/O	$2 \times \llbracket 180, 2 \rrbracket$
PL-15407ac6/P057	(15, 12)	$\llbracket 360, 4 \rrbracket$	5	Exact	—	0.28		$1 + x^2 y^2$	$1 + x^2 y^6 + x^4$	$1 + x^2 y^2$	$1 + x^4$	M	27	G/O	$2 \times \llbracket 180, 2 \rrbracket$
PL-2593a470/P061	(15, 12)	$\llbracket 360, 4 \rrbracket$	5	Exact	—	0.28		$1 + x^3 y^2$	$1 + x^2 y^6 + x^4$	0	$x^2 y^6$	M	1	G	$2 \times \llbracket 180, 2 \rrbracket$
PL-d096de3f/P054	(15, 12)	$\llbracket 360, 4 \rrbracket$	[5, 12]	Interval	—	[0.28, 1.60]		$1 + x y^5$	$1 + x y^6 + x^2$	0	$x y^6$	L4+I+B 23	G/O	conn.	
PL-28c4edd8/P055	(15, 12)	$\llbracket 360, 4 \rrbracket$	[5, 12]	Interval	—	[0.28, 1.60]		$1 + x^4 y$	$1 + x y^6 + x^2$	0	$1 + x^2$	L4+I+B 4	G/O	conn.	
PL-58200d48/P066	(15, 12)	$\llbracket 360, 4 \rrbracket$	[5, 12]	Interval	—	[0.28, 1.60]		$1 + x^6 y^5$	$1 + x y^6 + x^2$	$1 + x^6 y^5$	$1 + x^2$	L4+I+B 3	G	conn.	
PL-959e2913/P059	(15, 12)	$\llbracket 360, 4 \rrbracket$	10	Interval	Exact	1.11		$1 + y$	$1 + x + x^2$	$1 + y$	x	L4+I+B 126	G/O/N/F	conn.	
PL-b65c89f3/P067	(15, 12)	$\llbracket 360, 4 \rrbracket$	[5, 9]	Interval	—	[0.28, 0.90]		$1 + x^5 y^2$	$1 + x^2 y^6 + x^4$	$1 + x^5 y^2$	$1 + x^4$	L4+I+B 3	G/O	$2 \times \llbracket 180, 2 \rrbracket$	
PL-0b5dc15c/P065	(15, 12)	$\llbracket 360, 4 \rrbracket$	8	Interval	Exact	0.71		$1 + x^2 y$	$1 + x y^6 + x^2$	0	$1 + x^2$	L4+I+B 16	G/O	conn.	
PL-18401d67/P056	(15, 12)	$\llbracket 360, 4 \rrbracket$	[5, 8]	Interval	—	[0.28, 0.71]		$1 + x^4 y^5$	$1 + x^2 y^6 + x^4$	0	$x^2 y^6$	L4+I+B 10	G/O	conn.	
PL-9a84c818/P064	(15, 12)	$\llbracket 360, 4 \rrbracket$	[5, 7]	Interval	—	[0.28, 0.54]		$1 + x y^4$	$1 + x^2 y^6 + x^4$	0	$x^2 y^6$	L4+I+B 20	G/O	$2 \times \llbracket 180, 2 \rrbracket$	
PL-189e1cca/P063	(15, 12)	$\llbracket 360, 4 \rrbracket$	[5, 7]	Interval	—	[0.28, 0.54]		$1 + x y^4$	$1 + x^2 y^6 + x^4$	$1 + x y^4$	$1 + x^4$	L4+I+B 2	G	$2 \times \llbracket 180, 2 \rrbracket$	
PL-ce4ae900/P070	(15, 14)	$\llbracket 420, 4 \rrbracket$	10	Exact	—	0.95		$1 + y$	$1 + x + x^2$	$1 + y$	x	M	126	G/O/N/F	conn.
PL-7c740754/P070	(15, 14)	$\llbracket 420, 4 \rrbracket$	10	Exact	—	0.95		$1 + x + x^2$	$1 + y$	x	$1 + y$	M	126	G/O/N/F	conn.
PL-5f3d871a/P089	(18, 12)	$\llbracket 432, 24 \rrbracket$	4	Exact	—	0.89		$1 + x^3 y^2$	$1 + x^3 y^6 + x^6$	0	$x^3 y^6$	Exh.	9	G/O	$12 \times \llbracket 36, 2 \rrbracket$
PL-78dcc188/P090	(18, 12)	$\llbracket 432, 24 \rrbracket$	4	Exact	—	0.89		$1 + x^6 y^3$	$1 + x^2 y^6 + x^4$	$1 + x^6 y^3$	$1 + x^4$	Exh.	26	G/O	$6 \times \llbracket 72, 4 \rrbracket$
PL-3e07b603/P094	(18, 12)	$\llbracket 432, 24 \rrbracket$	4	Exact	—	0.89		$1 + x^6 y^3$	$1 + x^2 y^6 + x^4$	$1 + x^6 y^3$	$x^2 y^6$	Exh.	26	G/O	$6 \times \llbracket 72, 4 \rrbracket$
PL-88c05b17/P092	(18, 12)	$\llbracket 432, 24 \rrbracket$	3	Exact	—	0.50		$1 + x^3 y^6$	$1 + x y^6 + x^2$	$1 + x^3 y^6$	$1 + x^2$	Exh.	13	G/O	$12 \times \llbracket 36, 2 \rrbracket$
PL-38d2b569/P095	(18, 12)	$\llbracket 432, 24 \rrbracket$	3	Exact	—	0.50		$1 + x^6 y^4$	$1 + x^3 y^6 + x^6$	0	$x^3 y^6$	Exh.	23	G/O	$12 \times \llbracket 36, 2 \rrbracket$
PL-704d75b7/P091	(18, 12)	$\llbracket 432, 24 \rrbracket$	3	Exact	—	0.50		$1 + x^6 y^4$	$1 + x^3 y^6 + x^6$	$1 + x^6 y^4$	$x^3 y^6$	Exh.	14	G/O	$12 \times \llbracket 36, 2 \rrbracket$
PL-2dbeba05/P088	(18, 12)	$\llbracket 432, 24 \rrbracket$	3	Exact	—	0.50		$1 + x^6 y^6$	$1 + x^2 y^6 + x^4$	0	$x^2 y^6$	Exh.	17	G/O	$12 \times \llbracket 36, 2 \rrbracket$
PL-aff57094/P093	(18, 12)	$\llbracket 432, 24 \rrbracket$	3	Exact	—	0.50		$1 + x^6 y^6$	$1 + x^2 y^6 + x^4$	$1 + x^6 y^6$	$x^2 y^6$	Exh.	8	G	$12 \times \llbracket 36, 2 \rrbracket$
PL-aefa6773/P087	(18, 12)	$\llbracket 432, 12 \rrbracket$	8	Exact	—	1.78		$1 + x^3 y^3$	$1 + x y^6 + x^2$	$1 + x^3 y^3$	$x y^6$	M	14	G/O	$3 \times \llbracket 144, 4 \rrbracket$
PL-597f6b12/P086	(18, 12)	$\llbracket 432, 12 \rrbracket$	4	Exact	—	0.44		$1 + x^3 y$	$1 + x^3 y^6 + x^6$	0	$x^3 y^6$	Exh.	83	G/O/N	$3 \times \llbracket 144, 4 \rrbracket$
PL-cdea2b4a/P085	(18, 12)	$\llbracket 432, 12 \rrbracket$	4	Exact	—	0.44		$1 + x^3 y$	$1 + x^3 y^6 + x^6$	$1 + x^3 y$	$1 + x^6$	Exh.	5	G/O	$3 \times \llbracket 144, 4 \rrbracket$
PL-36c5ce3b/P079	(18, 12)	$\llbracket 432, 8 \rrbracket$	10	Exact	—	1.85		$1 + x^2 y^2$	$1 + x y^6 + x^2$	$1 + x^2 y^2$	$1 + x^2$	M	15	G/O	$2 \times \llbracket 216, 4 \rrbracket$
PL-ae4cc6ad/P076	(18, 12)	$\llbracket 432, 8 \rrbracket$	10	Exact	—	1.85		$1 + x^4 y^2$	$1 + x y^6 + x^2$	0	$1 + x^2$	M	12	G/O	$2 \times \llbracket 216, 4 \rrbracket$
PL-cdaa4f44/P084	(18, 12)	$\llbracket 432, 8 \rrbracket$	8	Exact	—	1.19		$1 + x y$	$1 + x^2 y^6 + x^4$	$1 + x y$	$x^2 y^6$	M	2	G/O	$2 \times \llbracket 216, 4 \rrbracket$
PL-9f24e6b3/P082	(18, 12)	$\llbracket 432, 8 \rrbracket$	8	Exact	—	1.19		$1 + x^2 y^5$	$1 + x^2 y^6 + x^4$	$1 + x^2 y^5$	$1 + x^4$	M	26	G/O	$2 \times \llbracket 216, 4 \rrbracket$
PL-8346d4c6/P084	(18, 12)	$\llbracket 432, 8 \rrbracket$	8	Exact	—	1.19		$1 + x^4 y$	$1 + x^2 y^6 + x^4$	$1 + x^4 y$	$x^2 y^6$	M	1	O	$2 \times \llbracket 216, 4 \rrbracket$
PL-2b5f2b8d/P083	(18, 12)	$\llbracket 432, 8 \rrbracket$	7	Exact	—	0.91		$1 + x^2 y^4$	$1 + x y^6 + x^2$	0	$1 + x^2$	M	26	G/O	$4 \times \llbracket 108, 2 \rrbracket$
PL-c60edd0d/P081	(18, 12)	$\llbracket 432, 8 \rrbracket$	7	Exact	—	0.91		$1 + x^2 y^4$	$1 + x y^6 + x^2$	$1 + x^2 y^4$	$x y^6$	M	1	O	$4 \times \llbracket 108, 2 \rrbracket$
PL-55176200/P083	(18, 12)	$\llbracket 432, 8 \rrbracket$	7	Exact	—	0.91		$1 + x^2 y^4$	$1 + x^2 y^6 + x^4$	0	$1 + x^4$	M	2	G	$4 \times \llbracket 108, 2 \rrbracket$
PL-9fc3055a/P077	(18, 12)	$\llbracket 432, 8 \rrbracket$	7	Exact	—	0.91		$1 + x^2 y^4$	$1 + x^2 y^6 + x^4$	0	$x^2 y^6$	M	12	G/O	$4 \times \llbracket 108, 2 \rrbracket$
PL-232490ff/P080	(18, 12)	$\llbracket 432, 8 \rrbracket$	3	Exact	—	0.17		$1 + x^6 y^4$	$1 + x^2 y^6 + x^4$	0	$x^2 y^6$	Exh.	8	G/O	$4 \times \llbracket 108, 2 \rrbracket$
PL-d8385899/P078	(18, 12)	$\llbracket 432, 8 \rrbracket$	[5, 8]	Interval	—	[0.46, 1.19]		$1 + x^5 y^5$	$1 + x^2 y^6 + x^4$	0	$1 + x^4$	L4+I+B 17	G/O	$2 \times \llbracket 216, 4 \rrbracket$	
PL-71c11b15/P072	(18, 12)	$\llbracket 432, 4 \rrbracket$	[5, 12]	Interval	—	[0.23, 1.33]		$1 + x + x^2$	$1 + y$	x	$1 + y$	L4+I+B 105	G/O/N/F	conn.	
PL-1820300c/P075	(18, 12)	$\llbracket 432, 4 \rrbracket$	[5, 12]	Interval	—	[0.23, 1.33]		$1 + x y$	$1 + x y^6 + x^2$	0	$x y^6$	L4+I+B 40	G/O	conn.	
PL-3a6796de/P074	(18, 12)	$\llbracket 432, 4 \rrbracket$	[5, 12]	Interval	—	[0.23, 1.33]		$1 + x y$	$1 + x y^6 + x^2$	$1 + x y$	$1 + x^2$	L4+I+B 85	G/O	conn.	
PL-8f03e9a5/P071	(18, 12)	$\llbracket 432, 4 \rrbracket$	[5, 12]	Interval	—	[0.23, 1.33]		$1 + x^2 y$	$1 + x y^6 + x^2$	$1 + x^2 y$	$x y^6$	L4+I+B 15	G/O	conn.	
PL-1713aed5/P073	(18, 12)	$\llbracket 432, 4 \rrbracket$	[5, 12]	Interval	—	[0.23, 1.33]		$1 + x^2 y^5$	$1 + x y^6 + x^2$	0	$1 + x^2$	L4+I+B 21	G/O	conn.	

[1] J. Cruz-Benito, A. W. Cross, D. Kremer, and I. Faro, qcode-discovery: LLM-guided evolutionary discovery of quantum LDPC codes, <https://github.com/qiskit-community/qcode-discovery> (2026), source code and data.